1. A point charge of 20 nC is fixed at the origin of x-y plane. The work done in taking a second point charge of – 50 nC from point (4 cm, 0) to point (0, 5 cm) is :

(A) 45 μJ

(B) – 45 μJ

(C) 90 μJ

(D) – 90 μJ

1. A point charge of 20 nC is fixed at the origin of x-y plane. The work done in taking a second point charge of – 50 nC from point (4 cm, 0) to point (0, 5 cm) is :

(A) 45 μJ (B) – 45 μJ

(C) 90 μJ (D) – 90 μJ

CBSE 25 C

(A) 45 μJ

15. *Assertion (A) :* Electrostatic force is a conservative force.

Reason (R) : In an electrostatic field, the work done between two points per unit positive charge depends on the path followed.

CBSE 25 C

15. *Assertion (A) :* Electrostatic force is a conservative force.

Reason (R) : In an electrostatic field, the work done between two points per unit positive charge depends on the path followed.

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(C) Assertion (A) is true, but reason (R) is false

31. (a) (i) A parallel plate capacitor A of capacitance C is charged by a battery to a potential 'V'. The battery is disconnected and an uncharged identical capacitor B is connected across it. Calculate for the capacitor A the new value of the :

(I) charge

(II) potential difference

(III) energy stored

Justify your answers.

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(ii) Draw the pattern of electric field lines due to :

(I) positively charged conducting sphere, and

(II) an electric dipole. *5*

(i) (I) Charge will become half

Charge will flow from capacitors A (high potential) to capacitor B (low potential) till they achieve equilibrium.

Alternatively

$$q_A = cv \qquad q_B = 0$$

$$V_{common} = \frac{q_A + q_B}{c_A + c_B} = \frac{cv}{2c}$$ ½

$$= \frac{v}{2}$$

$$q'_A = cv_{common} = \frac{cv}{2} = \frac{q_A}{2}$$ ½

(II) Potential difference will become half. ½

$$V_{common} = \frac{q_A + q_B}{c_A + c_B}$$

$$= \frac{cv + 0}{2c} = \frac{v}{2}$$ ½

(III) Energy stored will becomes one fourth ½

$$U_A = \frac{1}{2}CV^2$$

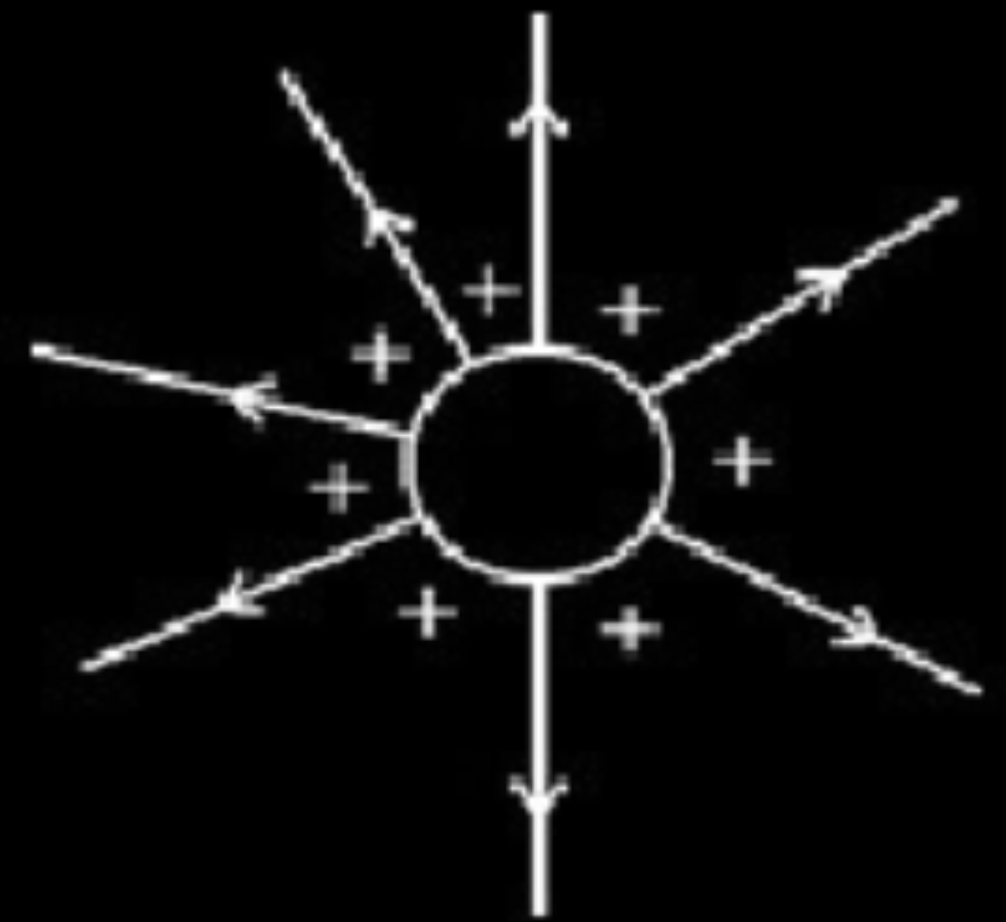
$$U'_A = \frac{1}{2}CV^2_{common}$$

$$= \frac{1}{2}C\left(\frac{V}{2}\right)^2$$

$$= \frac{U_A}{4}$$ ½

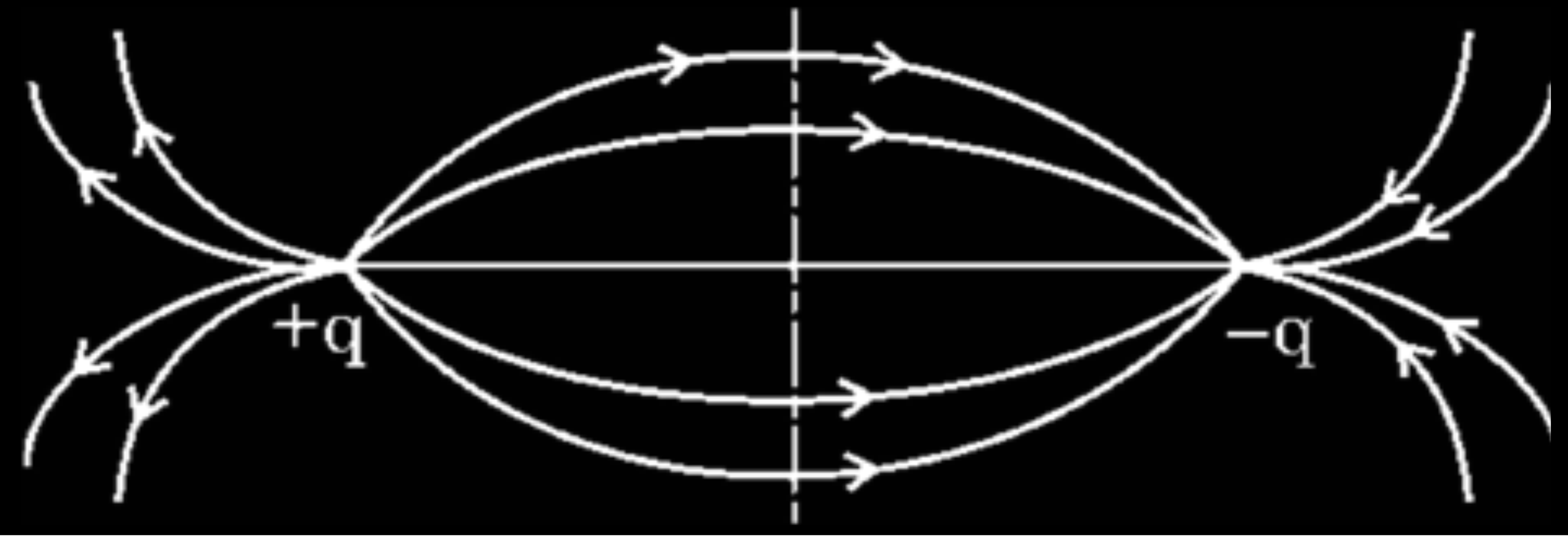
(ii)

(I)



1

(II)



1

1. A point charge of 10 nC is fixed at the origin of x-y plane. A second point charge of – 20 nC is moved from point (5 cm, 0) to point (0, 5 cm). The change in the potential energy of the system of the two charges is :

(A) 36 μJ (B) – 36 μJ

(C) 72 μJ (D) 0

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(D) Zero

18. A 50 pF capacitor is charged by a 100 V battery. The battery is disconnected and the capacitor is connected to another uncharged 50 pF capacitor. Find the energy stored in the combination. CBSE 25 C *2*

18. A 50 pF capacitor is charged by a 100 V battery. The battery is disconnected and the capacitor is connected to another uncharged 50 pF capacitor. Find the energy stored in the combination. CBSE 25 C *2*

$C_1 = C_2 = 50 \times 10^{-12}$ F	
$V_1 = 100$ V $\quad V_2 = 0$	
Common potential $V = \dfrac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$	½
$V = \dfrac{50\times10^{-12}\times100}{100\times10^{-12}} = 50V$	½
Energy stored $U = \dfrac{1}{2}(C_1 + C_2) V^2$	½
$U = \dfrac{1}{2} \times 100 \times 10^{-12} \times 50 \times 50$ J	
$U = 1.25 \times 10^{-7}$ J	½

Alternatively:

$q_1 = C_1V_1$

$= 50\times10^{-12}\times100$

$= 5\times10^{-9}$ C

$q_2 = 0$

$q = q_1 + q_2$

$q = 5\times10^{-9}$ C

$U = \dfrac{1}{2}\dfrac{q^2}{(C_1+C_2)} = \dfrac{1}{2}\dfrac{(5\times10^{-9})^2}{(100\times10^{-12})}$

$U = 1.25\times10^{-7}\ J$

(ii) A dipole is aligned parallel to the electric field. Find the work done in rotating it through an angle (I) $180°$, and (II) $90°$.

CBSE 25 C

(ii) A dipole is aligned parallel to the electric field. Find the work done in rotating it through an angle (I) 180°, and (II) 90°.

CBSE 25 C

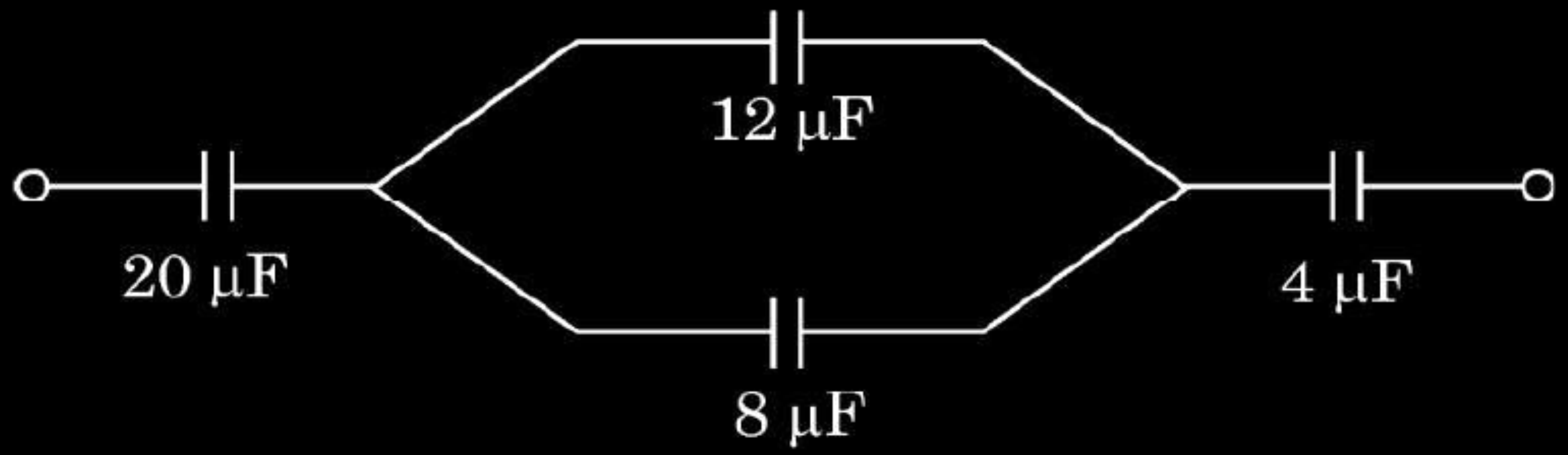
(ii) $W = U_2 - U_1 = -pE\cos 180° - (-pE\cos 0°)$	
$= pE + pE = 2pE$	½
$W = -pE\cos 90° - (-pE\cos 0°)$	½
$= 0 + pE = pE$	½
	½

(b) (i) A parallel plate capacitor of capacitance C is connected to a battery. Explain how the plates of the capacitor acquire equal and opposite charges. Hence obtain expression for the energy stored in the capacitor.

(ii) A parallel plate capacitor has + 4·0 μC and – 4·0 μC charges on its plates. Find the area of each plate so that an electric field of 1600 V/mm is produced between the plates. CBSE 25 C 5

(b) (i) As soon as we switch on the circuit the plate of the capacitor connected with positive terminal of battery loses electrons and gains positive charge. At the same time plate connected with negative terminal gains electrons and acquires negative charge till the potential difference between the capacitor plates is equal to potential difference between the terminals of battery.	1
Work done in charging the capacitor through charge dQ $dW = VdQ$	½
As, $V = \frac{Q}{C}$ $dW = \frac{Q}{C} dQ$	½
Total work done $\int dW = \int \frac{Q}{C} dQ$	½
$W = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad (\because Q = CV)$	½
(ii) Electric field , $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$	½
$A = \frac{Q}{E\varepsilon_0} = \frac{4.0 \times 10^{-6}}{1600 \times 10^3 \times 8.85 \times 10^{-12}} m^2$	1
$A = 0.28 m^2$	½

1. In the circuit shown, the charge on the left plate of 20 μF capacitor is – 50 μC. The charge on the right plate of the 12 μF capacitor is :

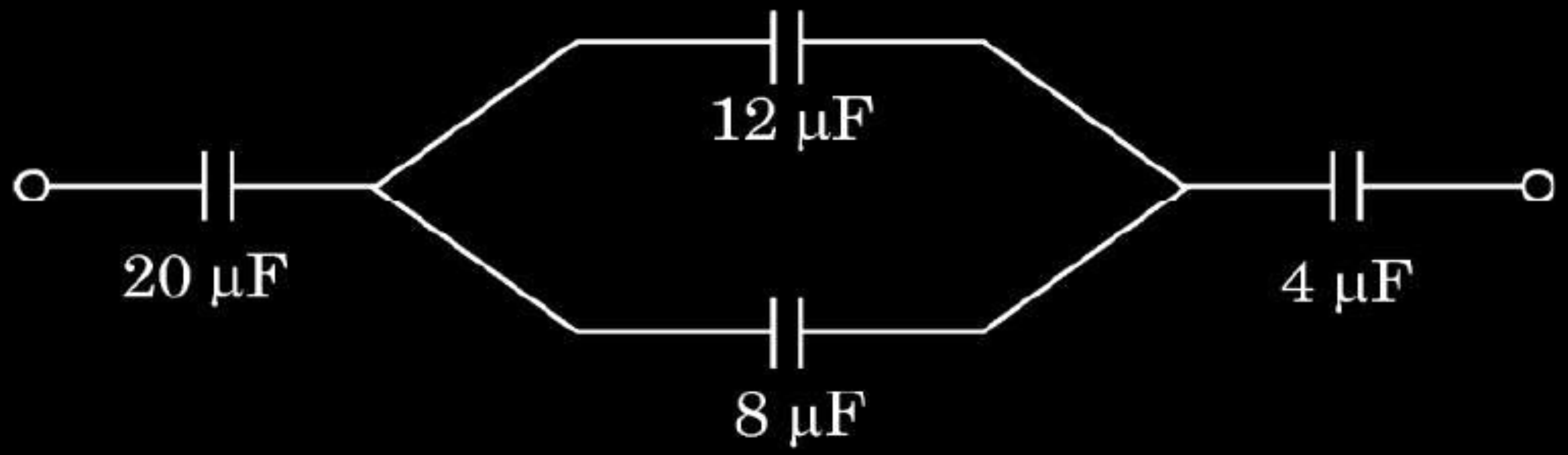


(A) 20 μC (B) – 20 μC

(C) 30 μC (D) – 30 μC

CBSE 25

1. In the circuit shown, the charge on the left plate of 20 μF capacitor is – 50 μC. The charge on the right plate of the 12 μF capacitor is :



(A) 20 μC (B) – 20 μC

(C) 30 μC (D) – 30 μC

CBSE 25

(C) 30 μC

33. (a) (i) A parallel plate capacitor with plate area A and plate separation d has a capacitance C_0. A slab of dielectric constant K having area A and thickness $\left(\frac{d}{4}\right)$ is inserted in the capacitor, parallel to the plates. Find the new value of its capacitance.

(ii) You are provided with a large number of 1 μF identical capacitors and a power supply of 1200 V. The dielectric medium used in each capacitor can withstand up to 200 V only. Find the minimum number of capacitors and their arrangement, required to build a capacitor system of equivalent capacitance of 2 μF for use with this supply. *5*

CBSE 25

(i) $C_0 = \frac{\varepsilon_0 A}{d}$	½
$C = \frac{\varepsilon_0 A}{(d-t) + \frac{t}{K}}$	½
$t = d/4$	
$C = \frac{\varepsilon_0 A}{\left(d - \frac{d}{4}\right) + \frac{d}{4K}} = \frac{\varepsilon_0 A}{d\left(\frac{3}{4} + \frac{1}{4K}\right)}$	1
	1
$= C_0 \frac{4K}{(3K+1)}$	½
(ii) Each capacitance can withstand 200V	
No. of capacitors in each row = $\frac{1200}{200} = 6$	½
Net capacitance of each row= 1/6 μF	
Number of rows = n	
$C_{eq} = C_1 + C_2 + \_\_\_\_\_\_ + C_n$	
$C_{eq} = \frac{1}{6} + \frac{1}{6} + ------n$	
$2 = \frac{n}{6}$	½
$\therefore n = 12$	
Total no. of capacitors in the arrangement = 6 × 12 = 72	½

Alternatively: When dielectric is inserted, the electric field between the plates is $E = E_0/K$

The potential difference will be

$$V = E_0\left(\frac{3d}{4}\right) + E\left(\frac{d}{4}\right)$$

$$= E_0\left(\frac{3d}{4}\right) + \frac{E_0}{K}\left(\frac{d}{4}\right)$$

$$= V_0\left(\frac{3}{4} + \frac{1}{4K}\right)$$

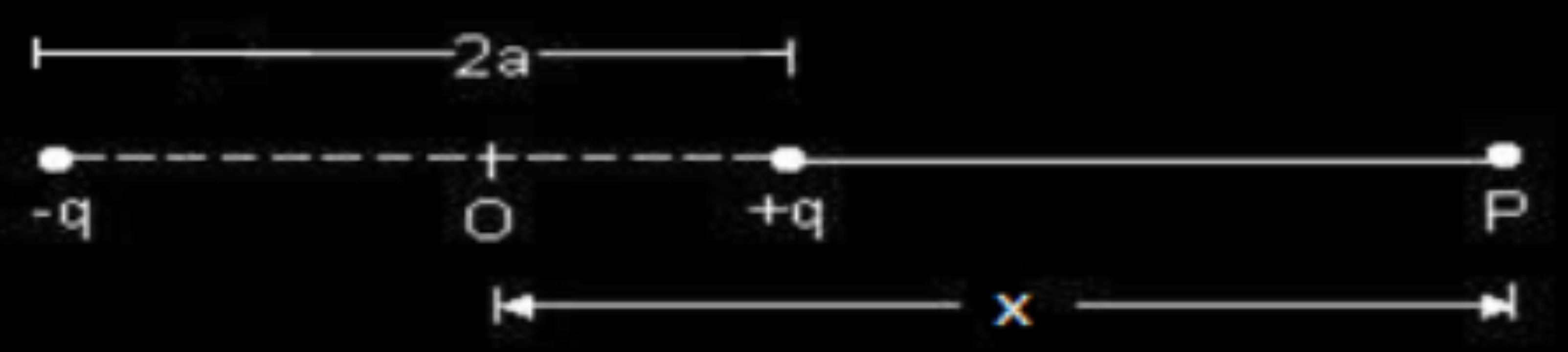
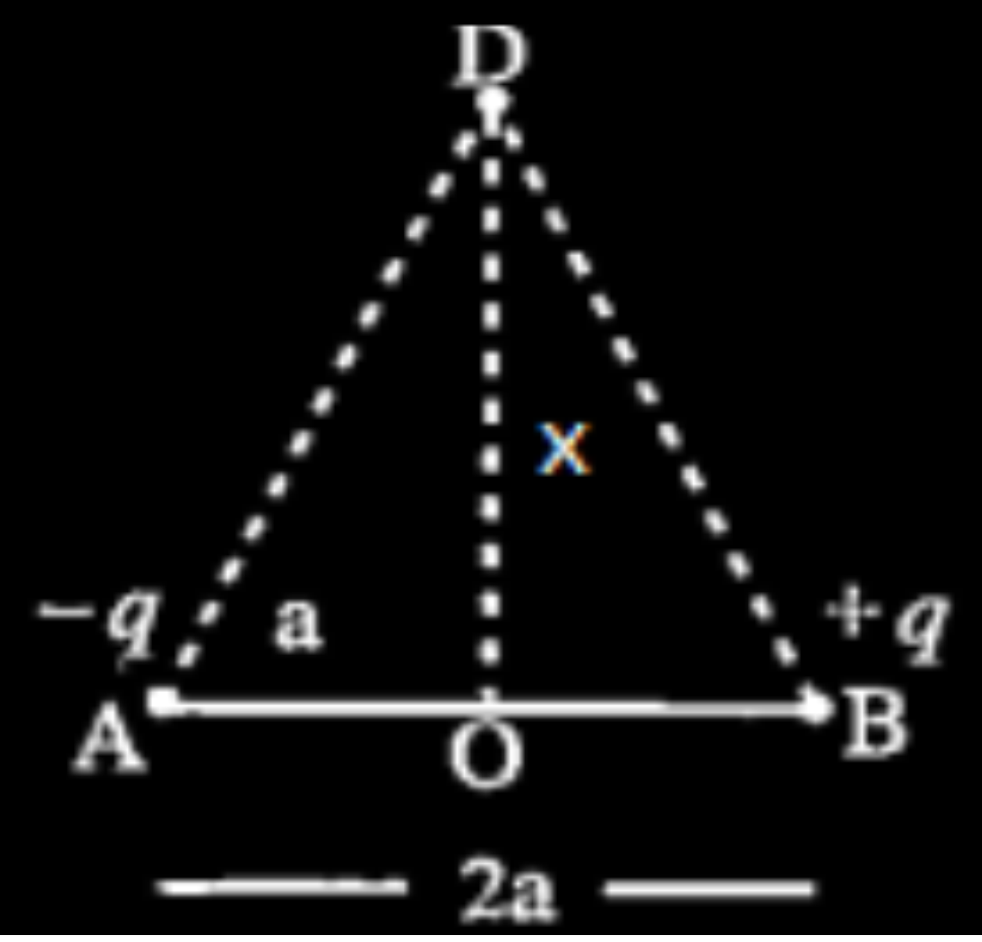
$$V = V_0\left(\frac{3K+1}{4K}\right)$$

$$C = \frac{Q_0}{V} = \left(\frac{4K}{3K+1}\right)\frac{Q_0}{V_0}$$

$$C = C_0\left(\frac{4K}{3K+1}\right)$$

(b) (i) An electric dipole of dipole moment p consists of point charges q and – q, separated by 2a. Derive an expression for electric potential in terms of its dipole moment at a point at a distance x (>> a) from its centre and lying (I) along its axis, and (II) along its bisector line.

CBSE 25

I. Along its axis	
$V_- = \frac{-kq}{x+a}$	
	
$V_+ = \frac{kq}{x-a}$	½
$V = V_- + V_+$	
$= kq\left(\frac{-1}{x+a} + \frac{1}{x-a}\right)$	
$= kq\frac{2a}{(x^2-a^2)} = \frac{kp}{x^2-a^2}$	½
$x \gg a \quad \therefore V = \frac{kp}{x^2}$	½
II. Along the bisector line	
	
$V_- = \frac{kq}{\sqrt{x^2+a^2}}$	½
$V_+ = \frac{-kq}{\sqrt{x^2+a^2}}$	½
$V = V_- + V_+$	½
$= 0$	

1. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively :

(A) $E = 0, \ V = 0$

(B) $E = 0, \ V = V_0$ (a constant)

(C) $E \neq 0, \ V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

CBSE 25

1. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively :

(A) $E = 0, \ V = 0$

(B) $E = 0, \ V = V_0$ (a constant)

(C) $E \neq 0, \ V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

CBSE 25

(B) $E = 0$, $V = V_0$ (a constant)

28. Two point charges of $-5\ \mu C$ and $2\ \mu C$ are located in free space at $(-4\text{ cm}, 0)$ and $(6\text{ cm}, 0)$ respectively.

(a) Calculate the amount of work done to separate the two charges at infinite distance.

(b) If this system of charges was initially kept in an electric field $E = \dfrac{A}{r^2}$, where $A = 8 \times 10^4\ \text{N C}^{-1}\ \text{m}^2$, calculate the electrostatic potential energy of the system. CBSE 25 *3*

(a) $U=\frac{kq_1q_2}{r}$	½
$=\frac{9\times10^{9}\times\left(-5\times10^{-6}\right)\times\left(2\times10^{-6}\right)}{10\times10^{-2}}$	
$=-0.9\,J$	
W = U(infinity) – U(r) = 0.9J	½
(b) $E=\frac{-dV}{dr}$	½
$dV=\frac{-A}{r^{2}}dr$	
$V=\frac{A}{r}$	
$U=q_1\,V_1+q_2\,V_2+\frac{kq_1q_2}{r}$	½
$=\left(-5\times10^{-6}\right)\left(\frac{8\times10^{4}}{4\times10^{-2}}\right)+\left(2\times10^{-6}\right)\left(\frac{8\times10^{4}}{6\times10^{-2}}\right)-0.9$	½
$=-10+2.67-0.9$	
$=-8.24\,J$	½

1. Two horizontal plates, separated by 1 cm, are arranged one above the other. A particle of mass 5 mg and charge 2 nC is released in air between the plates. The potential difference that should be applied to the plates so that the particle remains suspended between them, is :

(A) 250 V (B) 200 V

CBSE 25

(C) 100 V (D) 50 V

1. Two horizontal plates, separated by 1 cm, are arranged one above the other. A particle of mass 5 mg and charge 2 nC is released in air between the plates. The potential difference that should be applied to the plates so that the particle remains suspended between them, is :

(A) 250 V (B) 200 V

CBSE 25

(C) 100 V (D) 50 V

(A) 250 V

29. A capacitor is a system of two conductors separated by an insulator. In practice, the two conductors have charges Q and $-$ Q with potential difference $V = V_1 - V_2$ between them. The ratio $\frac{Q}{V}$ is a constant, denoted by C and is called the capacitance of the capacitor. It is independent of Q or V. It depends only on the geometrical configuration (shape, size, separation) of the two conductors and the medium separating the conductors. When a parallel plate capacitor is charged, the electric field E_0 is localised between the plates and is uniform throughout. When a slab of a dielectric is inserted between the charged plates (charge density σ), the dielectric is polarised by the field. Consequently opposite charges appear on the faces of the slab, near the plates, with surface charge density of magnitude σ_p. For a linear dielectric σ_p is proportional to E_0. Introduction of a dielectric changes the electric field, and hence, the capacitance of a capacitor, and hence, the energy stored in the capacitor.

Like resistors, capacitors can also be arranged in series or in parallel or in a combination of series and parallel.

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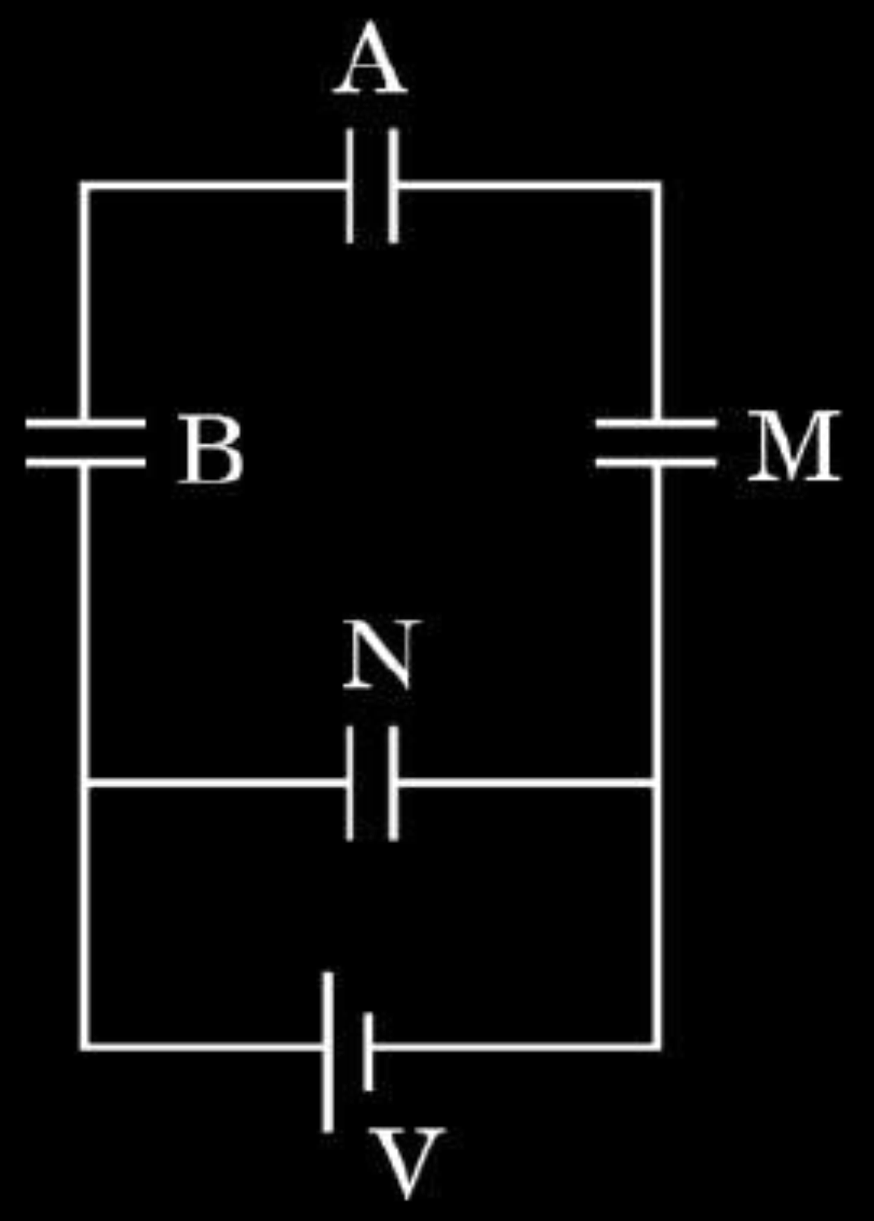
(i) (a) Three capacitors A, B and M, each of capacitance C are connected to a capacitor N of capacitance 2C and a battery as shown in the figure. If the charges on A and N are Q and Q′ respectively, then $\frac{Q'}{Q}$ is : *1*



(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) 3 (D) 6

(b) A slab (area A and thickness $\frac{d}{2}$) of dielectric constant K is inserted in a parallel plate capacitor of plate area A and plate separation d. If C and C_0 are the capacitances of the capacitors with and without the dielectric, then $\frac{C}{C_0}$ is : *1*

(A) $\frac{K+1}{2K}$ (B) $\frac{2K}{K+1}$

(C) $\frac{K}{K-1}$ (D) $\frac{K-1}{K}$

(ii) An electric field E is established between the plates of an air filled parallel plate capacitor, with charges Q and – Q. V is the volume of the space enclosed between the plates. The energy stored in the capacitor is : *1*

(A) $\frac{1}{2} \varepsilon_0 E^2$ (B) $\varepsilon_0 Q^2 E$

(C) $\frac{1}{2} \varepsilon_0 E^2 V$ (D) $\varepsilon_0 E Q V$

(iii) A slab (area A and thickness d_1) of a linear dielectric of dielectric constant K is inserted between charged plates (charge density σ) of a parallel plate capacitor [plate area A and plate separation d ($> d_1$)] and opposite charges with charge density of magnitude σ_p appear on the faces of the slab. The dielectric constant K is given by : *1*

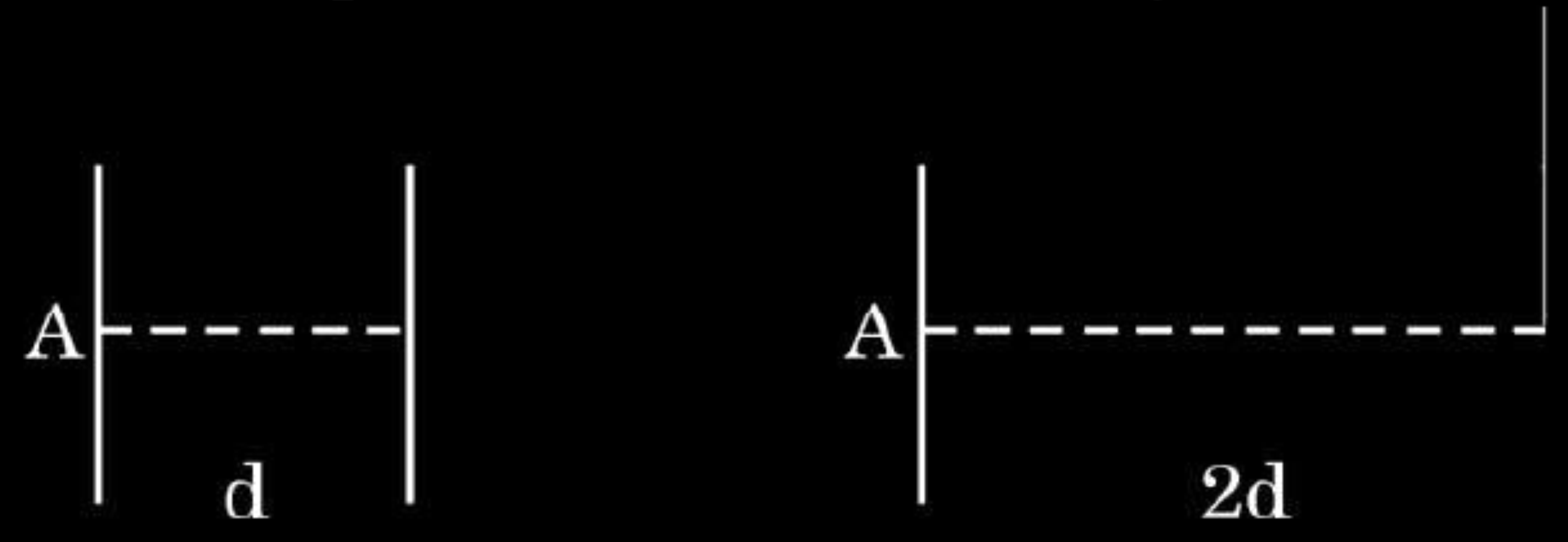
(A) $\dfrac{\sigma + \sigma_p}{\sigma}$

(B) $\dfrac{\sigma}{\sigma - \sigma_p}$

(C) $\dfrac{\sigma + \sigma_p}{\sigma_p}$

(D) $\dfrac{\sigma}{\sigma_p}$

(iv) Consider a capacitor of capacitance C, with plate area A and plate separation d, filled with air [Fig. (a)]. The distance between the plates is increased to 2d and one of the plates is shifted as shown in Fig. (b). The capacitance of the new system now is : *1*



(a) (b)

(A) $\frac{C}{4}$ (B) $\frac{C}{2}$

(C) 2C (D) 4C

31. (a) (i) The electric field in a region is given by $\vec{E} = 40x\ \hat{i}$ N/C. Find the amount of work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0).

(ii) A charge Q is distributed over two concentric hollow spheres of radii r and R (> r) such that their surface charge densities are equal. Find :

(I) the electric field, and

(II) the potential

at their common centre. CBSE 25 5

(a)

(i) $V = -\int \vec{E} \cdot \vec{dr}$

$= -\int 40\, x\, dx$

$= -20\, x^2$ ½

Potential at A (0, 3m), $V_A = 0$

Potential at B (5m, 0), V_B = -500 V ½

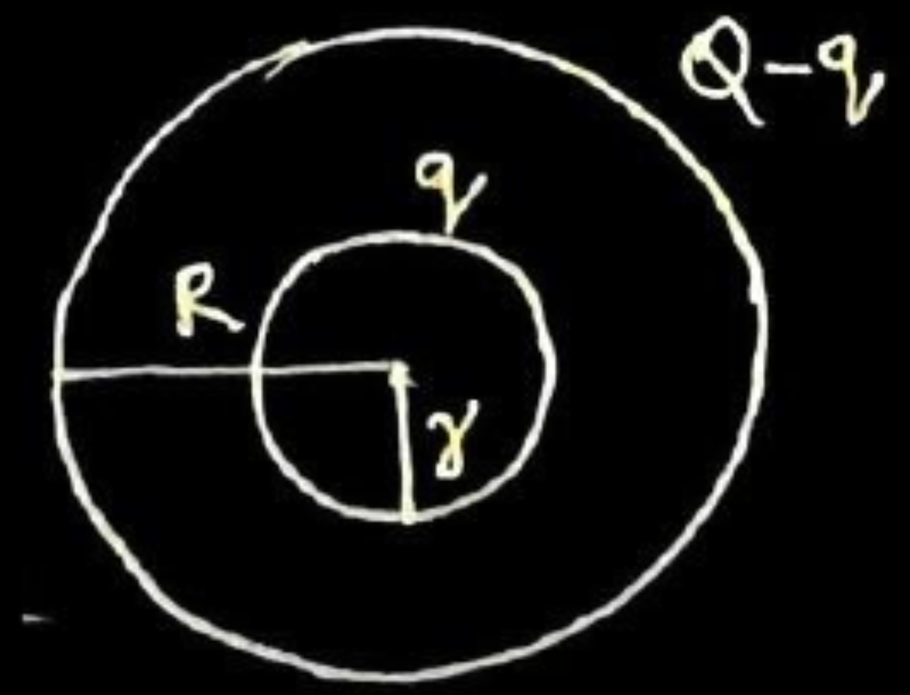
Work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0)

$W = q(V_B - V_A)$ ½

$= 1(-500 - 0)$

$W = -500\,J$ ½

(ii) (I) Electric field at the common centre will be zero as the charge enclosed by the inner sphere is zero. 1



Alternatively: $q_{en} = 0$

$\phi_E = 0$

$\oint \vec{E}.\vec{ds} = 0$

$E = 0$

(II) $\because$ Surface charge densities are equal

$$\frac{q}{4\pi r^2} = \frac{Q-q}{4\pi R^2}$$

$$q = \frac{Qr^2}{R^2 + r^2}$$

Potential at common centre

$$V = \frac{kq}{r} + \frac{k(Q-q)}{R}$$

$$V = \frac{k}{r}\frac{Qr^2}{(R^2+r^2)} + \frac{k}{R}\left[Q - \frac{Qr^2}{(R^2+r^2)}\right]$$

$$V = \frac{kQ(R+r)}{R^2 + r^2}$$

29. A power supply is connected across the two plates of a capacitor. Consequently, one plate becomes positively charged and the other plate becomes negatively charged. Thus a uniform electric field is established between the two plates. If a charged particle is released between these two plates, it experiences a force. The potential difference between the plates affects its charge and the energy stored in the capacitor. **$4 \times 1 = 4$**

(i) A proton and an electron are released from rest in the space between the two plates of a charged capacitor. Assume that the proton and the electron do not interact with each other. The acceleration of the proton compared with that of the electron is **1**

(A) same in both magnitude and direction.

(B) same in magnitude but in opposite direction.

(C) greater in magnitude and opposite in direction.

(D) lesser in magnitude but in opposite direction.

CBSE 25

29. A power supply is connected across the two plates of a capacitor. Consequently, one plate becomes positively charged and the other plate becomes negatively charged. Thus a uniform electric field is established between the two plates. If a charged particle is released between these two plates, it experiences a force. The potential difference between the plates affects its charge and the energy stored in the capacitor. **$4 \times 1 = 4$**

(i) A proton and an electron are released from rest in the space between the two plates of a charged capacitor. Assume that the proton and the electron do not interact with each other. The acceleration of the proton compared with that of the electron is **1**

(A) same in both magnitude and direction.
(B) same in magnitude but in opposite direction.
(C) greater in magnitude and opposite in direction.
(D) lesser in magnitude but in opposite direction.

CBSE 25

(D) Lesser in magnitude but in opposite direction.

(ii) An electron is released between the two parallel plates of a charged capacitor. Which of the following statements is incorrect ? 1

(A) The electric field between the plates is directed from the positively charged plate to the negatively charged plate.

(B) The magnitude of electric field is the same at all points between the plates.

(C) The acceleration of the electrons will be constant between the plates.

(D) The potential increases in the direction of the electric field between the plates.

(ii) An electron is released between the two parallel plates of a charged capacitor. Which of the following statements is <u>incorrect</u> ? **1**

(A) The electric field between the plates is directed from the positively charged plate to the negatively charged plate.

(B) The magnitude of electric field is the same at all points between the plates.

(C) The acceleration of the electrons will be constant between the plates.

(D) The potential increases in the direction of the electric field between the plates.

(ii) (D) The potential increases in the direction of the electric field between the plates.

(iii) The potential difference between the two plates of a capacitor is V. An electron of mass m and charge – e is released from rest near the negative plate. The maximum speed gained by the electron is **1**

(A) $2\ eVm$ (B) $\sqrt{\frac{em}{2\ V}}$ (C) $\sqrt{\frac{2\ eV}{m}}$ (D) $\sqrt{\frac{2\ me}{V}}$

(iii) The potential difference between the two plates of a capacitor is V. An electron of mass m and charge – e is released from rest near the negative plate. The maximum speed gained by the electron is **1**

(A) $2\,eVm$ (B) $\sqrt{\dfrac{em}{2V}}$ (C) $\sqrt{\dfrac{2eV}{m}}$ (D) $\sqrt{\dfrac{2me}{V}}$

(C) $\sqrt{\dfrac{2ev}{m}}$

(iv) (a) The electric field between two parallel plates of a charged capacitor is 200 V/m. The plates are 5 mm apart. The potential difference between the two plates is 1

(A) 20 V (B) 10 V

(C) 5.0 V (D) 1.0 V

(iv) (a) The electric field between two parallel plates of a charged capacitor is 200 V/m. The plates are 5 mm apart. The potential difference between the two plates is **1**

(A) 20 V (B) 10 V

(C) 5.0 V (D) 1.0 V

(iv) (a) (D) 1.0 V

(b) A proton is located at a point between the above plates. The force acting on the proton is **1**

(A) 1.6×10^{-19} N (B) 2×10^{9} N

(C) 3.2×10^{-17} N (D) 4×10^{9} N

(b) A proton is located at a point between the above plates. The force acting on the proton is **1**

(A) 1.6×10^{-19} N (B) 2×10^{9} N

(C) 3.2×10^{-17} N (D) 4×10^{9} N

(C) 3.2×10^{-17} N

31. (a) (A) Two points charges q_1 and q_2 in air are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in a region of uniform external field $\vec{E}$. Obtain an expression for the electrostatic potential energy of the system.

(B) A potential difference of 480 V is applied between two large horizontal plates, kept one above the other, 3.0 cm apart. A charged particle of mass 4×10^{-11} kg, when released at rest between the plates, remains stationary in the region. If the upper plate is at positive potential, find

(i) nature of the charge on the particle

(ii) the magnitude of the charge on the particle. **5**

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(a) Expression for potential energy.	
Work done in bringing the charge q_1, from ∞ to r_1, $W_1 = q_1 V(\vec{r}_1)$	½
Work done in bringing the charge q_2, from ∞ to r_2 is not only against the external field but also against the field due to q_1	½
$$W_2 = q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_o \vec{r}_{12}}$$	1
Where r_{12} is the distance between q_1 and q_2	
Thus, potential energy of the system = total work done in assembling the system	½
$$U = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_o \vec{r}_{12}}$$	½
(B) (i) As charge particle at rest which emplies that electrostatic force balances the gravitational force.	
So, nature of charge is negative.	1
(ii) $mg = qE = q\frac{V}{l}$	½
$$q = \frac{4\times10^{-11}\times10\times3\times10^{-2}}{480}$$	
$q = 2.5\times10^{-14}$ C	½

3. Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be :

(A) $\dfrac{Q^2}{\pi \varepsilon_0 l}$

(B) $\dfrac{\sqrt{2}\, Q^2}{\pi \varepsilon_0 l}$

(C) $\dfrac{Q^2}{2\pi \varepsilon_0 l}$

(D) Zero

CBSE 25

3. Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be :

(A) $\dfrac{Q^2}{\pi \varepsilon_0 l}$ (B) $\dfrac{\sqrt{2}\,Q^2}{\pi \varepsilon_0 l}$

(C) $\dfrac{Q^2}{2\pi \varepsilon_0 l}$ (D) Zero

CBSE 25

(B) $\dfrac{\sqrt{2}Q^2}{\pi\varepsilon_0 l}$

4. A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C. If the sheet partly occupies the space between the plates, the capacitance :

(A) remains C
(B) becomes greater than C
(C) becomes less than C
(D) becomes zero

CBSE 25

4. A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C. If the sheet partly occupies the space between the plates, the capacitance :

(A) remains C
(B) becomes greater than C
(C) becomes less than C
(D) becomes zero

CBSE 25

(B) becomes greater than C

22. (a) "There is a limit to the amount of charge that can be stored on a given capacitor." Explain.

(b) A capacitor is charged by a battery to a potential difference V. It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor. CBSE 25 *3*

22. (a) "There is a limit to the amount of charge that can be stored on a given capacitor." Explain.

(b) A capacitor is charged by a battery to a potential difference V. It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor. CBSE 25 *3*

(a) When a conductor holds a large amount of charge its potential is also high. If electric field becomes high enough, the atoms or molecules of surrounding medium gets ionized. A breakdown occurs in medium and charge on conductor get neutralized or leaks away.	1
(b) The common potential after the connection of two capacitors	
$Given V_1 = V \ , V_2 = 0$	
$C_1 = C_2 = C$	
$= \dfrac{CV}{2C}$	
$= \dfrac{V}{2}$	½
$U_i = \dfrac{1}{2}CV^2$	½

$U_f = \dfrac{1}{2}C'V'^2$	
$= \dfrac{1}{2}\times 2C\times\left(\dfrac{V}{2}\right)^2$	
$= \dfrac{1}{4}CV^2$	½
$\dfrac{U_f}{U_i} = \dfrac{1}{2}$	½

26. (a) Two small solid metal balls A and B of radii R and 2R having charge densities 2σ and 3σ respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire. CBSE 25 *3*

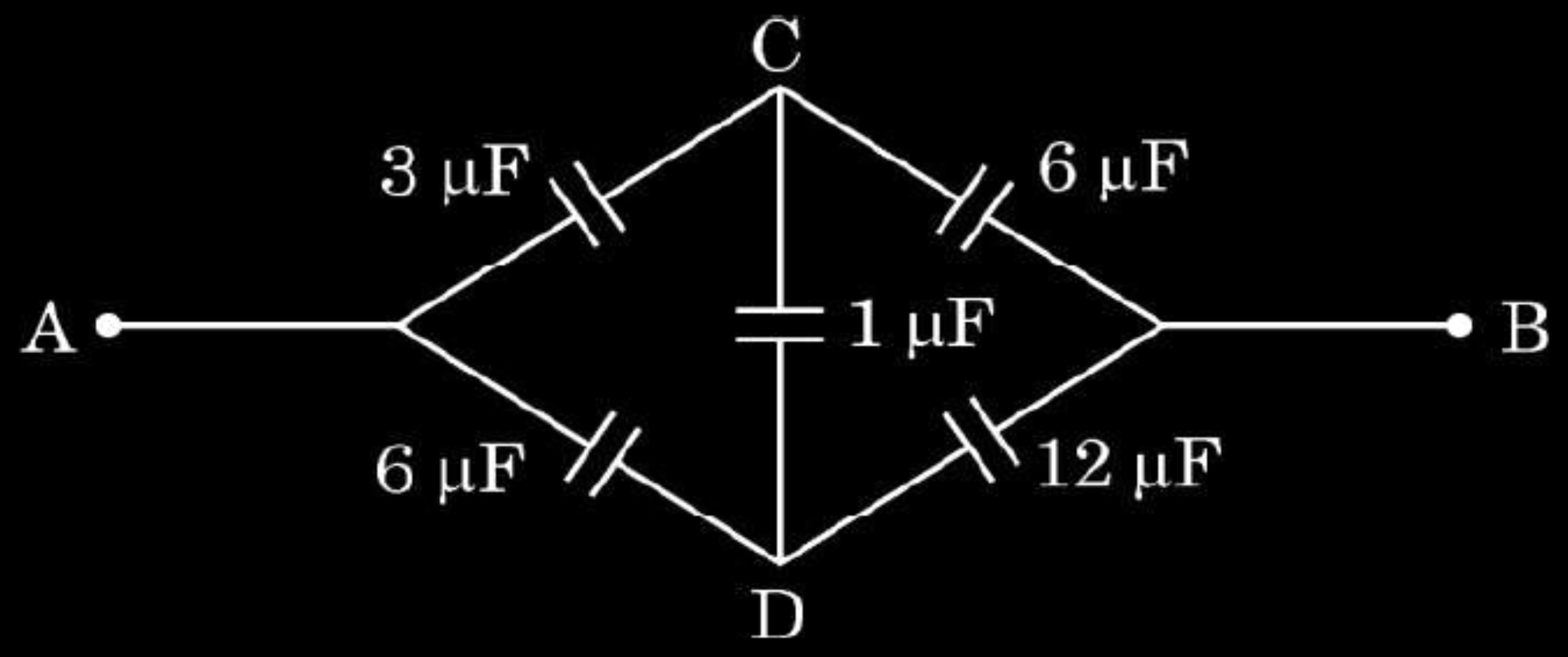
26. (a) Two small solid metal balls A and B of radii R and 2R having charge densities 2σ and 3σ respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire. *3*

CBSE 25

For ball A $q_1 = 2\sigma \times 4\pi R^2$ $= 8\pi R^2 \sigma$ For ball B $q_2 = 3\sigma \times 4\pi (2R)^2$ $= 48\pi R^2 \sigma$	½
Total charge (Q) $= q_1 + q_2$ $= 56\pi R^2 \sigma$	½
When balls A and B are connected by a wire, their potentials will be equal Let q be the charge on ball A and (Q – q) be the charge on the ball B after connecting wire. $\frac{Kq}{R} = \frac{K(Q-q)}{2R}$	½
$2q = Q - q$ $q = \frac{Q}{3}$ $= \frac{56\pi R^2 \sigma}{3}$	½

$Q - \frac{Q}{3} = \frac{112\pi R^2 \sigma}{3}$ $\sigma_A = \frac{\frac{56\pi R^2 \sigma}{3}}{4\pi R^2}$ $= \frac{14}{3}\sigma$	½
$\sigma_B = \frac{\frac{112\pi R^2 \sigma}{3}}{4\pi (2R)^2}$ $= \frac{7}{3}\sigma$	½

2. The effective capacitance of the network between points A and B shown in the figure is :



(A) $\frac{4}{3}$ μF (B) 2 μF

(C) $\frac{3}{2}$ μF (D) 6 μF

CBSE 25

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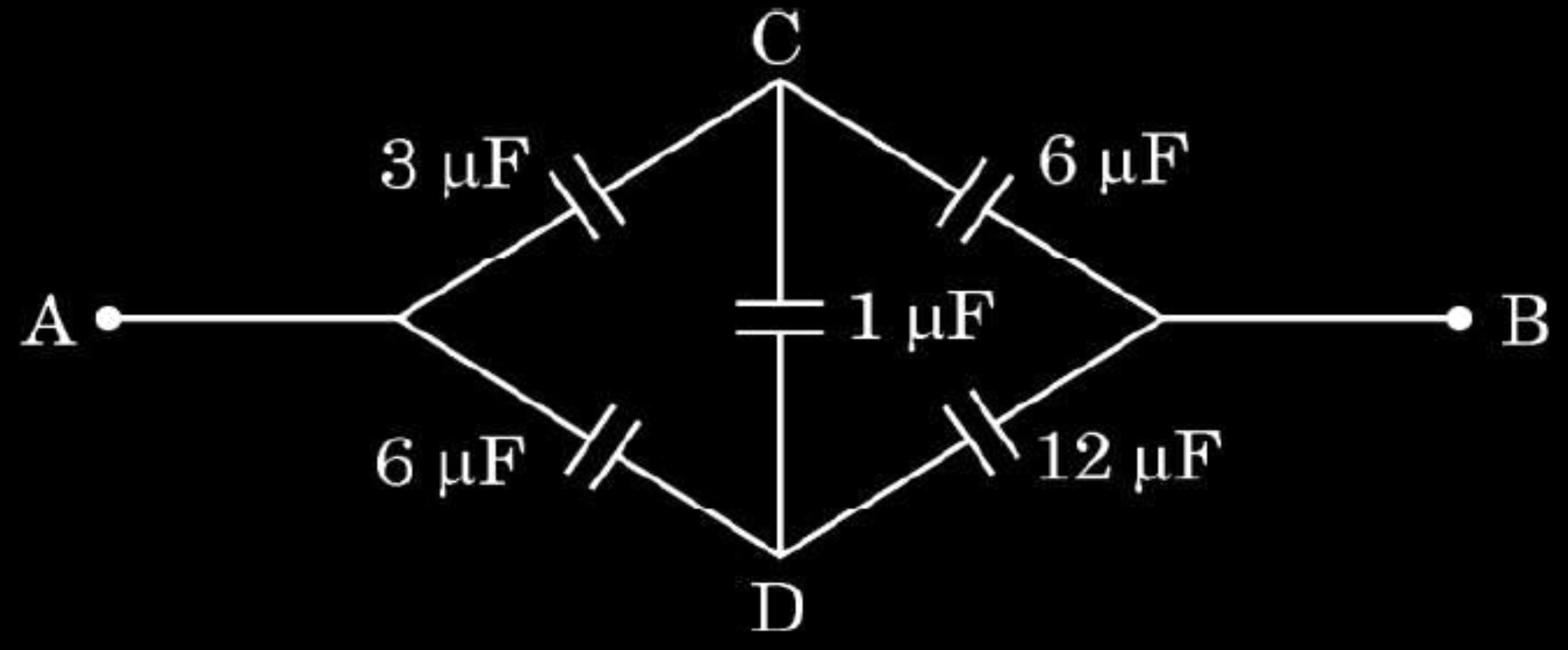

(A) $\frac{4}{3}$ µF

(B) 2 µF

(C) $\frac{3}{2}$ µF

(D) 6 µF

CBSE 25

(D) 6 μF

3. The electric field at a point in a region is given by $\vec{E} = \alpha \frac{\vec{r}}{|\vec{r}|^3}$, where α is a constant and r is the distance of the point from the origin. The magnitude of potential of the point is :

(A) $\frac{\alpha}{r}$

(B) $\frac{\alpha r^2}{2}$

(C) $\frac{\alpha}{2r^2}$

(D) $-\frac{\alpha}{r}$

CBSE 25

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CBSE 25

(A) $\frac{\alpha}{r}$

24. A capacitor of plate area A and plate separation d is charged by a battery to voltage V. The battery is disconnected and plates are slowly pulled apart till the separation becomes 2d. Find the value of : *3*

(a) potential difference between the plates,

(b) electric field between the plates,

(c) work done in pulling the plates apart.

CBSE 25

24. A capacitor of plate area A and plate separation d is charged by a battery to voltage V. The battery is disconnected and plates are slowly pulled apart till the separation becomes 2d. Find the value of : *3*

(a) potential difference between the plates,

(b) electric field between the plates,

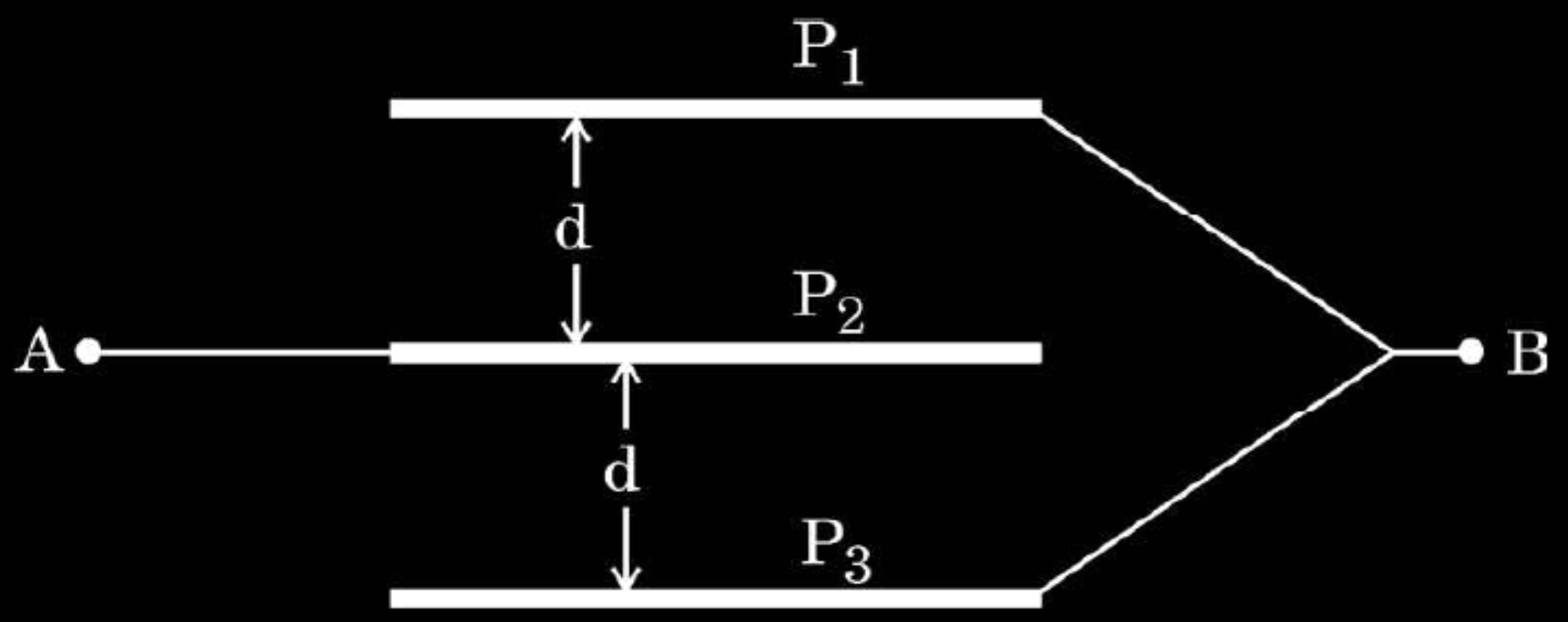
(c) work done in pulling the plates apart.

CBSE 25

As battery is dis connected, charge remain same	
$C' = \frac{C}{2}$	
(i) $q = q'$	
$CV = C'V'$	½
$CV = \frac{C}{2}V'$	
$V' = 2V$	½

(ii) $E' = \frac{V'}{d'}$	½
$= \frac{2V}{2d}$	
$= \frac{V}{d} = E$	½
(iii) $W = U' - U$	½
$= \frac{1}{2}\times\frac{C}{2}\times(2V)^2 - \frac{1}{2}CV^2$	
$= \frac{1}{2}CV^2$	½

30. A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between P_1 and P_2 and P_2 and P_3 is completely filled with mica sheets of dielectric constant 'K'. The plate P_2 is connected to point A and other plates P_1 and P_3 are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.



CBSE 25

(i) The capacitance of the system between A and B will be : *1*

(A) $\frac{\varepsilon_0 K L^2}{d}$ (B) $\frac{\varepsilon_0 K L^2}{2d}$

(C) $\frac{2\varepsilon_0 K L^2}{d}$ (D) $\frac{2\varepsilon_0 K d}{L^2}$

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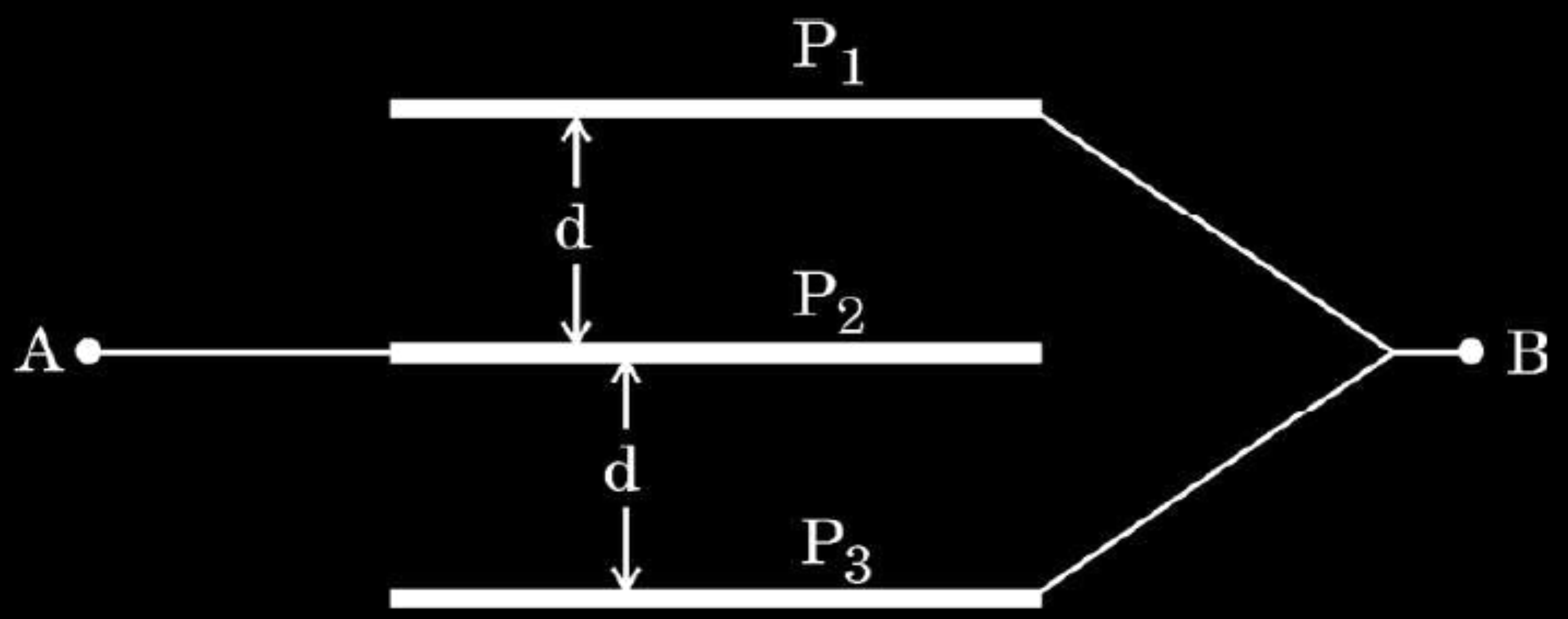

CBSE 25

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(C) $\dfrac{2\varepsilon_o K L^2}{d}$

(ii) The charge on plate P_1 is : *1*

(A) $\frac{\varepsilon_0 VKL^2}{2d}$

(B) $\frac{\varepsilon_0 VKL^2}{d}$

(C) $\frac{2\varepsilon_0 VKL^2}{d}$

(D) $\frac{\varepsilon_0 VKL^2}{4d}$

(ii) The charge on plate P_1 is : *1*

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(C) $\dfrac{2\varepsilon_0 VKL^2}{d}$ (D) $\dfrac{\varepsilon_0 VKL^2}{4d}$

(B) $\dfrac{\varepsilon_o VKL^2}{d}$

(iii) The electric field in the region between P_1 and P_2 is : *1*

(A) $\frac{V}{d}$

(B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$

(D) $\frac{d}{V}$

(iii) The electric field in the region between P_1 and P_2 is : *1*

(A) $\frac{V}{d}$ (B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$ (D) $\frac{d}{V}$

(A) $\frac{V}{d}$

(iv) (a) The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be : *1*

(A) $\frac{d}{K}$ (B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$ (D) $\frac{d}{4K}$

(iv) (a) The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be : *1*

(A) $\frac{d}{K}$ (B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$ (D) $\frac{d}{4K}$

(C) $\frac{d}{2K}$

(b) If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be : *1*

(A) $\frac{\varepsilon_0 VKL^2}{4d}$ (B) $\frac{\varepsilon_0 VKL^2}{2d}$

(C) $\frac{\varepsilon_0 VKL^2}{d}$ (D) Zero

(b) If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be : *1*

(A) $\frac{\varepsilon_0 VKL^2}{4d}$ (B) $\frac{\varepsilon_0 VKL^2}{2d}$

(C) $\frac{\varepsilon_0 VKL^2}{d}$ (D) Zero

(D) Zero

(ii) Three point charges of – 2 nC, – 1 nC, and + 5 nC are kept at the vertices A, B and C of an equilateral triangle of side 0·2 m. Find the total amount of work done in shifting the charges from A to A_1, B to B_1 and C to C_1. Here A_1, B_1 and C_1 are the midpoints of sides AB, BC and CA, respectively.

CBSE 25

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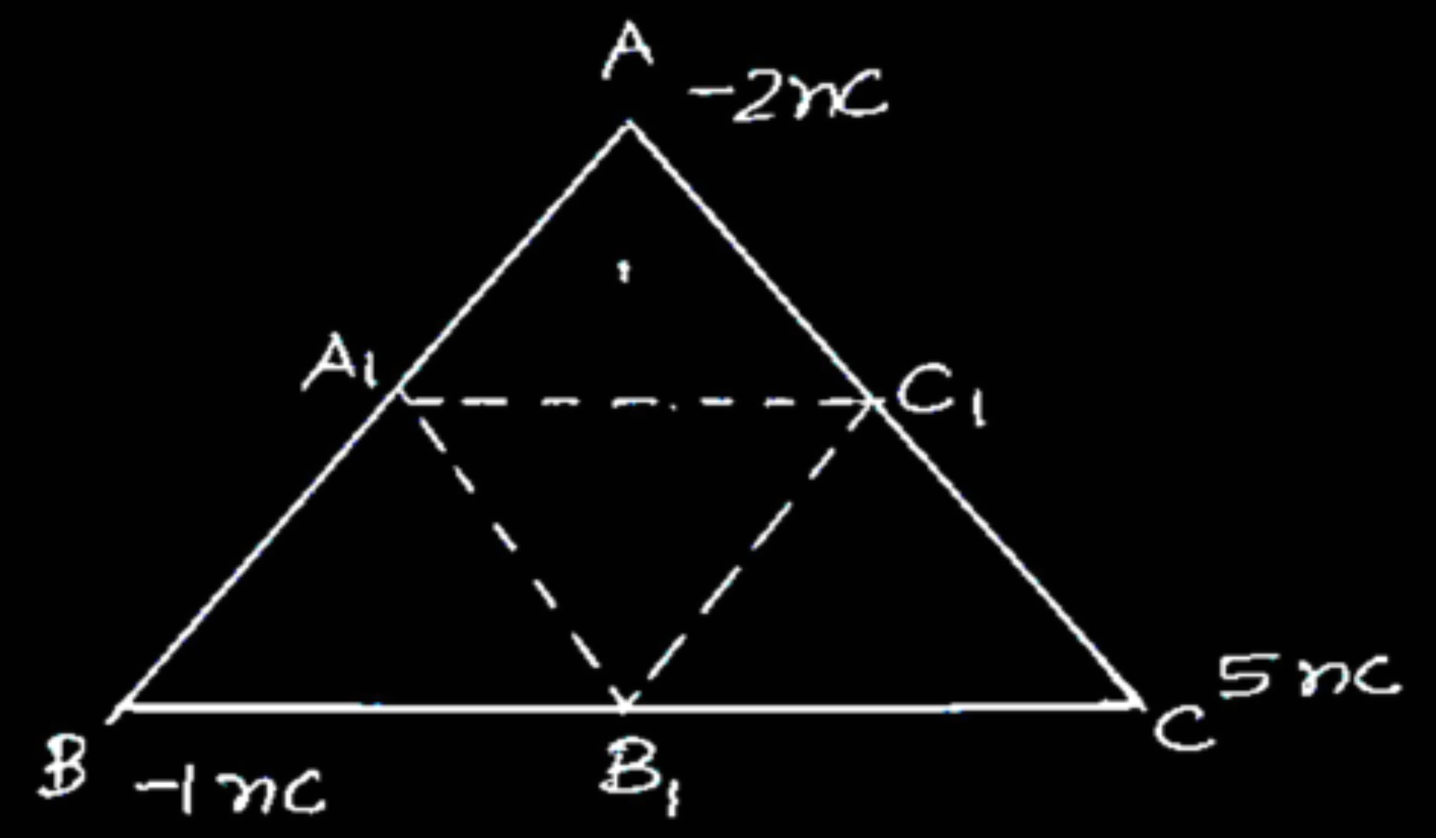
CBSE 25

(ii) 	
Initial electrostatic potential energy of the system	
$U_1=\frac{1}{4\pi\varepsilon_o}(\frac{q_A q_B}{AB}+\frac{q_C q_A}{AC}+\frac{q_C q_B}{BC})$	
$=\frac{9\times10^9}{0.2}[(-2\times-1)+(-2\times5)+(-1\times5)]\times10^{-18}$	½
$U_1= -5.85\times10^{-7}J$	
$U_2=\frac{1}{4\pi\varepsilon_o}(\frac{q_{A_1} q_{B_1}}{A_1B_1}+\frac{q_{C_1} q_{A_1}}{A_1C_1}+\frac{q_{c_1} q_{B_1}}{B_1C_1})$	½
$U_2= -11.7\times10^{-7}J$	
$W=U_2-U_1=-5.85\times10^{-7}J$	½

(ii) A point charge of $+2\,nC$ is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be kept at $(0, 0, -6m)$ so that the potential due to the system becomes zero at $(0, 0, 2m)$. CBSE 25

(ii) A point charge of + 2 nC is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be kept at (0, 0, – 6m) so that the potential due to the system becomes zero at (0, 0, 2m).

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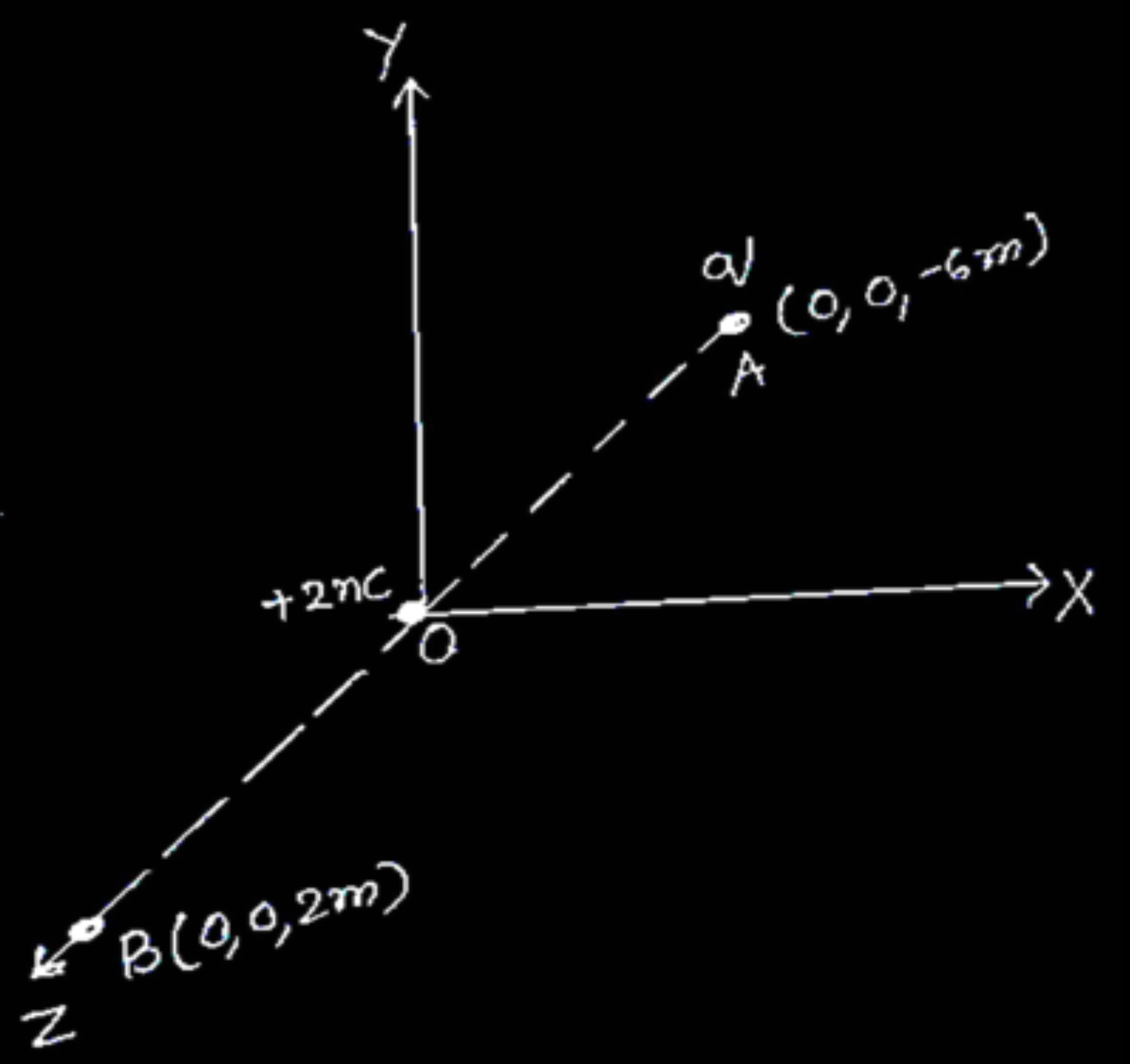
(ii)



Let the charge is kept at A be q

Potential at point B due to charge at the origin O and charge (q) at A

$$V = V_1 + V_2$$

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{2\times10^{-9}}{2} + \frac{q}{6+2}\right]$$ ½

$$\frac{1}{4\pi\varepsilon_0}\left[10^{-9} + \frac{q}{8}\right] = 0$$ ½ ½

$$q = -8\times10^{-9}C$$ ½

3. Which one of the following statements is correct ?

Electric field due to static charges is **1**

(A) conservative and field lines do not form closed loops.

(B) conservative and field lines form closed loops.

(C) non-conservative and field lines do not form closed loops.

(D) non-conservative and field lines form closed loops.

CBSE 25

3. Which one of the following statements is correct ?

Electric field due to static charges is **1**

(A) conservative and field lines do not form closed loops.
(B) conservative and field lines form closed loops.
(C) non-conservative and field lines do not form closed loops.
(D) non-conservative and field lines form closed loops.

CBSE 25

(A) conservative and field lines do not form closed loops.

30. A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates. **$4 \times 1 = 4$**

(i) The electric field between the plates of a parallel plate capacitor is E. Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be :

(A) E (B) 2E

(C) $\frac{E}{4}$ (D) $\frac{E}{2}$

CBSE 25

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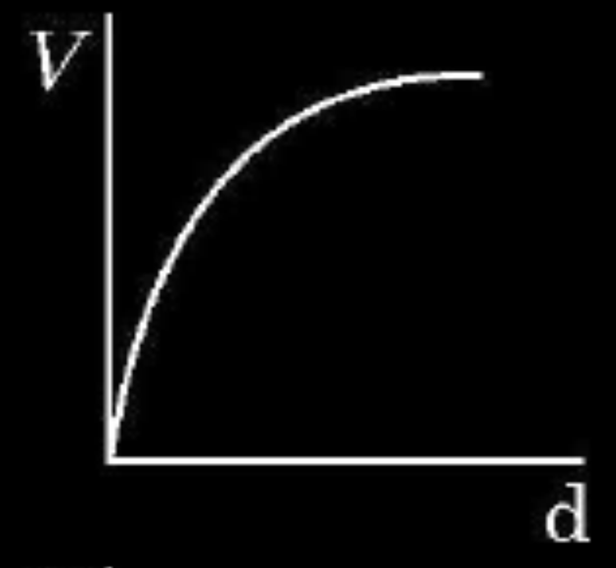
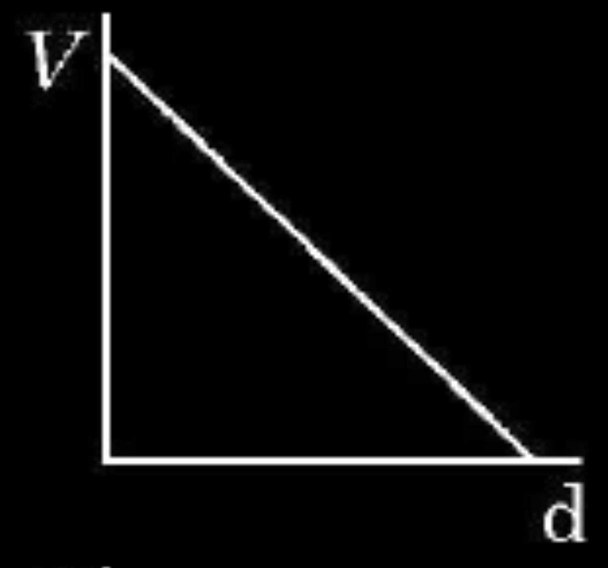
(C) $\frac{E}{4}$ (D) $\frac{E}{2}$

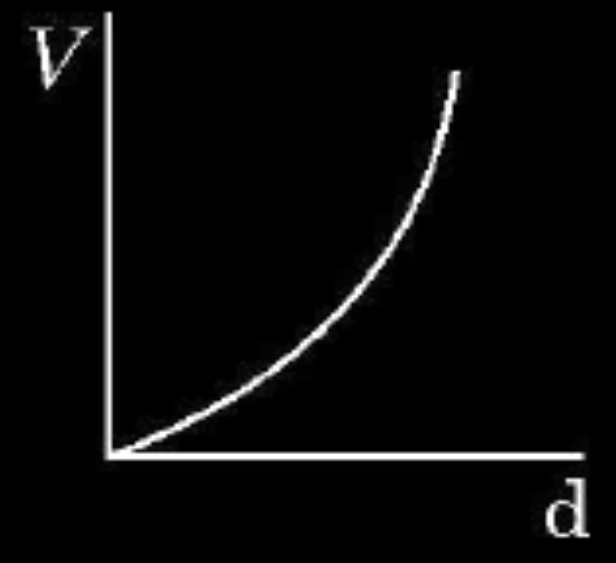
CBSE 25

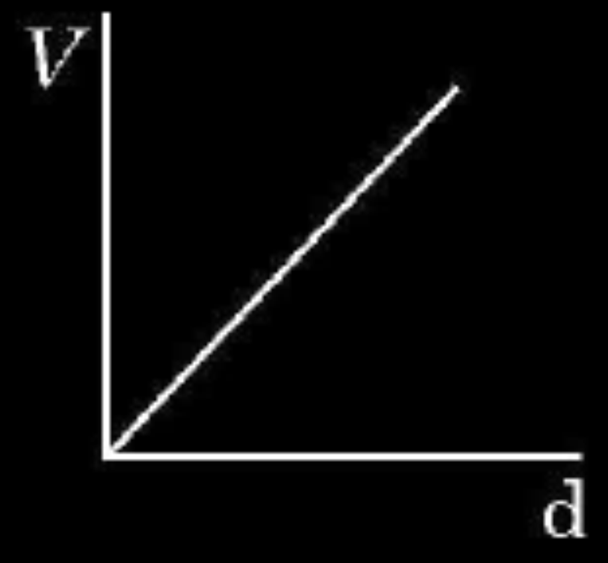
(C) $\frac{E}{4}$

(ii) A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly depict the potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field ?

(A) 

(B) 

(C) 

(D) 

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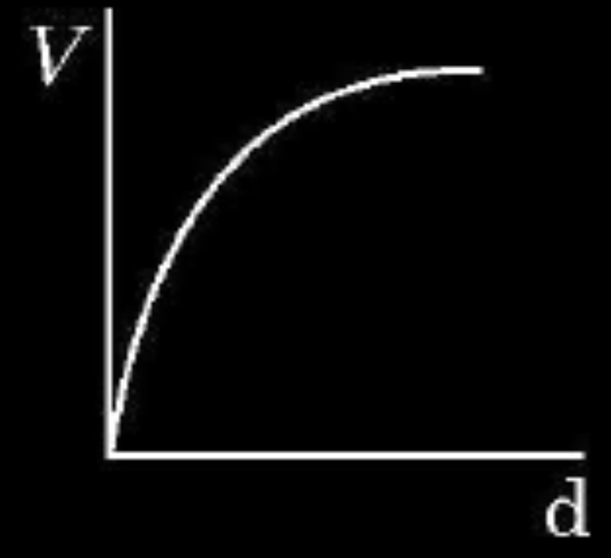
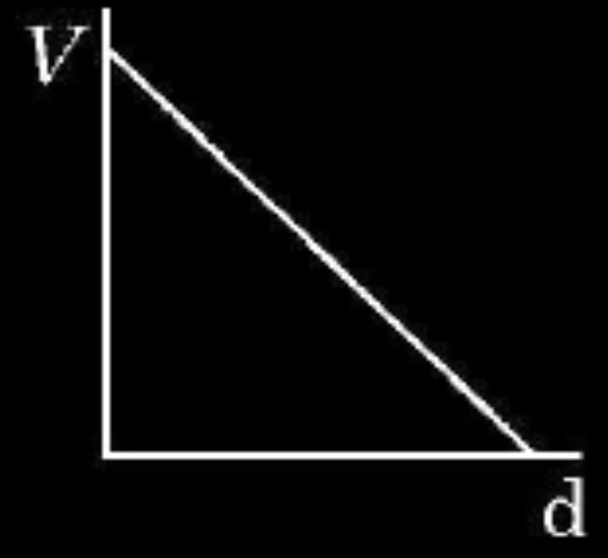
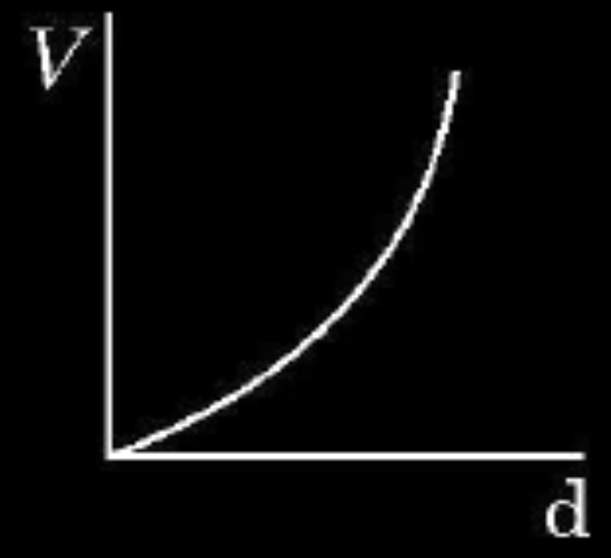
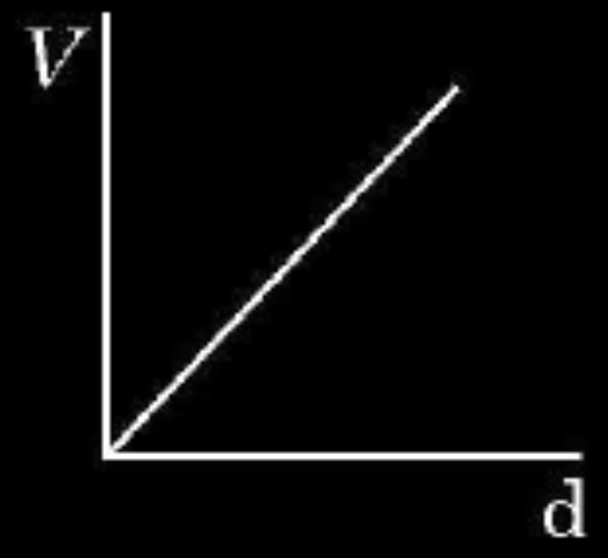
(A) 

(B) 

(C) 

(D) 

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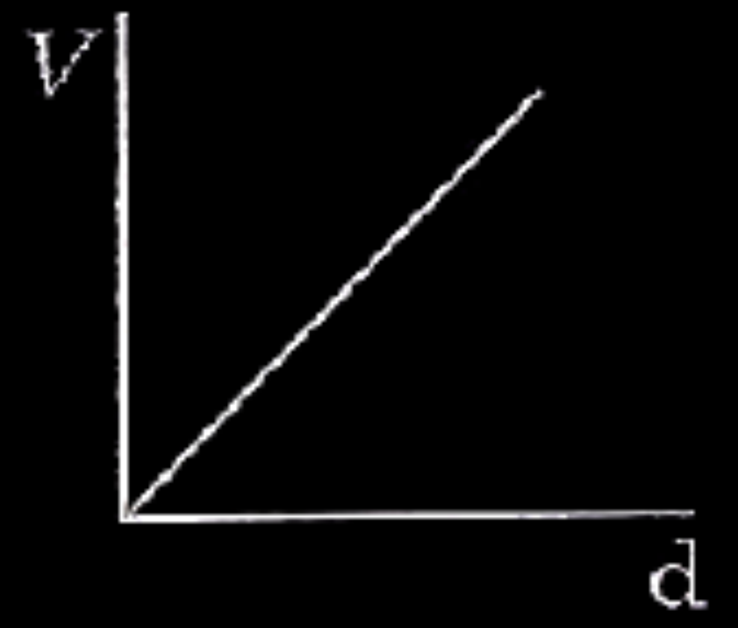
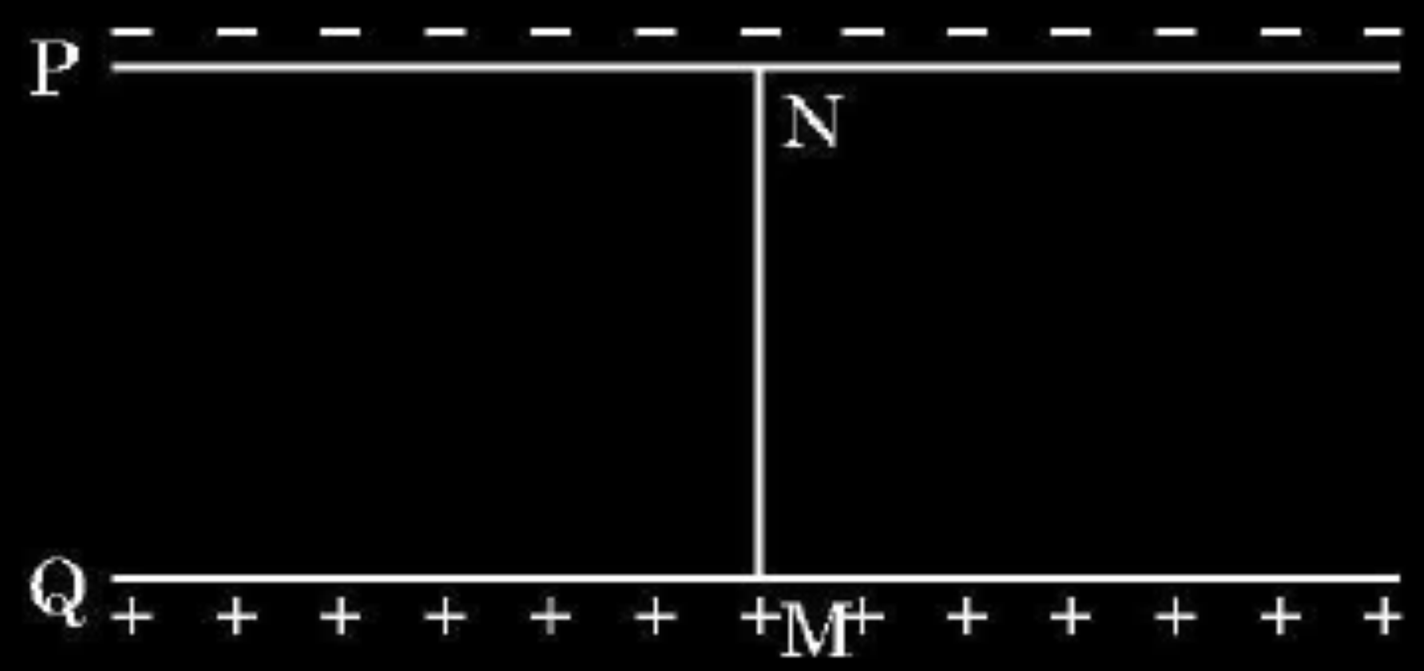
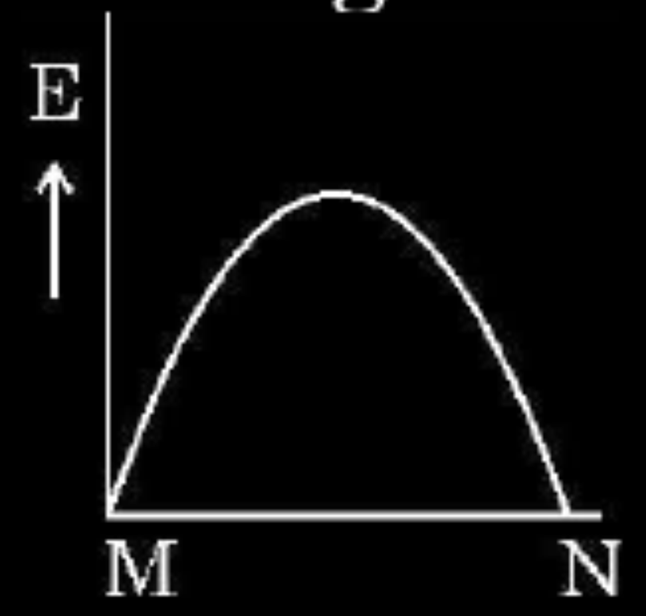
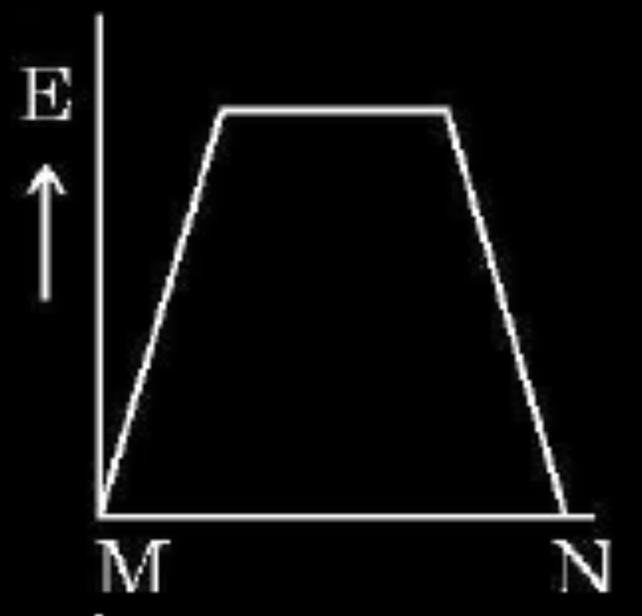
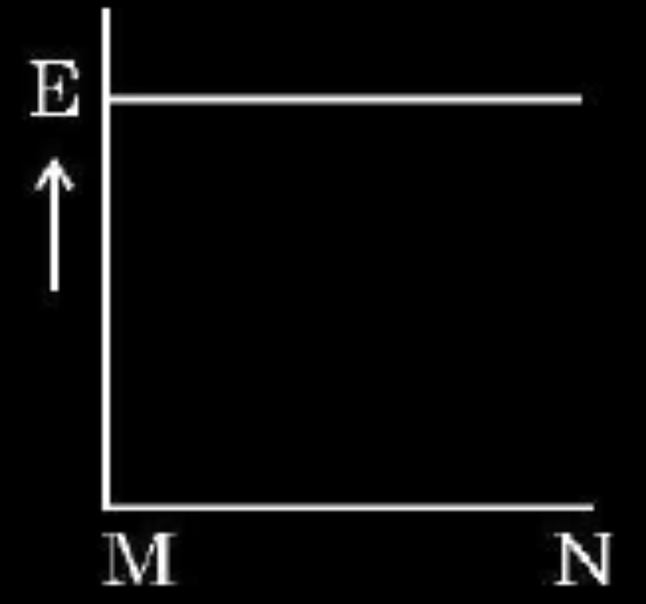


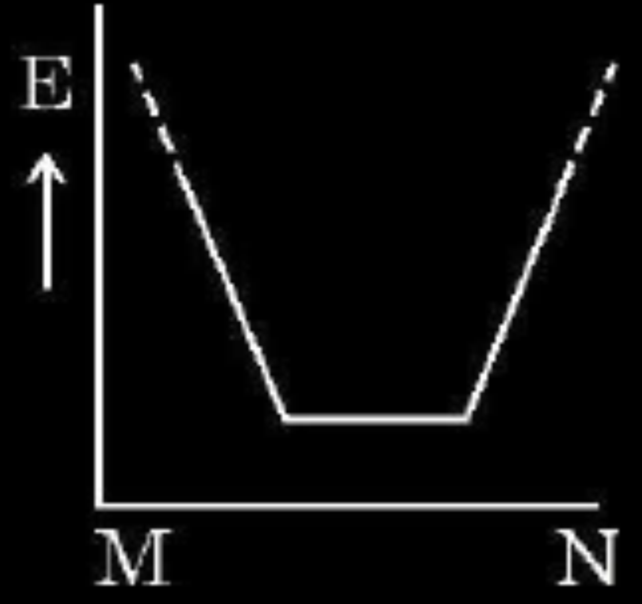
(iii) 

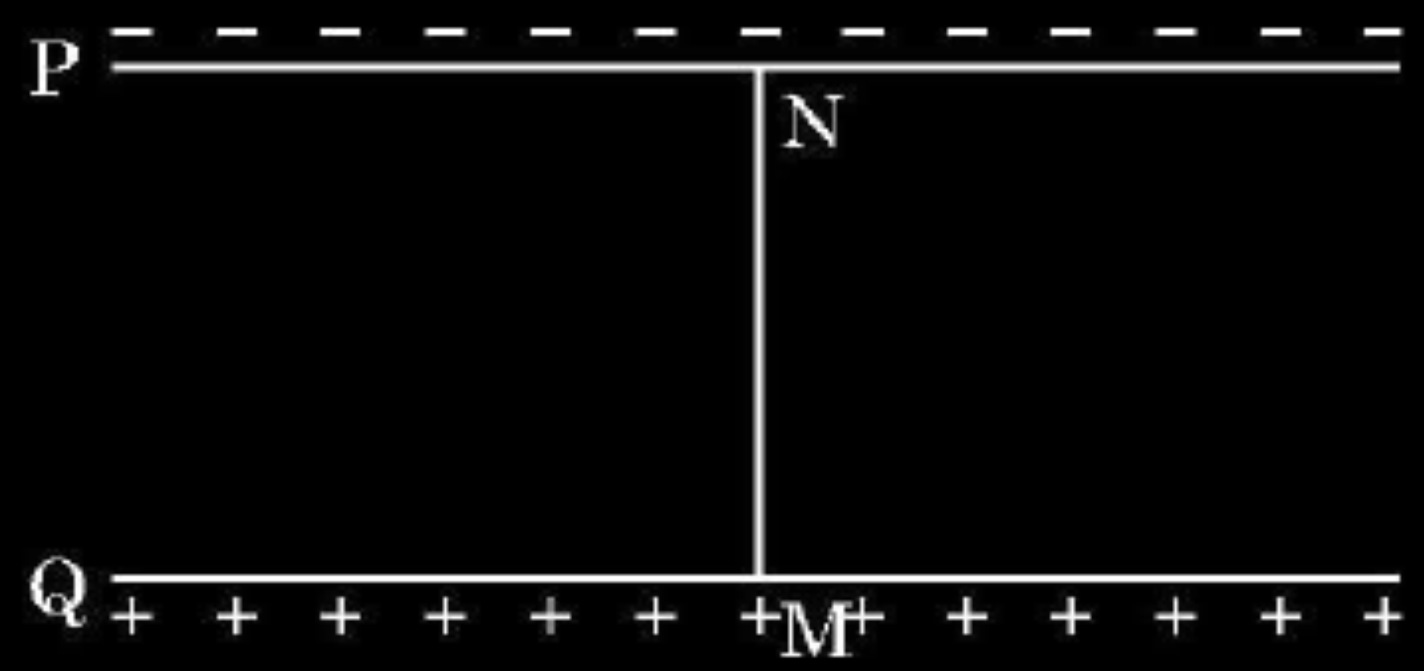
In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN ?

(A) 

(B) 

(C) 

(D) 

(iii) 

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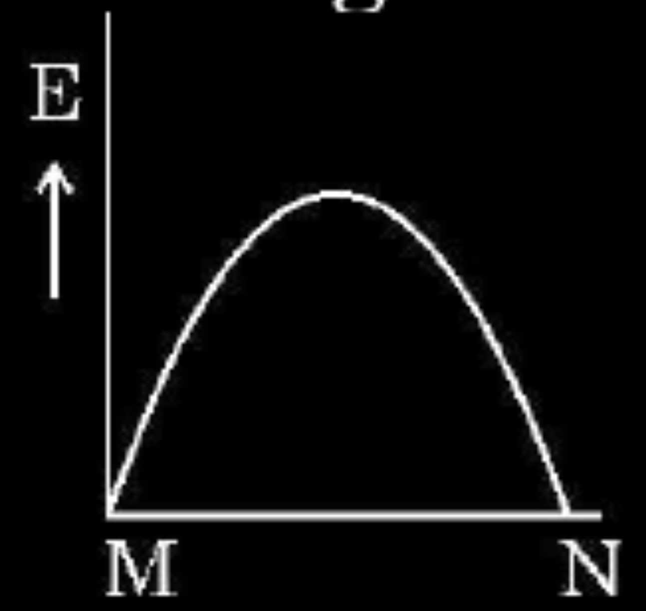
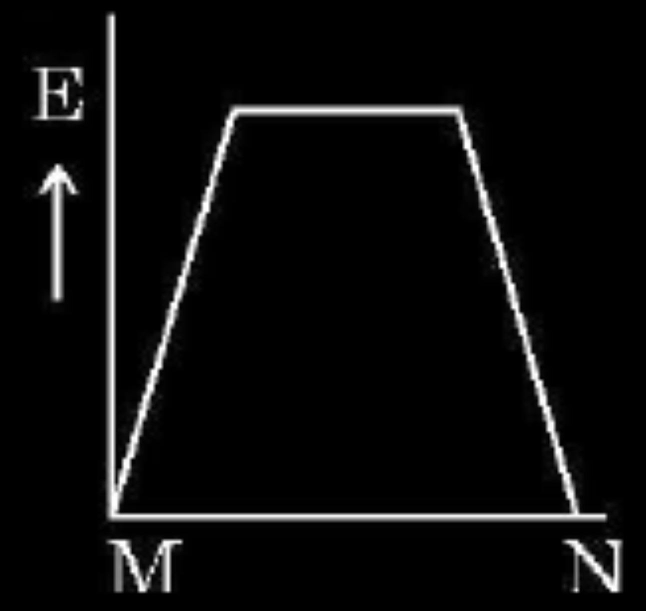
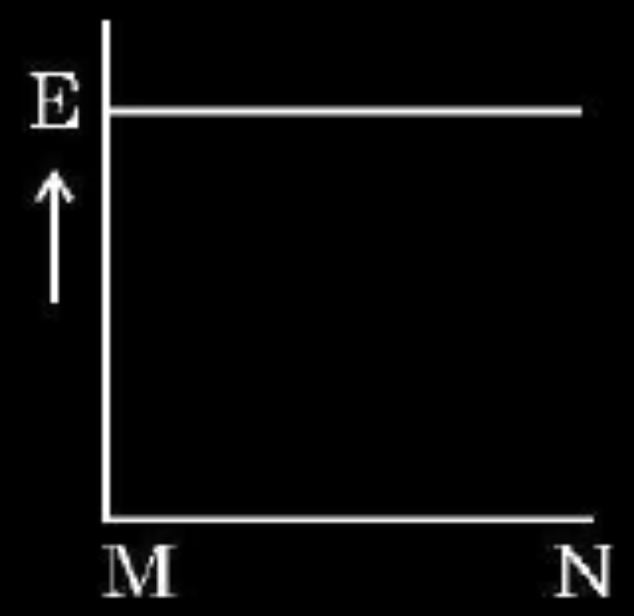
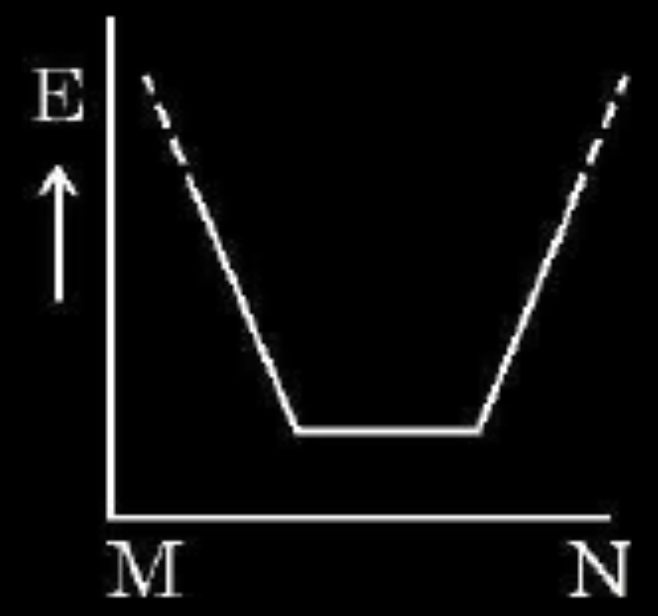
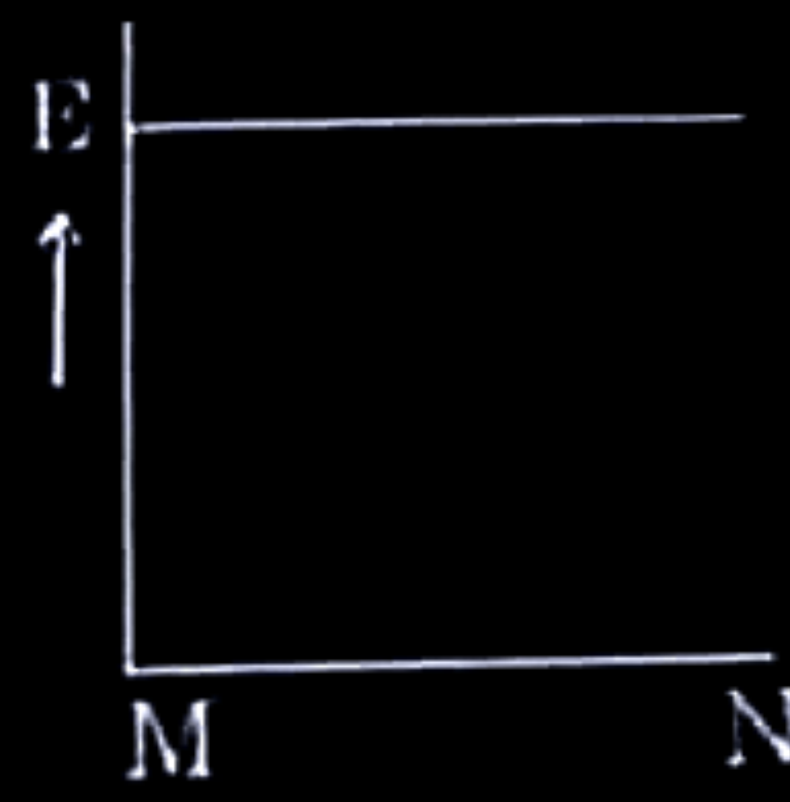
(A) 

(B) 

(C) 

(D) 

(C) 

(iv) Three parallel plates are placed above each other with equal displacement $\vec{d}$ between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is –

(A) $(\vec{E}_1 + \vec{E}_2) \cdot \vec{d}$

(B) $(\vec{E}_1 - \vec{E}_2) \cdot \vec{d}$

(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

(D) $\dfrac{d(E_1 + E_2)}{2}$

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(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

(D) $\dfrac{d(E_1 + E_2)}{2}$

(A) $(\vec{E}_1+\vec{E}_2).\vec{d}$

(iv) A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C. The new value of its capacitance becomes

(A) C

(B) $\frac{C}{K}$

(C) CK

(D) $C\left(1 + \frac{1}{K}\right)$

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(C) CK

(D) $C\left(1 + \frac{1}{K}\right)$

(C) CK

33. (a) (i) A small conducting sphere A of radius r charged to a potential V, is enclosed by a spherical conducting shell B of radius R. If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B. **5**

(ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along $+x$ axis. If the potential at the origin (0, 0) is 220 V, find the potential at a point (4m, 3m).

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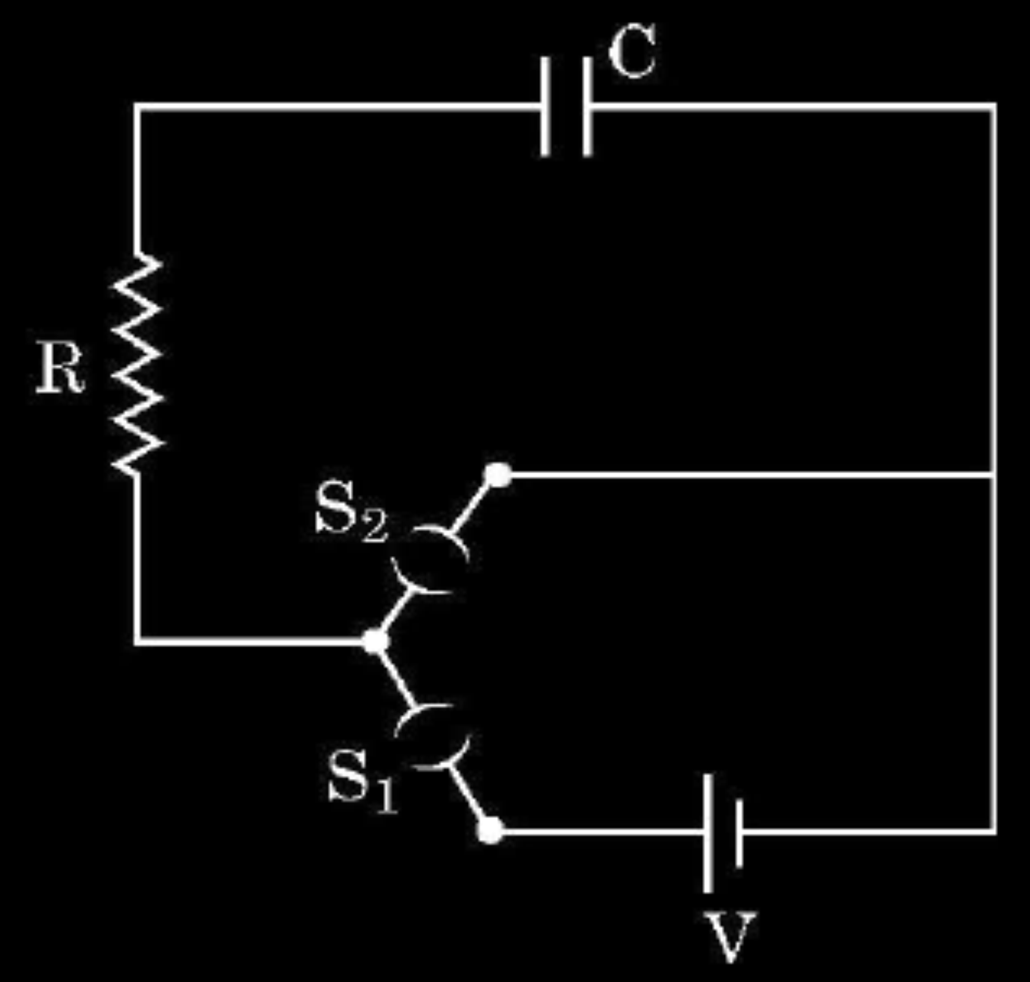
(i) Potential on sphere A = V = $\dfrac{Q}{4\pi\varepsilon_o r}$ Charge on sphere A = $4\pi\varepsilon_0 r\,V$	½
The charge is transferred to shell B. Potential on shell B = $\dfrac{1}{4\pi\varepsilon_0} \text{ x } \dfrac{4\pi\varepsilon_0 rV}{R}$	½
Potential on shell B = $\dfrac{rV}{R}$ Potential on sphere A = Potential on shell B	1
(ii) Characteristics of equipotential surfaces: - (Any two) - Potential at all points on the surface is same. - Equipotential surface is normal to the direction of the electric field. - The work done in moving a charge on an equipotential surface is zero.	½+½
$V_0 - V = E\, d = 50 \text{ x } 4$	½
$V_0 - V = 200$ V	½
V = 220 V – 200 V	½
V = 20 V	½

30. A circuit consisting of a capacitor C, a resistor of resistance R and an ideal battery of emf V, as shown in figure is known as RC series circuit. $4 \times 1 = 4$

CBSE 25



As soon as the circuit is completed by closing key S_1 (keeping S_2 open) charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference V_c (= q/C) across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged (Q = VC). During this process of charging, the charge q on the capacitor changes with time t as $q = Q[1 - e^{-t/RC}]$

The charging current can be obtained by differentiating it and using $\frac{d}{dx}(e^{mx}) = me^{mx}$.

Consider the case when R = 20 kΩ, C = 500 μF and V = 10 V.

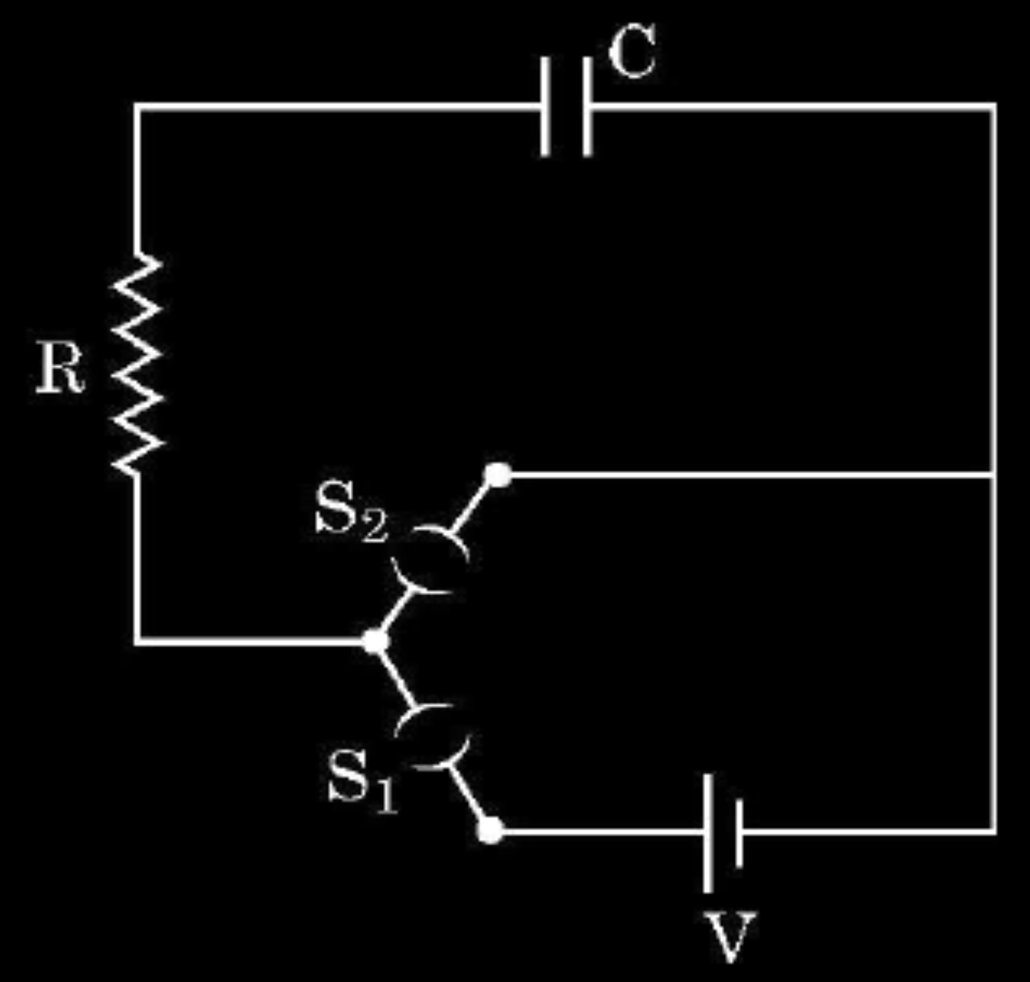
(i) The final charge on the capacitor, when key S_1 is closed and S_2 is open, is

(A) 5 μC (B) 5 mC

(C) 25 mC (D) 0.1 C

30. A circuit consisting of a capacitor C, a resistor of resistance R and an ideal battery of emf V, as shown in figure is known as RC series circuit. $4 \times 1 = 4$

CBSE 25



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(i) The final charge on the capacitor, when key S_1 is closed and S_2 is open, is

(A) 5 μC (B) 5 mC

(C) 25 mC (D) 0.1 C

(B) 5mC

(ii) For sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor ?

(A) Zero (B) 5 mC

(C) 2.5 mC (D) 5 μC

(ii) For sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor ?

(A) Zero (B) 5 mC

(C) 2.5 mC (D) 5 μC

(A) zero

(iii) The dimensional formula for RC is

(A) $[M L^2 T^{-3} A^{-2}]$ (B) $[M^0 L^0 T^{-1} A^0]$

(C) $[M^{-1} L^{-2} T^4 A^2]$ (D) $[M^0 L^0 T A^0]$

(iii) The dimensional formula for RC is

(A) $[M\,L^2\,T^{-3}\,A^{-2}]$ (B) $[M^0\,L^0\,T^{-1}\,A^0]$

(C) $[M^{-1}\,L^{-2}\,T^4\,A^2]$ (D) $[M^0\,L^0\,T\,A^0]$

(D) $[M^0L^0TA^0]$

(iv) The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds, is

(A) $\frac{1}{2\sqrt{e}}$ mA (B) $\sqrt{e}$ mA (C) $\frac{1}{\sqrt{e}}$ mA (D) $\frac{1}{2e}$ mA

(iv) The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds, is

(A) $\frac{1}{2\sqrt{e}}$ mA (B) $\sqrt{e}$ mA (C) $\frac{1}{\sqrt{e}}$ mA (D) $\frac{1}{2e}$ mA

iv) (A) $\frac{1}{2\sqrt{e}} mA$

Note: 1 mark for this part may be given to all the students who have attempted other parts of the question.

(iv) The key S_1 is closed and S_2 is open. The initial value of charging current in the resistor, is

(A) 5 mA (B) 0.5 mA (C) 2 mA (D) 1 mA

(iv) The key S_1 is closed and S_2 is open. The initial value of charging current in the resistor, is

(A) 5 mA (B) 0.5 mA (C) 2 mA (D) 1 mA

(B) 0.5 mA

32. (a) (i) Two point charges 5 μC and –1 μC are placed at points (–3 cm, 0, 0) and (3 cm, 0, 0) respectively. An external electric field $\vec{E} = \frac{A}{r^2}\hat{r}$ where $A = 3 \times 10^5$ Vm is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field. **5**

(ii) A system of two conductors is placed in air and they have net charge of +80μC and –80μC which causes a potential difference of 16 V between them.

(1) Find the capacitance of the system.

(2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, what will be the potential difference between the two conductors ?

(3) If the charges on two conductors are changed to +160 μC and –160 μC, will the capacitance of the system change ? Give reason for your answer.

CBSE 25

(i) $\vec{E} = \dfrac{3\times10^5}{r^2}\hat{r}$ (Given) $dV = -\vec{E}.d\vec{r}$

$V = 3 \times 10^5 / r$

Electrostatic energy of the system in the absence of the field

$U_i = \dfrac{Kq_1q_2}{r_{12}}$ ½

Electrostatic energy in the presence of the field

$U_f = \dfrac{Kq_1q_2}{r_{12}} + q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$

$\Delta U = U_f - U_i = q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$ ½

$\Delta U = \dfrac{5\times10^{-6}\times3\times10^5}{3\times10^{-2}} - \dfrac{1\times10^{-6}\times3\times10^5}{3\times10^{-2}}$ ½

$= 40\ J$ ½

ii) 1) $C = \dfrac{Q}{V} = \dfrac{80}{16} = 5\mu F$ 1

2) $C' = KC$

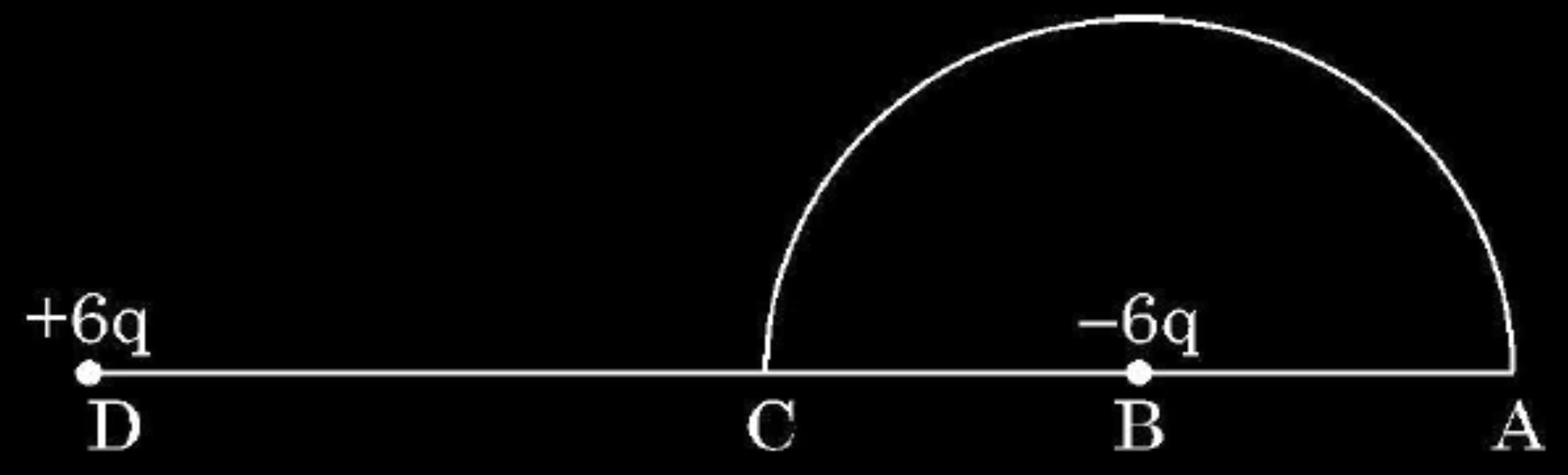
$= 3\times5\mu F = 15\ \mu F$ ½

$V' = \dfrac{Q}{C'} = \dfrac{80\mu C}{15\mu F} = 5.33V$ ½

3) No, ½

The capacitance of the system depends on its geometry. ½

(ii) A charge –6 μC is placed at the centre B of a semicircle of radius 5 cm, as shown in the figure. An equal and opposite charge is placed at point D at a distance of 10 cm from B. A charge +5 μC is moved from point 'C' to point 'A' along the circumference. Calculate the work done on the charge.



CBSE 25

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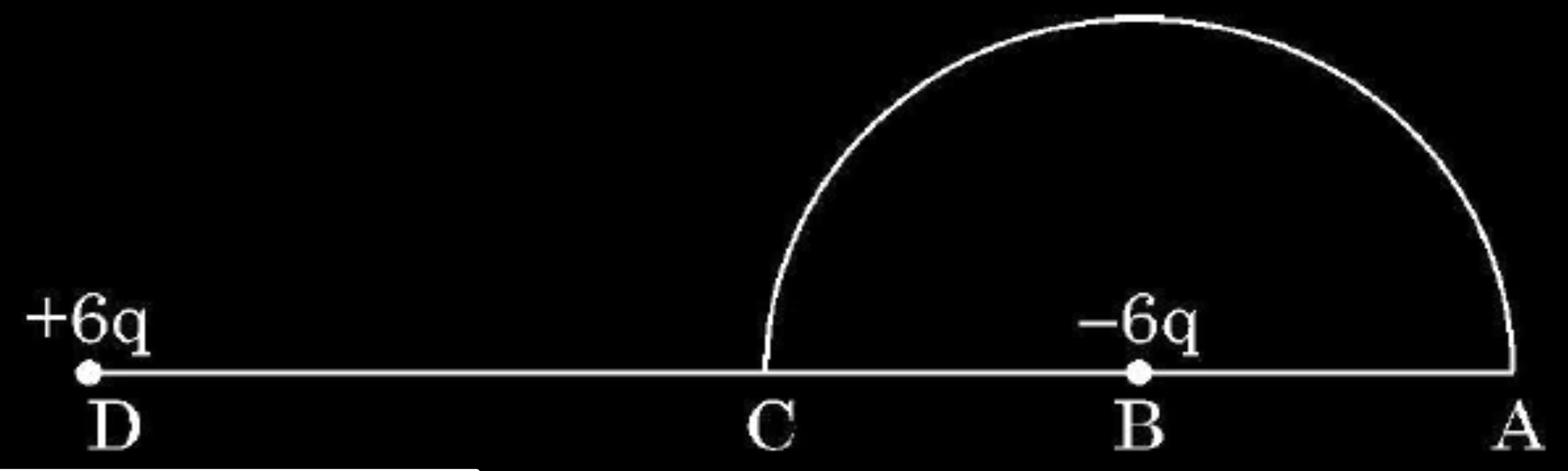
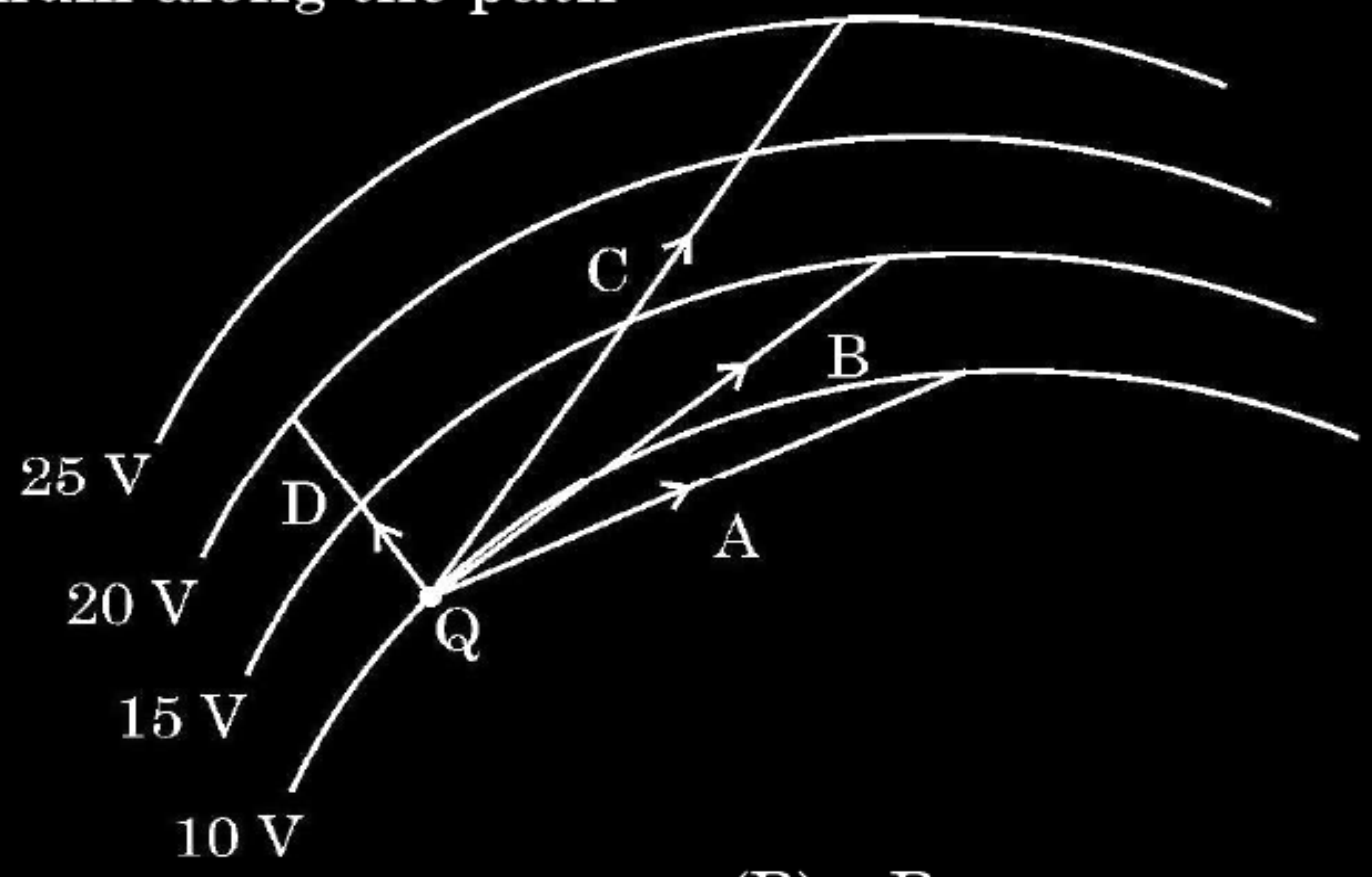


CBSE 25

ii) $V_c = \left[\dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}} - \dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}}\right]$	1
$= 0$	
$V_A = \left[\dfrac{k\times 6\times 10^{-6}}{15\times 10^{-2}} - \dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}}\right]$	½
$= \dfrac{k\times 6\times 10^{-6}}{10^{-2}}\left[\dfrac{1-3}{15}\right]$	
$= -\dfrac{9\times 10^{9}\times 6\times 10^{-6}\times 2}{15\times 10^{-2}}$	
$= -7.2\times 10^{5}$ V	½
$W = q[V_A - V_c]$	½
$= 5\times 10^{-6}[-7.2\times 10^{5} - 0]$	
$W = -3.6$ J	½

1. In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path 1



(A) A
(B) B
(C) C
(D) D

CBSE 25

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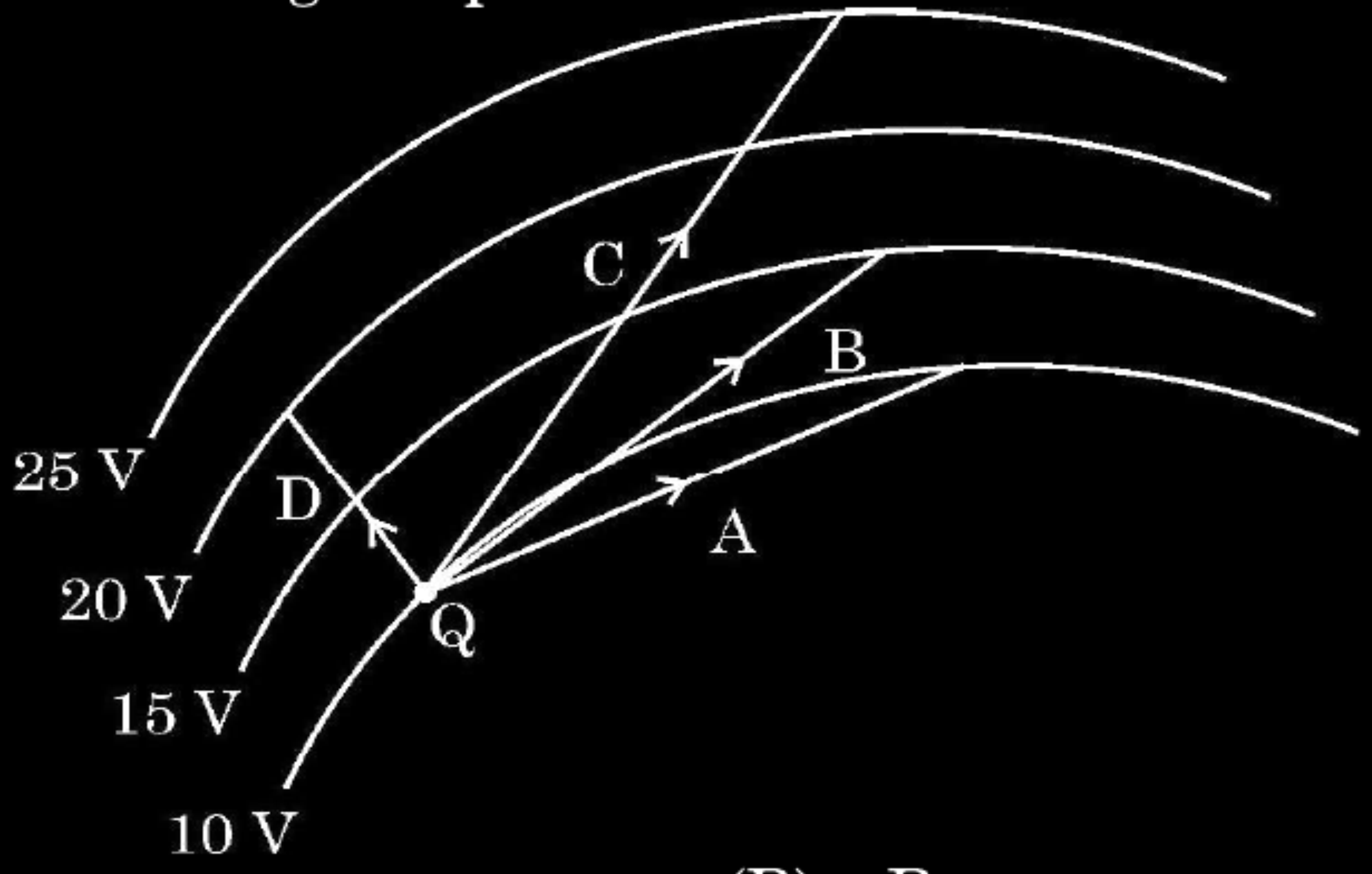

(A) A (B) B

(C) C (D) D

CBSE 25

(C) C

23. Two point charges of 10 μC and 20 μC are located at points (– 4 cm, 0, 0) and (5 cm, 0, 0) respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6\ NC^{-1}\ m^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. *3*

CBSE 24 C

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CBSE 24 C

Electrostatic potential energy of the system :	
$U = \frac{kq_1q_2}{r_{12}} + q_1V_1 + q_2V_2$	1
$\frac{kq_1q_2}{r_{12}} = \frac{9\times10^9\times10\times10^{-6}\times20\times10^{-6}}{9\times10^{-2}} = 20J$	½
$q_1V_1 = q_1\frac{A}{r_1} = \frac{10\times10^{-6}\times2\times10^6}{4\times10^{-2}} = 500J$	½
$q_2V_2 = q_2\frac{A}{r_2} = \frac{20\times10^{-6}\times2\times10^6}{5\times10^{-2}} = 800J$	½
$U = (20+500+800)\,J$ $U = 1320\,J$	½

3. A conducting sphere of radius R is given a charge Q. Consider three points A, B and C — A at the centre, B at a distance R/2 from the centre and C on the surface of the sphere. The electric potentials at these points are such that :

(A) $V_A = V_B = V_C$

(B) $V_A = V_B \neq V_C$

(C) $V_A \neq V_B \neq V_C$

(D) $V_A \neq V_B = V_C$

CBSE 24 C

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(C) $V_A \neq V_B \neq V_C$

(D) $V_A \neq V_B = V_C$

CBSE 24 C

(A) $V_A = V_B = V_C$

4. The capacitance of a parallel plate capacitor is 10 μF when the distance between its plates is 8 cm. If the distance between the plates is halved, the capacitance will become :

(A) 10 μF (B) 15 μF (C) 20 μF (D) 40 μF

CBSE 24 C

4. The capacitance of a parallel plate capacitor is 10 μF when the distance between its plates is 8 cm. If the distance between the plates is halved, the capacitance will become :

(A) 10 μF (B) 15 μF (C) 20 μF (D) 40 μF

CBSE 24 C

(C) 20μF

1. Two particles A and B of the same mass but having charges q and 4q respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

CBSE 24 C

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CBSE 24 C

(D) 4

12. A parallel plate capacitor is charged by a battery. The battery is then disconnected and the plates of the charged capacitor are then moved farther apart. In the process :

(A) the charge on the capacitor increases.

(B) the potential difference across the plates decreases.

(C) the capacitance of the capacitor increases.

(D) the electrostatic energy stored in the capacitor increases. CBSE 24 C

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(B) the potential difference across the plates decreases.

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(D) the electrostatic energy stored in the capacitor increases. CBSE 24 C

(D) The electrostatic energy stored in the capacitor increases

1. Three points charges + q, – 4q and + 2q are kept at the vertices of an equilateral triangle of side ‘a’. The electrostatic potential energy of this system of charges is : CBSE 24 C

(A) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{q^2}{a^2}\right)$

(B) $+\frac{1}{4\pi\varepsilon_0}\left(\frac{8q^2}{a^2}\right)$

(C) $+\frac{1}{4\pi\varepsilon_0}\left(\frac{6q^2}{a}\right)$

(D) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{10q^2}{a}\right)$

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CBSE 24 C

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29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.

(i) The shape of equipotential surface due to an isolated point charge is : *1*

(A) ellipsoidal, with charge at its one foci

(B) plane surface not passing through point charge

(C) spherical with charge at its centre

(D) cylindrical with charge at its axis

CBSE 24 C

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(D) cylindrical with charge at its axis

CBSE 24 C

(C) Spherical with charge at its centre

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are : *1*

(A) planes parallel to the XY plane

(B) concentric spherical surfaces

(C) planes parallel to the YZ plane

(D) planes parallel to the XZ plane

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are : *1*

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(A) Planes parallel to the XY plane

(iii) Which of the following statements is ***not*** true for equipotential surfaces ? *1*

(A) Potentials at two equipotential surfaces are different.

(B) No work is done in moving a charge on an equipotential surface.

(C) Equipotential surfaces always enclose some volume.

(D) Two equipotential surfaces do not intersect each other.

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(D) Two equipotential surfaces do not intersect each other.

(C) Equipotential surfaces always enclose some volume

(iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : *1*

(A) $90°$ (B) $45°$

(C) $30°$ (D) $0°$

(iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : *1*

(A) $90°$ (B) $45°$

(C) $30°$ (D) $0°$

(D) 0^0

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : *1*

(A) remain stationary

(B) move from A to B

(C) move from B to A

(D) move around Q in a circular path between the equipotential surfaces A and B

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : *1*

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(B) move from A to B

(C) move from B to A

(D) move around Q in a circular path between the equipotential surfaces A and B

(B) move from A to B

(b) (i) Obtain an expression for the capacitance of a parallel plate capacitor.

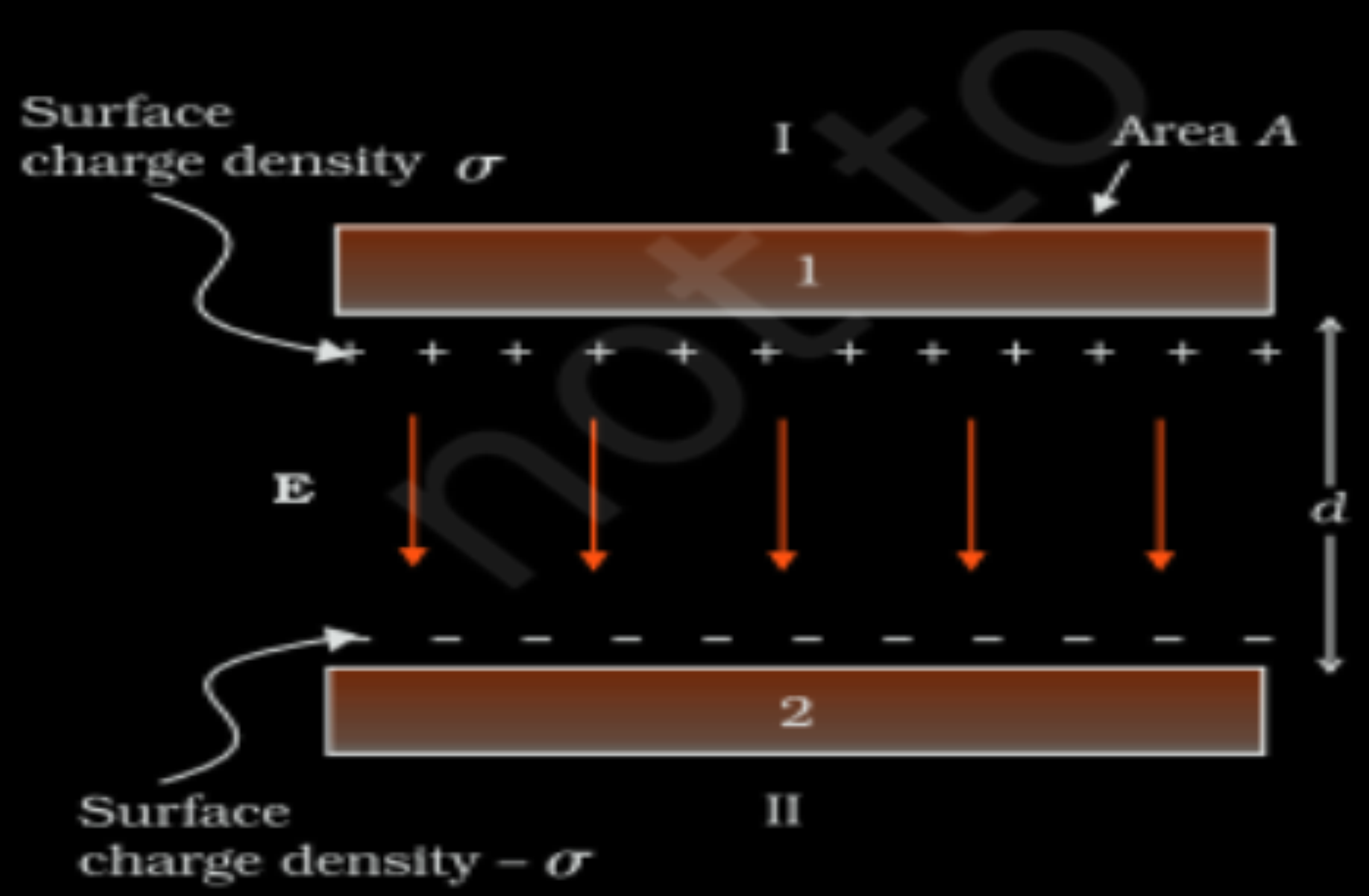
(ii) Two identical plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a distance d in air and a dielectric of dielectric constant K is filled between them. Write the expressions for : *5*

(1) the potential difference between the plates

(2) capacitance of the capacitor so formed

(3) energy stored in this capacitor

CBSE 24 C



Electric field in between the plates of the capacitor

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A} ----------- (1)$$ ½

The potential difference between the plates of the capacitor ½

V=Ed

The capacitance of the parallel plate capacitor

$$C = \frac{Q}{V}$$

$$C = \frac{\varepsilon_0 A}{d}$$ 1

(ii)

(1) Potential difference when dielectric of constant K is filled $= \frac{\sigma d}{\varepsilon_0 K}$ 1

Alternatively:-

$$V' = \frac{V}{K}$$

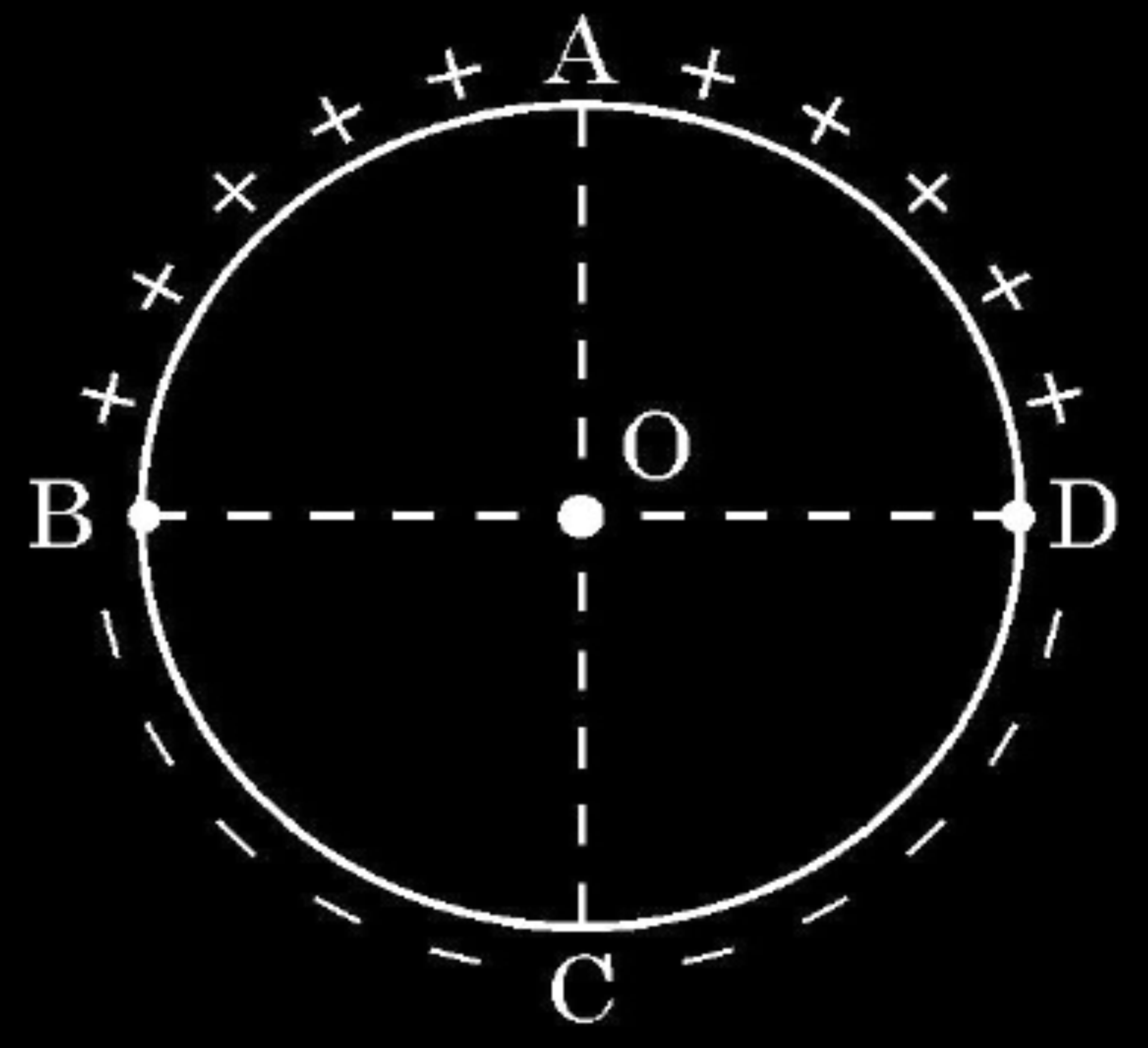
(2) Capacitance of the capacitor formed $= \frac{KA\varepsilon_0}{d}$ 1

$C' = KC$

(3) Energy stored in the capacitor $= \frac{U}{K}$ 1

15. **Assertion (A)** : Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.

Reason (R) : It is so because the net potential at O is not zero.



CBSE 24

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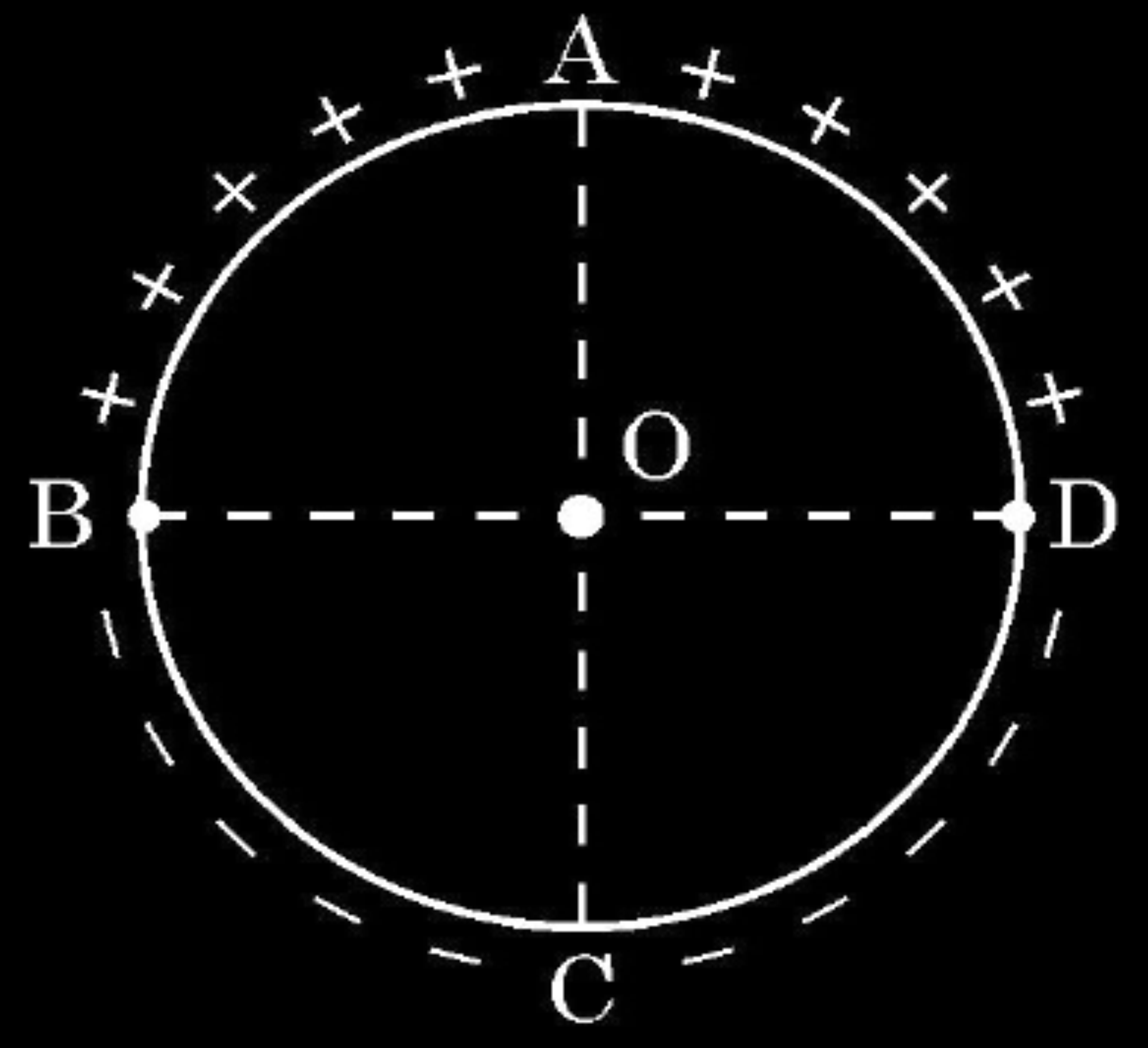

CBSE 24

(C) Assertion (A) is true but Reason (R) is false.

30. Dielectrics play an important role in design of capacitors. The molecules of a dielectric may be polar or non-polar. When a dielectric slab is placed in an external electric field, opposite charges appear on the two surfaces of the slab perpendicular to electric field. Due to this an electric field is established inside the dielectric.

The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel.

(i) Which of the following is a polar molecule ? CBSE 24

(A) O_2 (B) H_2

(C) N_2 (D) HCl

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(i) Which of the following is a polar molecule ? CBSE 24

(A) O_2 (B) H_2

(C) N_2 (D) HCl

(D) HCl

(ii) Which of the following statements about dielectrics is correct ?

(A) A polar dielectric has a net dipole moment in absence of an external electric field which gets modified due to the induced dipoles.

(B) The net dipole moments of induced dipoles is along the direction of the applied electric field.

(C) Dielectrics contain free charges.

(D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

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(B) The net dipole moment of induced dipoles is along the direction of the applied electric field.

(iii) When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it :

(A) increases and the electric field inside it also increases.

(B) decreases and the electric field also decreases.

(C) decreases and the electric field increases.

(D) increases and the electric field decreases.

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(B) decreases and the electric field also decreases.

(C) decreases and the electric field increases.

(D) increases and the electric field decreases.

(B) decreases and the electric field also decreases.

(iv) (a) An air-filled capacitor with plate area A and plate separation d has capacitance C_0. A slab of dielectric constant K, area A and thickness $\left(\frac{d}{5}\right)$ is inserted between the plates. The capacitance of the capacitor will become

(A) $\left[\frac{4K}{5K+1}\right]C_0$

(B) $\left[\frac{K+5}{4}\right]C_0$

(C) $\left[\frac{5K}{4K+1}\right]C_0$

(D) $\left[\frac{K+4}{5K}\right]C_0$

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(C) $\left[\frac{5K}{4K+1}\right]C_0$ (D) $\left[\frac{K+4}{5K}\right]C_0$

(C) $\left[\frac{5K}{4K+1}\right]C_0$

(iv) (b) Two capacitors of capacitances $2\ C_0$ and $6\ C_0$ are first connected in series and then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is

(A) $\frac{1}{4}$ (B) $\frac{1}{6}$

(C) $\frac{2}{15}$ (D) $\frac{3}{16}$

(iv) (b) Two capacitors of capacitances $2\ C_0$ and $6\ C_0$ are first connected in series and then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is

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(C) $\frac{2}{15}$ (D) $\frac{3}{16}$

(D) $\frac{3}{16}$

33. (a) (i) Draw equipotential surfaces for an electric dipole.

(ii) Two point charges q_1 and q_2 are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in an external electric field $\vec{E}$. Obtain an expression for the potential energy of the system.

(iii) The dipole moment of a molecule is 10^{-30} Cm. It is placed in an electric field $\vec{E}$ of 10^5 V/m such that its axis is along the electric field. The direction of $\vec{E}$ is suddenly changed by $60°$ at an instant. Find the change in the potential energy of the dipole, at that instant.

CBSE 24

1

(ii) Work done in bringing a charge q_1 from infinity to $\vec{r}_1$:

$$W_1 = q_1 V(\vec{r}_1) \qquad \text{-------------(1)}$$ ½

Work done in bringing a charge q_2 from infinity to $\vec{r}_2$ against the external field :

$$W_2 = q_2 V(\vec{r}_2) \qquad \text{-------------(2)}$$ ½

Work done on q_2 against the field due to q_1:

$$W_{12} = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} \qquad \text{-----------------(3)}$$ ½

Potential energy of the system = Total work done

$$= q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$$ ½

(iii) Change in Potential energy = Work done

$W = pE\,[\cos\theta_0 - \cos\theta_1]$ 1

$W = 10^{-30} \times 10^{5}\,[\cos 0^0 - \cos 60^0]$ ½

$W = 5.0 \times 10^{-26}$ J ½

1. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be **1**

(A) 2 pE (B) –2 pE

(C) pE (D) zero

CBSE 24

1. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be **1**

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CBSE 24

(A) 2 pE

23. Two conducting spherical shells A and B of radii R and 2R are kept far apart and charged to the same charge density σ. They are connected by a wire. Obtain an expression for final potential of shell A. **3**

CBSE 24

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CBSE 24

Charge on spherical Shell A $q_A = 4\pi R^2\sigma$	½
Charge on speherical shell B $q_B = 4\pi(2R)^2\sigma = 16\pi R^2\sigma$	½
After connecting by a wire , their potentials will become equal after sharing of charge. $\frac{1}{4\pi\varepsilon_0}\frac{q'_A}{R} = \frac{1}{4\pi\varepsilon_0}\frac{q'_B}{2R}$ $q'_B = 2q'_A$	½
From conservation of charge $q_A + q_B = q'_A + q'_B$ $4\pi R^2\sigma + 16\pi R^2\sigma = 3q'_A$	½
$q'_A = \frac{20\pi R^2\sigma}{3}$	½
Final potential of Shell A $V_A = \frac{1}{4\pi\varepsilon_0}\frac{q'_A}{R}$ $V_A = \frac{1}{4\pi\varepsilon_0}\frac{20\pi R^2\sigma}{3R}$ $V_A = \frac{5\sigma R}{3\varepsilon_0}$	½

Alternatively

Charge on spherical shell A

$q_A = 4\pi R^2\sigma$

Charge on spherical shell B

$q_B = 4\pi(2R)^2\sigma = 16\pi R^2\sigma$

After connecting by a wire , their potential will become equal after sharing of charges

Therefore the potential of shell A

$$V_A = V_{common} = \frac{q_A + q_B}{C_A + C_B}$$

$$= \frac{4\pi R^2\sigma + 16\pi R^2\sigma}{4\pi\varepsilon_0 R + 4\pi\varepsilon_0 (2R)}$$

$$= \frac{5\sigma R}{3\varepsilon_0}$$

31. (a) (i) A dielectric slab of dielectric constant 'K' and thickness 't' is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.

(ii) Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. If the total energy stored in the combination in the two cases are 40 mJ and 250 mJ respectively, find the capacitance of the capacitors. **5**

CBSE 24

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CBSE 24

a) (i)

Electric field in air between plates

$$E_0 = \frac{\sigma}{\varepsilon_0}$$ ½

Electric field inside the dielectric

$$E = \frac{\sigma}{\varepsilon_0 K}$$ ½

Potential difference between the plate

$$V = E_0(d-t) + Et$$ ½

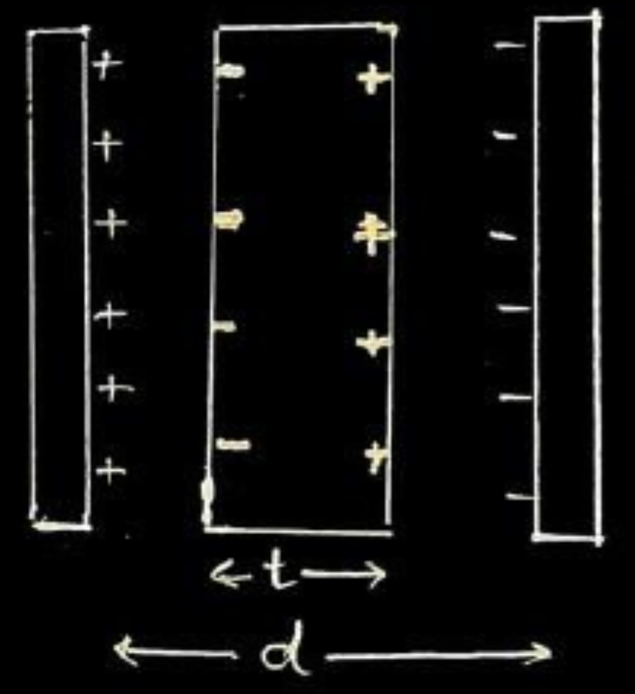
$$V = \frac{\sigma}{\varepsilon_0}\left[d-t+\frac{t}{K}\right]$$

$$V = \frac{q}{A\varepsilon_0}\left[d-t+\frac{t}{K}\right]$$ ½

Capacitance

$$C = \frac{q}{V}$$ ½

$$C = \frac{A\varepsilon_0}{d-t+\frac{t}{K}}$$



$$C = \frac{A\varepsilon_0}{d-t\left(1-\frac{1}{K}\right)}$$ ½

ii) Total energy stored in series combination

$$\frac{1}{2}\left(\frac{C_1C_2}{C_1+C_2}\right)V^2 = 40\times10^{-3}\text{ J}\ldots\ldots(1)$$ ½

Energy stored in parallel combination

$$\frac{1}{2}(C_1+C_2)V^2 = 250\times10^{-3}\text{ J}\ldots\ldots(2)$$ ½

Substituting value of V=100 V in eq (1) and (2) , on solving

$C_1 = 4\times10^{-5}$ F or $40\,\mu$F ½

$C_2 = 1\times10^{-5}$ F or $10\,\mu$F ½

1. Three point charges, each of charge q are placed on vertices of a triangle ABC, with AB = AC = 5L, BC = 6L. The electrostatic potential at midpoint of side BC will be **1**

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$

(B) $\frac{8q}{36\pi\varepsilon_0 L}$

(C) $\frac{5q}{24\pi\varepsilon_0 L}$

(D) $\frac{1}{16}\frac{q}{\pi\varepsilon_0 L}$

CBSE 24

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CBSE 24

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$

23. A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. Find (i) The charge density on the outer surface of the shell and (ii) the potential at a distance of (R/2) from the centre of the shell. 3

CBSE 24

23. A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. Find (i) The charge density on the outer surface of the shell and (ii) the potential at a distance of (R/2) from the centre of the shell. **3**

CBSE 24

(i) When a point charge Q is placed at the centre of the shell, a charge (–Q) is induced at its inner surface, consequently a net charge on outer surface of the shell = q + Q	1
Charge density on outer surface of the shell $\sigma = \frac{charge}{Area}$	½
$= \frac{q+Q}{4\pi R^2}$	
(ii) Potential due to shell at a distance of (R/2) from the centre of the shell	
$V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$	½
Potential due to charge Q at a distance of (R/2) from the centre of the shell	
$V_2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{Q}{R/2}$	½
Net potential at a distance of (R/2) from the centre of the shell	
$V = V_2 + V_2$	
$V = \frac{1}{4\pi\varepsilon_0 R}(q+2Q)$	½

1. Two charges + q each are kept '2a' distance apart. A third charge – 2q is placed midway between them. The potential energy of the system is – **1**

(A) $\frac{q^2}{8\pi\varepsilon_0 a}$ (B) $-\frac{6q^2}{8\pi\varepsilon_0 a}$

(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$ (D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

CBSE 24

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(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$ (D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

CBSE 24

(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$

23. The electric field in a region is given by

$$\vec{E} = (10x + 4)\,\hat{i}$$

where x is in m and E is in N/C. Calculate the amount of work done in taking a unit charge from

(i) (5 m, 0) to (10 m, 0)

(ii) (5 m, 0) to (5 m, 10 m) CBSE 24 **3**

(i)

$$\Delta V = -\int_{x_1}^{x_2} E\,dx \qquad \tfrac{1}{2}$$

$$\Delta V = -\int_{5}^{10} (10x+4)dx = -\left[\frac{10x^2}{2} + 4x\right]_5^{10}$$

$$= -395 \text{ V} \qquad \tfrac{1}{2}$$

$$W = q\Delta V = -395 \times 1 \qquad \tfrac{1}{2}$$

$$= -395 \text{ J} \qquad \tfrac{1}{2}$$

(ii)

$$\Delta V = -\int_{x_1}^{x_2} E\,dx$$

$$\Delta V = -\int_{5}^{5} (10x+4)dx = 0 \qquad \tfrac{1}{2}$$

$$W = q.\Delta V = 0 \qquad \tfrac{1}{2}$$

Alternatively

If a student writes, displacement is perpendicular to electric field then

$$\Delta V = 0 \qquad \tfrac{1}{2}$$

$$W = q.\Delta V = 0 \qquad \tfrac{1}{2}$$

Award full credit for part (ii)

1. The plates P_1 and P_2 of a 2 μF capacitor are at potentials 25 V and – 25 V respectively. The charge on plate P_1 will be :

(A) 0·02 mC (B) 0·1 mC

(C) 0·1 μC (D) 1 μC

CBSE 24

1. The plates P_1 and P_2 of a 2 μF capacitor are at potentials 25 V and – 25 V respectively. The charge on plate P_1 will be :

(A) 0·02 mC (B) 0·1 mC

(C) 0·1 μC (D) 1 μC

CBSE 24

(B) 0.1mC

2. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5$ V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :

(A) $-1 \cdot 6 \times 10^{-18}$ J

(B) $1 \cdot 6 \times 10^{-18}$ J

(C) Zero

(D) $0 \cdot 8 \times 10^{-18}$ J

CBSE 24

2. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5$ V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :

(A) $-1 \cdot 6 \times 10^{-18}$ J (B) $1 \cdot 6 \times 10^{-18}$ J

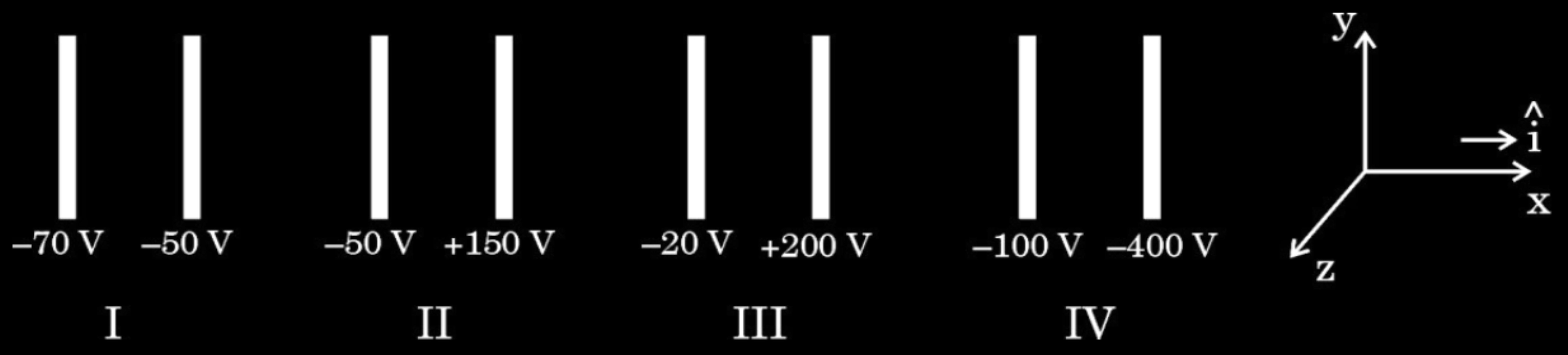
(C) Zero (D) $0 \cdot 8 \times 10^{-18}$ J

CBSE 24

(B) 1.6×10^{-18} J

29. The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2·0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates.

CBSE 24



(i) For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$? *1*

(A) I (B) II

(C) III (D) IV

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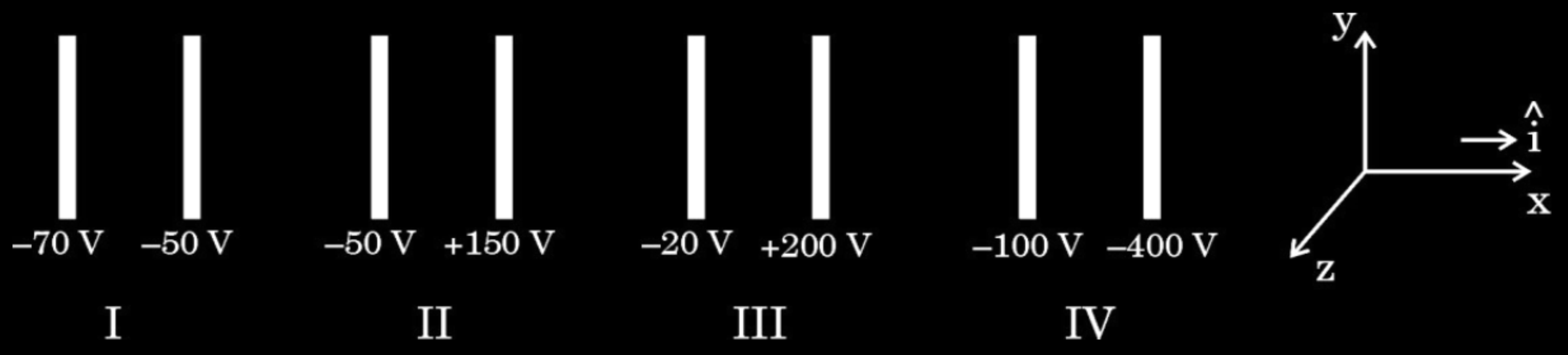
CBSE 24



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(A) I (B) II

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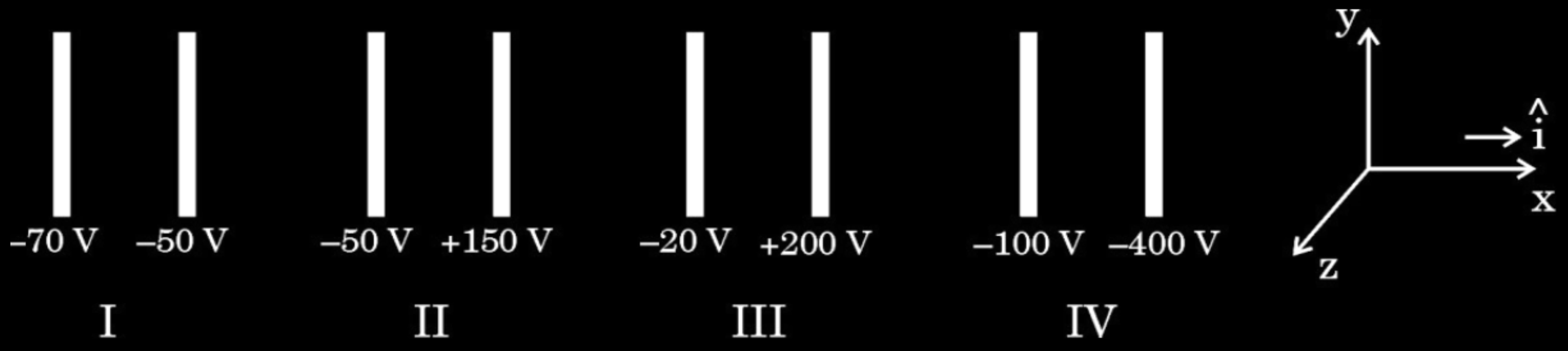
(D) IV



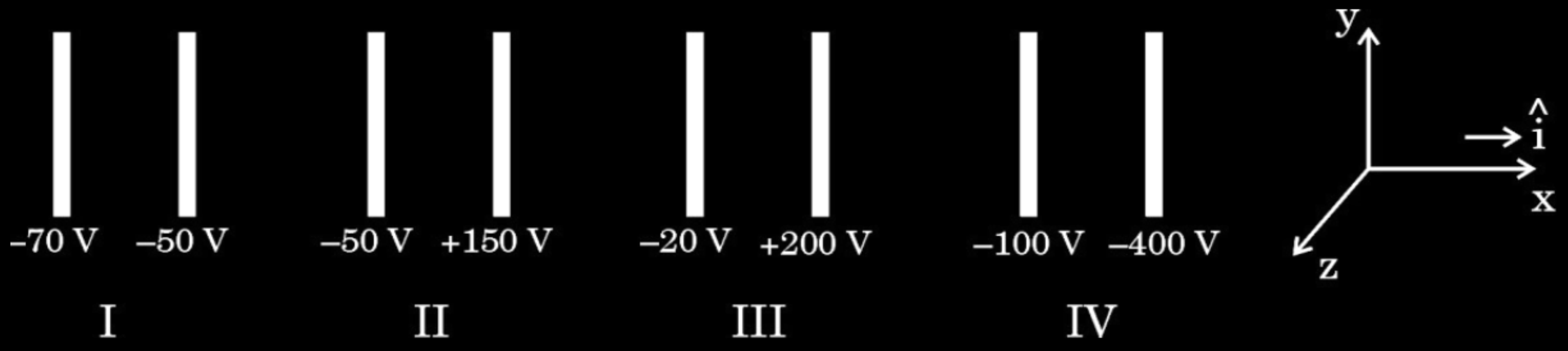
(ii) An electron is released midway between the plates of pair IV. It will : *1*

(A) move along $\hat{i}$ at constant speed

(B) move along $-\hat{i}$ at constant speed

(C) accelerate along $\hat{i}$

(D) accelerate along $-\hat{i}$



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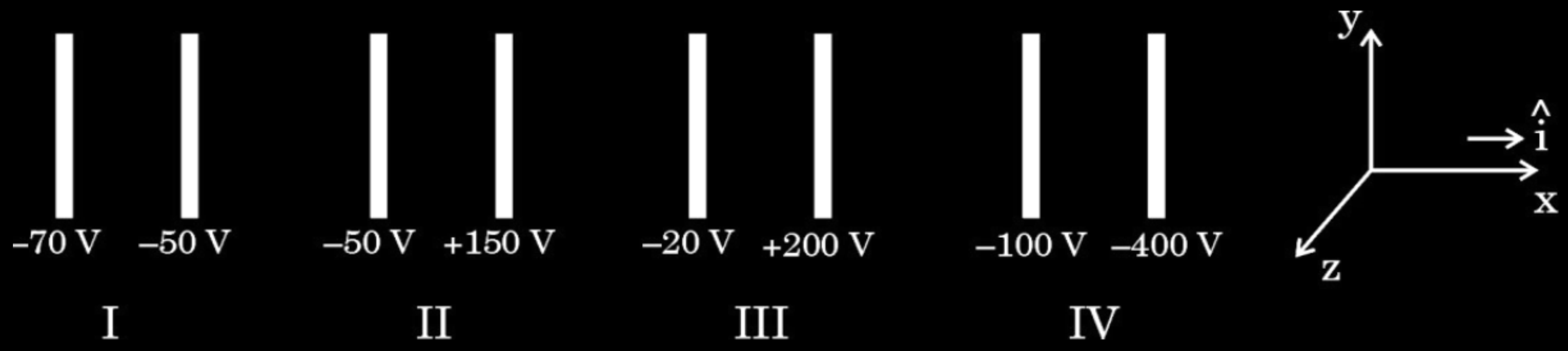
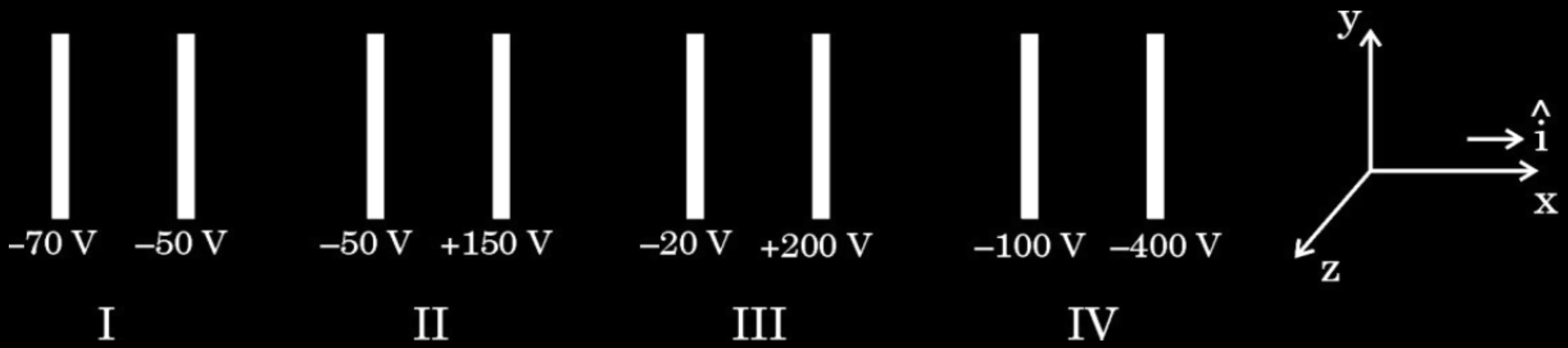


(iii) Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point ($0 \le x \le 2$ cm) between the plates of that set can be expressed as : *1*

(A) $V = V_0 + \alpha x$ (B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$ (D) $V = V_0 + \alpha x^{3/2}$

where α is a constant, positive or negative.



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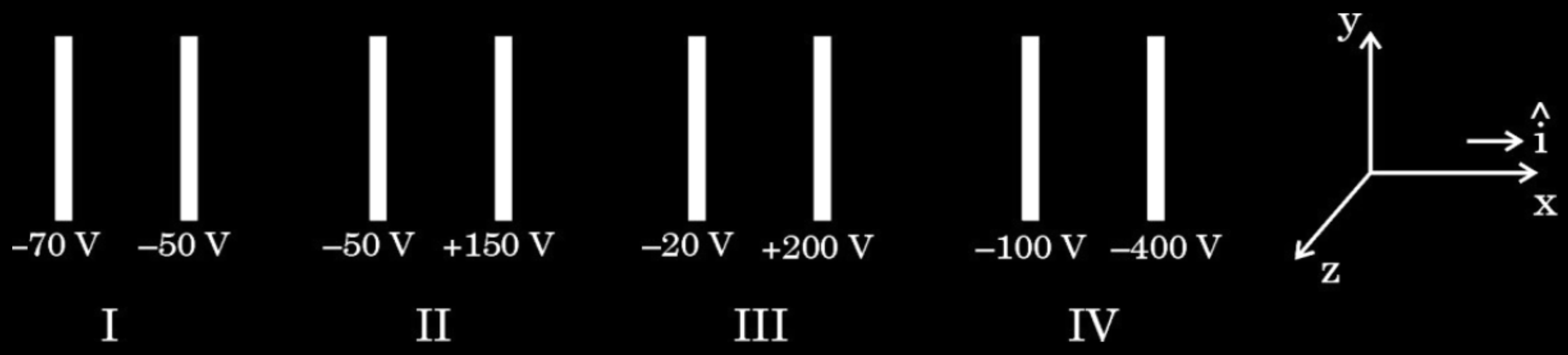
where α is a constant, positive or negative.

(iii) (A) $V = V_0 + \alpha x$



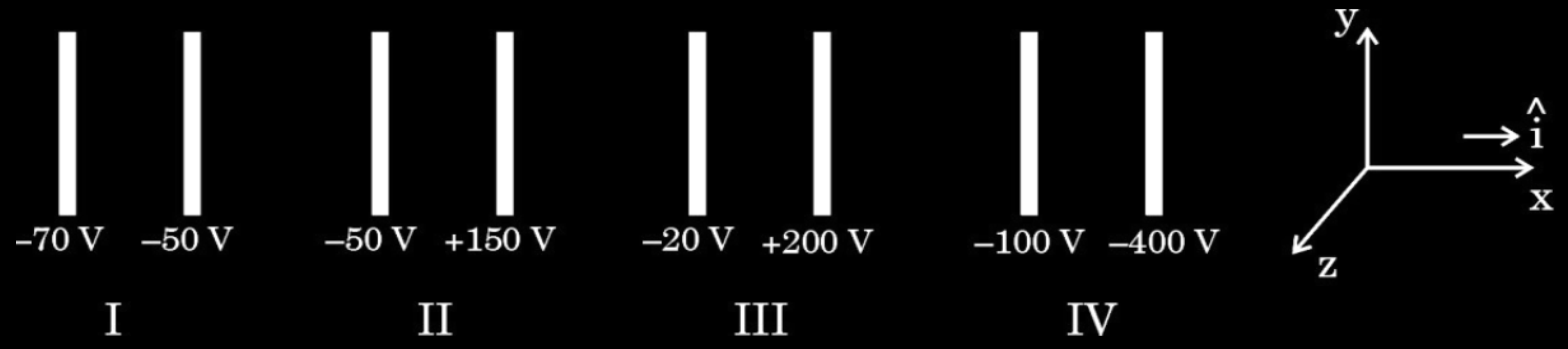
(iv) (a) Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then : *1*

(A) $E_1 > E_2 > E_3 > E_4$

(B) $E_3 > E_4 > E_1 > E_2$

(C) $E_4 > E_3 > E_2 > E_1$

(D) $E_2 > E_3 > E_4 > E_1$



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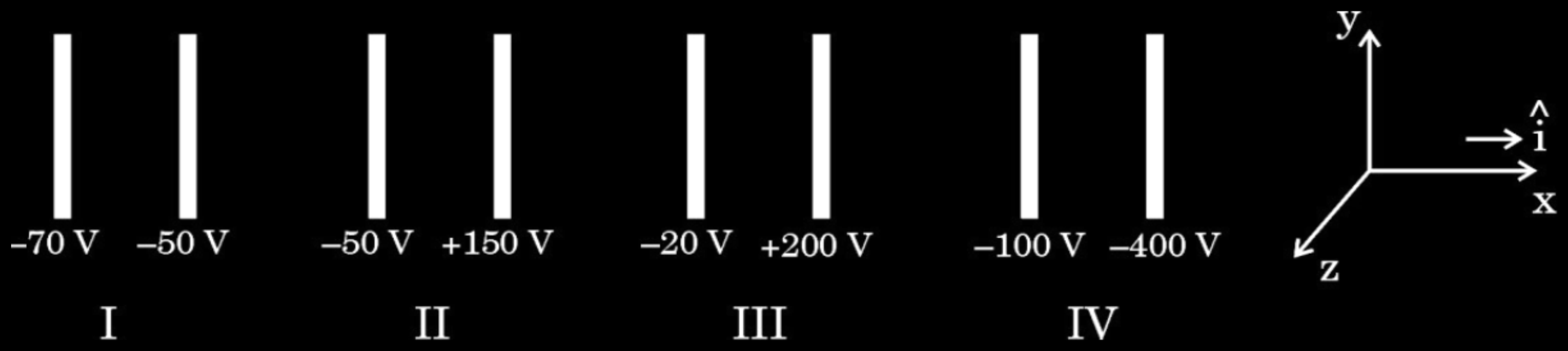
(A) $E_1 > E_2 > E_3 > E_4$ (B) $E_3 > E_4 > E_1 > E_2$

(C) $E_4 > E_3 > E_2 > E_1$ (D) $E_2 > E_3 > E_4 > E_1$

(a) (C) $E_4 > E_3 > E_2 > E_1$

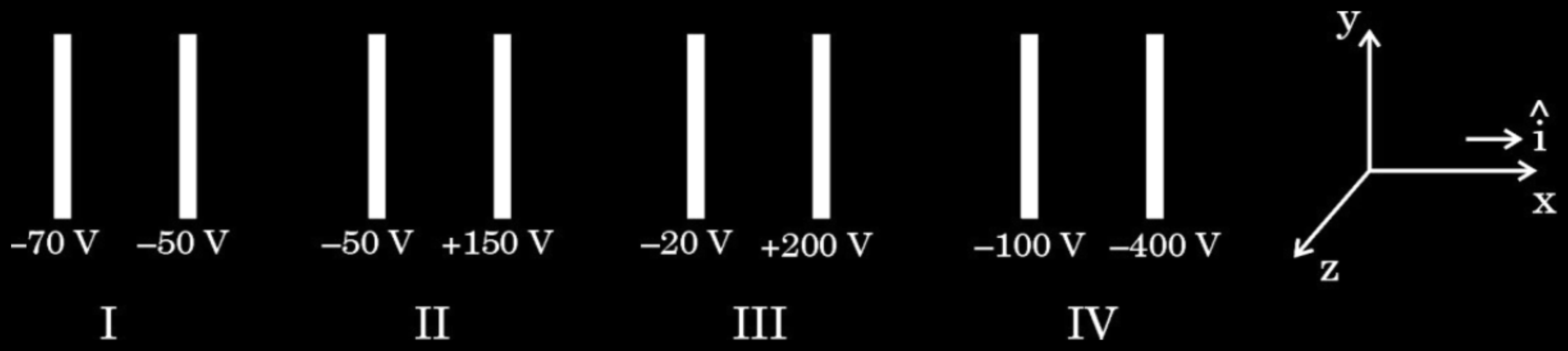


(b) An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about :

(Take (e/m) $=1 \cdot 76 \times 10^{11}$ C/kg) *1*

(A) $1 \cdot 3 \times 10^5$ m/s (B) $2 \cdot 6 \times 10^6$ m/s

(C) $6 \cdot 5 \times 10^5$ m/s (D) $5 \cdot 2 \times 10^7$ m/s



(b) An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about :
(Take $(e/m) = 1{\cdot}76 \times 10^{11}$ C/kg) *1*

(A) $1{\cdot}3 \times 10^5$ m/s (B) $2{\cdot}6 \times 10^6$ m/s

(C) $6{\cdot}5 \times 10^5$ m/s (D) $5{\cdot}2 \times 10^7$ m/s

(B) $2.6\text{x}10^6$ m/s

31. (a) (i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.

(ii) A charge of 6 μC is given to a hollow metallic sphere of radius 0·2 m. Find the potential at (i) the surface and (ii) the centre of the sphere. CBSE 24 *5*

When a dielectric slab is inserted between the plates of capacitance there is induced charge density σ_P which opposes the original charge density (σ) on the plate of capacitance.	½
Electric field with dielectric medium is	
$E = \frac{(\sigma - \sigma_P)}{\varepsilon_0}$	½
$V = E \times d = \frac{(\sigma - \sigma_P)}{\varepsilon_0} d$	½
$(\sigma - \sigma_P) = \frac{\sigma}{K}$	½
$V = \frac{\sigma\, d}{\varepsilon_0 K} = \frac{Qd}{A\varepsilon_0 K}$	½
$C = \frac{Q}{V} = \frac{K\varepsilon_0 A}{d}$	½
(ii) Electric potential due to a point charge	
$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$	½
(i) At the surface	
$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = \frac{9\times10^9\times6\times10^{-6}}{0.2}$	½
$V = 2.7 \times 10^5$ V	½
(ii) Since electric field inside the hollow sphere is zero, hence V remains constant throughout the volume.	
$V = 2.7 \times 10^5$ V	½

1. Consider a group of charges q_1, q_2, q_3 ... such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

CBSE 24

1. Consider a group of charges q_1, q_2, q_3 ... such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

CBSE 24

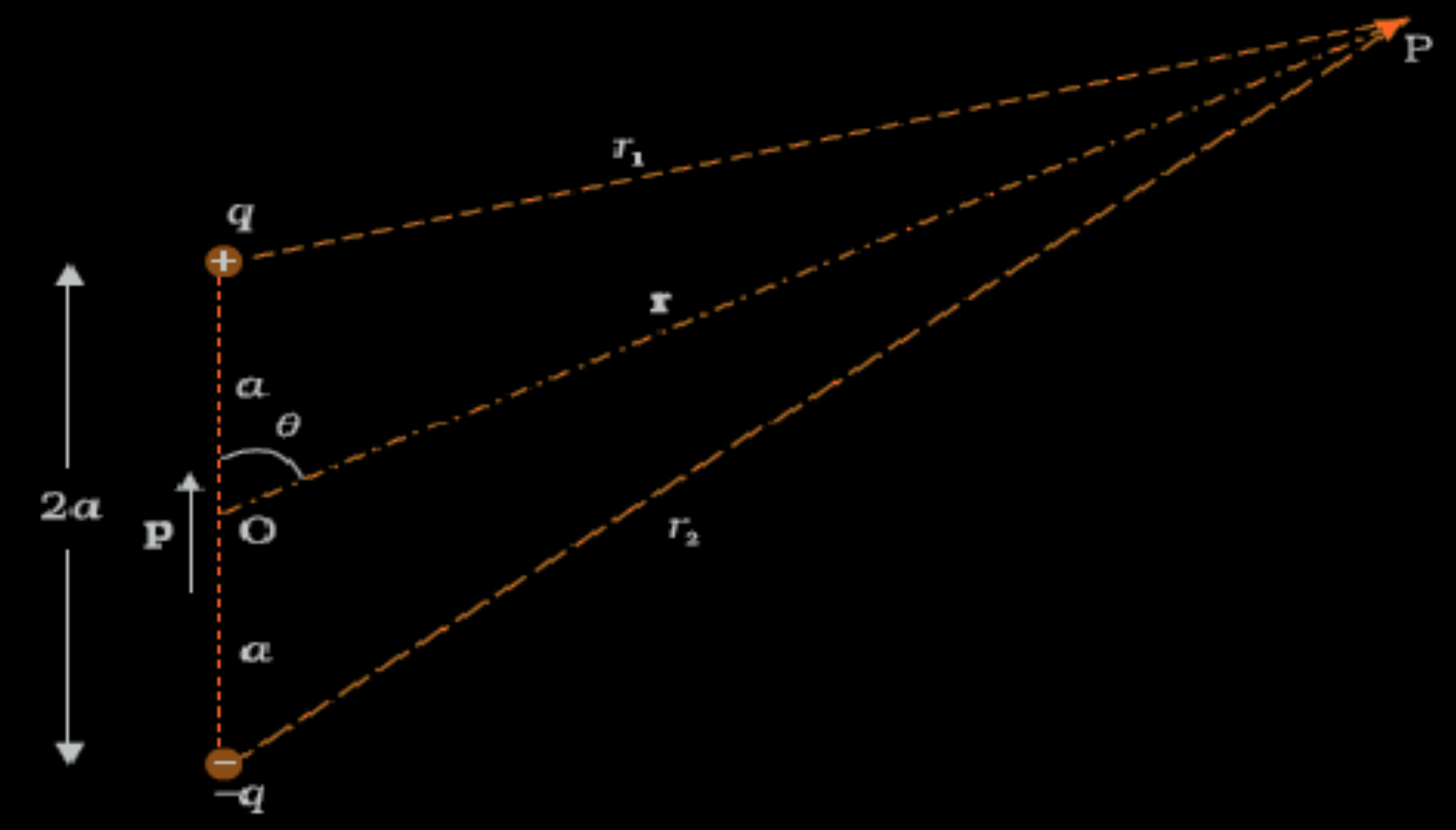
(B) Spherical surface

32. (a) (i) Obtain an expression for the electric potential due to a small dipole of dipole moment $\vec{p}$, at a point $\vec{r}$ from its centre, for much larger distances compared to the size of the dipole.

(ii) Three point charges q, 2q and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n. *5*

CBSE 24

(i)



½

Potential due to the dipole is the sum of potentials due to charges q and -q

$$V=\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_1}-\frac{q}{r_2}\right) \quad \text{------------------------------------(1)}$$ ½

By geometry

$$r_1^2=r^2+a^2-2ar\cos\theta$$

$$r_2^2=r^2+a^2+2ar\cos\theta$$ ½

For r >> a, retaining terms only up to first order in a/r

$$r_1^2=r^2\left(1-\frac{2a\cos\theta}{r}+\frac{a^2}{r^2}\right)$$

$$\cong r^2\left(1-\frac{2a\cos\theta}{r}\right)$$

Similarly

$$r_2^2\cong r^2\left(1+\frac{2a\cos\theta}{r}\right)$$ ½

Using the binomial theorem and retaining terms up to the first order in a/r

$$\frac{1}{r_1}\cong\frac{1}{r}\left(1-\frac{2a\cos\theta}{r}\right)^{-1/2}$$

$$\cong\frac{1}{r}\left(1+\frac{a\cos\theta}{r}\right) \quad \text{------------------------------------(2)}$$

$$\frac{1}{r_2}\cong\frac{1}{r}\left(\frac{1+2a\cos\theta}{r}\right)^{-1/2} \quad \text{------------------------------(3)}$$

$$\cong\frac{1}{r}\left(1-\frac{a\cos\theta}{r}\right)$$ ½

Using eqn. (1) (2), (3) and p = 2qa

$$V=\frac{q}{4\pi\varepsilon_0}\frac{2a\cos\theta}{r^2}$$ ½

$$=\frac{\mathrm{p}\cos\theta}{4\pi\varepsilon_0 r^2}$$

(ii) Consider the side of equilateral triangle as ‘a’ ½

Potential energy $=\text{U}=\frac{kq_1q_2}{a}+\frac{kq_2q_3}{a}+\frac{kq_1q_3}{a}$ ½

According to question

$$U=\frac{k(q)(2q)}{a}+\frac{k(2q)(nq)}{a}+\frac{k(q)(\text{n}\,q)}{a}=0$$ ½

$$=\frac{2q^2}{a}+\frac{2nq^2}{a}+\frac{\text{n}\,q^2}{a}=0$$

$$2+2n+n=0$$ ½

$$3n=-2$$

$$n=-\frac{2}{3}$$ ½

1. The capacitance of a parallel plate capacitor having a medium of dielectric constant K = 4 in between the plates is C. If this medium is removed, then the capacitance of the capacitor becomes :

(A) 4C

(B) C

(C) $\frac{C}{4}$

(D) 2C

CBSE 24

1. The capacitance of a parallel plate capacitor having a medium of dielectric constant K = 4 in between the plates is C. If this medium is removed, then the capacitance of the capacitor becomes :

(A) 4C (B) C

(C) $\frac{C}{4}$ (D) 2C

CBSE 24

(C) $\frac{C}{4}$

22. An air-filled parallel plate capacitor with plate separation 1 mm has a capacitance of 20 pF. It is charged to $4 \cdot 0$ μC. Calculate the amount of work done to pull its plates to a separation of 5 mm. Assume the charge on the plates remains the same. CBSE 24 *3*

22. An air-filled parallel plate capacitor with plate separation 1 mm has a capacitance of 20 pF. It is charged to 4·0 μC. Calculate the amount of work done to pull its plates to a separation of 5 mm. Assume the charge on the plates remains the same. CBSE 24 *3*

$C=\frac{A\varepsilon_\circ}{d}$; $C'=\frac{A\varepsilon_\circ}{d'}$ $\Rightarrow \frac{C'}{C}=\frac{d}{d'} \Rightarrow C'=\frac{1}{5}\times 20$ $C' = 4\text{pF}$	1
Work done = change in energy $= \frac{1}{2}\frac{Q^2}{C'}-\frac{1}{2}\frac{Q^2}{C}$ $= \frac{Q^2}{2}\left[\frac{1}{4}-\frac{1}{20}\right]\times 10^{12}$	1
On solving ; Work done $= \frac{8}{5}J$ $=1.6J$	1

32. (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum ? CBSE 24

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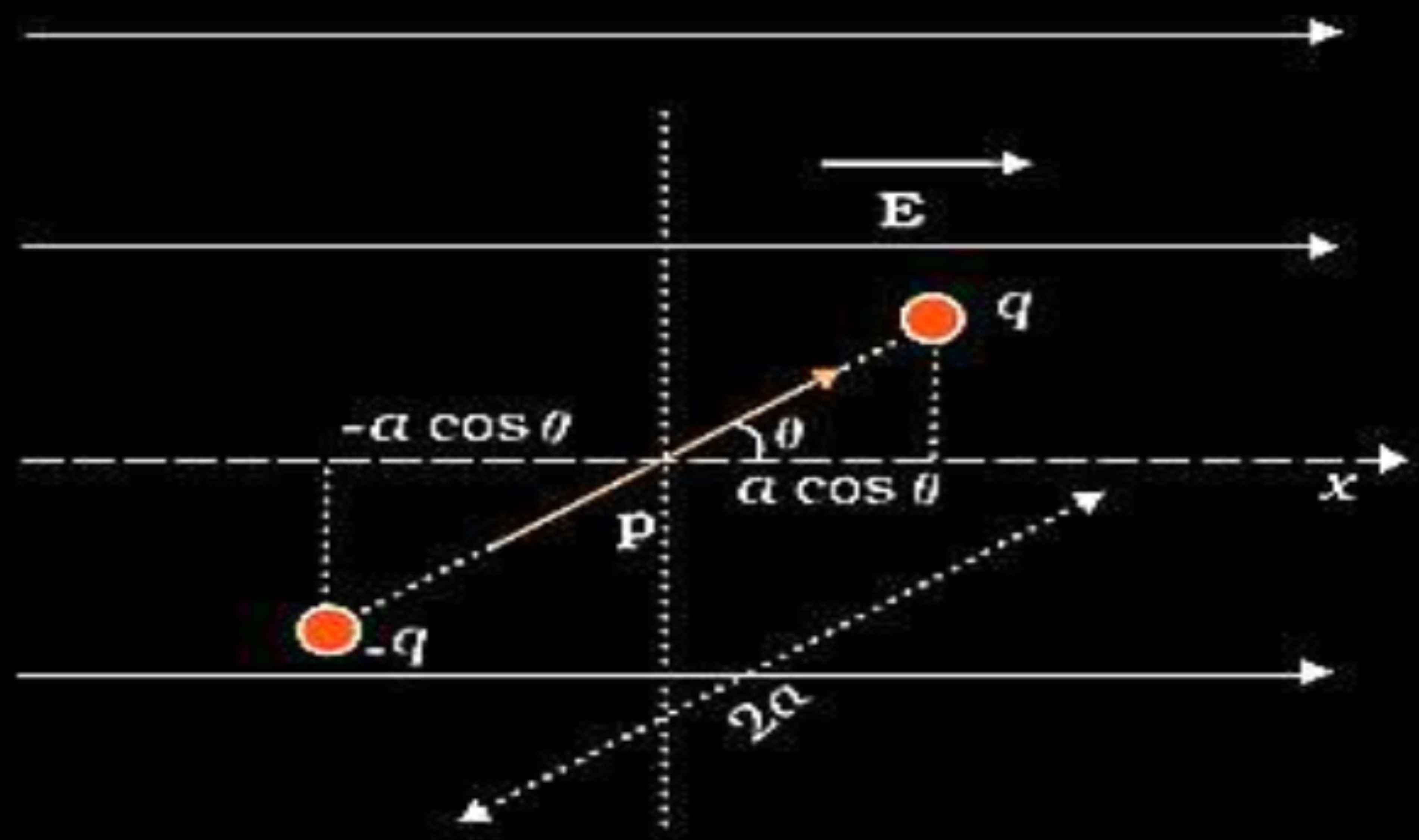
CBSE 24

The amount of work done in rotating the dipole from $\theta = \theta_0$ to $\theta = \theta_1$ by the external torque	
$W = \int_{\theta_0}^{\theta_1} \tau_{ext}\, d\theta$	½
$= \int_{\theta_0}^{\theta_1} pE \sin\theta \; d\theta$	½
$W = pE(\cos\theta_0 - \cos\theta_1)$	
For $\theta_0 = \frac{\pi}{2}$ & $\theta_1 = \theta$	½
$= pE(\cos\frac{\pi}{2} - \cos\theta)$	
$U(\theta) = -pE \cos\theta$	
$= -\vec{p}.\vec{E}$	½

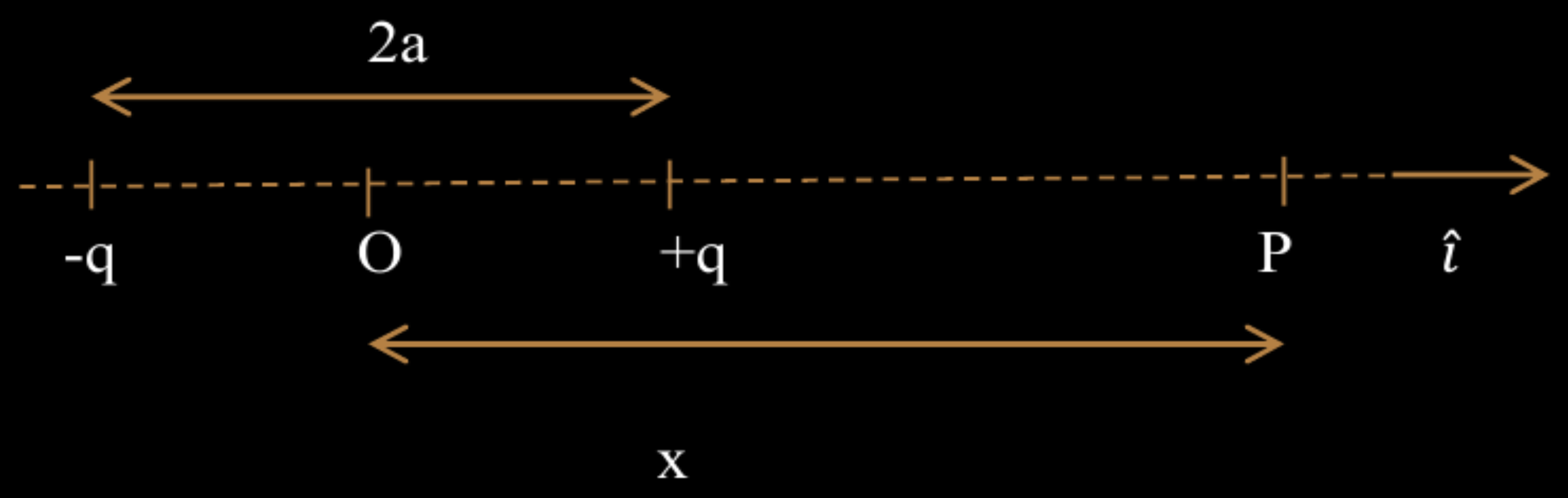


(1) Potential energy is maximum when: $\vec{p}$ is antiparallel to $\vec{E}$ Alternatively: $\theta = 180°$ or π radians	½
(2) Potential energy is minimum when: $\vec{p}$ is along to $\vec{E}$ Alternatively: $\theta = 0°$	½

(b) (i) An electric dipole (dipole moment $\vec{p} = p\,\hat{i}$), consisting of charges – q and q, separated by distance 2a, is placed along the x-axis, with its centre at the origin. Show that the potential V, due to this dipole, at a point x, (x >> a) is equal to $\frac{1}{4\pi\varepsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$.

(ii) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) C/m^2$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1. CBSE 24 *5*

(i)



½

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$ ½

$$V = V_{+q} - V_{-q}$$

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{(x-a)} - \frac{q}{(x+a)}\right]$$ ½

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{x+a-x+a}{(x^2-a^2)}\right]$$ ½

$$V = \frac{q}{4\pi\varepsilon_0}\frac{2a}{(x^2-a^2)} = \frac{p}{4\pi\varepsilon_0(x^2-a^2)}$$

As p is along x-axis, so

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\hat{i}}{(x^2-a^2)}$$

If x>>a

½

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\hat{i}}{x^2}$$

(ii)

Charge on sphere S_1 :

$$Q_1 = \text{surface charge density} \times \text{surface Area}$$

$$= \left(\frac{2}{\pi}\times 10^{-9}\right)\times 4\pi\,(1\times 10^{-2})^2$$

$$= 8\times 10^{-13}\,C$$ ½

Charge on sphere S_2 :

$$Q_2 = \text{surface charge density} \times \text{surface Area}$$

$$= \left(\frac{2}{\pi}\times 10^{-9}\right)\times 4\pi(3\times 10^{-2})^2$$

$$= 72\times 10^{-13}\,C$$ ½

When connected by a thin wire they acquire a common potential V and the charge remains conserved.

$$Q_1 + Q_2 = Q_1' + Q_2'$$ ½

$$= C_1V + C_2V$$

$$Q_1 + Q_2 = (C_1 + C_2)V$$

$$\text{Common potential}(V) = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$C_1 = 4\pi\varepsilon_0 r_1 = \frac{1}{9\times 10^9}\times 10^{-2} = \frac{1}{9}\times 10^{-11}F$$

$$C_2 = 4\pi\varepsilon_0 r_2 = \frac{1}{9\times 10^9}\times 3\times 10^{-2} = \frac{1}{3}\times 10^{-11}F$$

$$V = \frac{80\times 10^{-13}}{\left(\frac{1}{9}+\frac{1}{3}\right)\times 10^{-11}} = 1.8\,V$$ ½

$$Q_1' = C_1V = \frac{1}{9}\times 10^{-11}\times 1.8$$

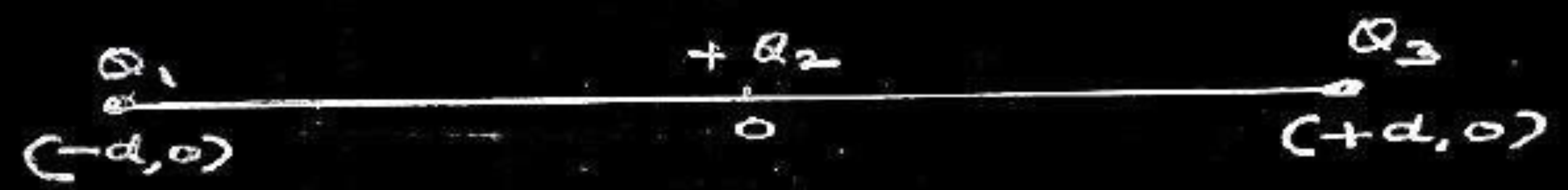
$$Q_1' = 2\times 10^{-12}\,C$$ ½

22. Three point charges Q_1, Q_2 and Q_3 are located in $x - y$ plane at points $(-d, 0)$, $(0, 0)$ and $(d, 0)$ respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ? *3*

CBSE 24

22. Three point charges Q_1, Q_2 and Q_3 are located in x – y plane at points (– d, 0), (0, 0) and (d, 0) respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ? *3*

CBSE 24



Nature of Q_1 will be negative.	1
Let , $Q_1 = Q_3 = q$	
$$\frac{1}{4\pi\varepsilon_0}\left[\frac{qQ_2}{d}+\frac{qq}{2d}+\frac{Q_2q}{d}\right]=0$$	½
$$\frac{1}{4\pi\varepsilon_0 d}\left[qQ_2+\frac{q^2}{2}+Q_2q\right]=0$$	
$$2qQ_2+\frac{q^2}{2}=0$$	½
$$2qQ_2=-\frac{q^2}{2}$$	
$$Q_1=q=-4Q_2$$	1

2. Ten capacitors, each of capacitance 1 μF, are connected in parallel to a source of 100 V. The total energy stored in the system is equal to :

(A) 10^{-2} J (B) 10^{-3} J

(C) $0{\cdot}5 \times 10^{-3}$ J (D) $5{\cdot}0 \times 10^{-2}$ J

CBSE 24

2. Ten capacitors, each of capacitance 1 μF, are connected in parallel to a source of 100 V. The total energy stored in the system is equal to :

(A) 10^{-2} J

(B) 10^{-3} J

(C) $0{\cdot}5 \times 10^{-3}$ J

(D) $5{\cdot}0 \times 10^{-2}$ J

CBSE 24

(D) 5.0×10^{-2} J

31. (a) (i) A capacitor of capacitance C is charged to a potential difference V by a battery. The battery is disconnected and the separation between the plates is halved. How will the following quantities be affected ?

(1) Capacitance of the capacitor

(2) Electric field between the plates

(3) Energy stored in the capacitor

Justify your answer in each case.

(ii) A capacitor of 20 μF is charged to 30 V. It is then connected to an uncharged 30 μF capacitor. Calculate the charges on each capacitor in equilibrium. *5*

CBSE 24

(1)Capacitance becomes two times / increases.	½
<u>Justification</u>: $C = \varepsilon_0 \frac{A}{d}$	
d is reduced to $\frac{d}{2}$.	½
Hence $C' = 2(\varepsilon_0 \frac{A}{d}) = 2C$	½
(2)Electric field remains constant.	½
<u>Justification</u>: Electric field between the plates $E = \frac{\sigma}{\varepsilon_0}$	½
When battery is removed σ is constant, hence E is constant.	
(3)Energy becomes half / decreases.	
Energy $U_E = \frac{1}{2}\frac{q^2}{C}$	½
C becomes two times, hence U_E becomes half.	
(ii) Common potential on the capacitors = V'	
Total Charge	
$q = q_1 + q_2$,	½
$q = C_1V_1 + C_2V_2 = 20 \times 30 = 600\mu C$ ....(1)	
$\frac{q_1}{q_2} = \frac{C_1V'}{C_2V'} = \frac{2}{3}$ ....(2)	½
Solving (1) and (2)	
$q_1 = 240\mu C$	½
$q_2 = 360\mu C$	½

(b) (i) Obtain an expression for electric potential at a distance r from a point charge Q.

(ii) Two point charges of 10 μC and – 5 μC are placed at (–3 cm, 0) and (6 cm, 0) in an external electric field $E = \frac{A}{r^2}$, where $A = 1{\cdot}8 \times 10^5 \text{ N C}^{-1} \text{ m}^2$. Calculate the electrostatic energy of the system. CBSE 24 *5*

(i) At some intermediate point P' the electrostatic force on unit positive test charge $= \dfrac{Q}{4\pi\varepsilon_0 r'^2}$	½
Work done in bringing a unit charge against this force is $\Delta W = -\dfrac{Q}{4\pi\varepsilon_0 r'^2}\Delta r'$	½
Total work done (W) by the external force is $W = -\int_{\infty}^{r}\dfrac{Q}{4\pi\varepsilon_0 r'^2}dr' = \dfrac{Q}{4\pi\varepsilon_0 r}$	1
This work done is the potential at point P due to charge Q. $V(r) = \dfrac{Q}{4\pi\varepsilon_0 r}$	1

(ii) $V = -\int \vec{E}.d\vec{r}$ $U_1 = q_1V_1 = \dfrac{A\times10\times10^{-6}}{3\times10^{-2}}$	½
$= \dfrac{1.8\times10^5\times10\times10^{-6}}{3\times10^{-2}}$ $= 60J$ $U_2 = q_2V_2 = \dfrac{A\times(-5)\times10^{-6}}{6\times10^{-2}}$	½
$= \dfrac{1.8\times10^5\times(-5)\times10^{-6}}{6\times10^{-2}}$ $U_2 = -15J$ $U_{12} = \dfrac{Kq_1q_2}{r_{12}} = \dfrac{9\times10^9\times10\times10^{-6}\times(-5\times10^{-6})}{9\times10^{-2}} = -5J$	½
Total Energy= $U_1+U_2+U_3$=60-15-5= 40J	½

17. *Assertion (A) :* The electrostatic field $\vec{E}$ is a conservative field.

Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

CBSE 23 C

17. *Assertion (A) :* The electrostatic field $\vec{E}$ is a conservative field.

Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

CBSE 23 C

(c) Assertion (A) is true, but Reason (R) is false.

(b) (i) How will the capacitance of a parallel plate capacitor change if :

(1) the plates area is doubled ?

(2) the separation between the plates is doubled ?

(ii) The effective capacitance of three capacitors of the same capacitance connected in series is 1 μF. Find the : 5

(1) effective capacitance if they are connected in parallel.

(2) ratio of energy stored in the parallel combination of the capacitors to that in the series combination, if the combinations are connected to the same source one by one.

CBSE 23 C

(i) (1) $C \propto A$ Capacitance gets doubled	1
(2) $C \propto \frac{1}{d}$ Capacitance reduces to half	1
(ii) $\frac{C}{3} = 1\mu F$ ______________(given) $C = 3\mu F$	
(1)Effective capacitance in parallel $C_P = 3C = 3 \times 3\mu F$ $C_P = 9\mu F$	1½
(2) $U = \frac{1}{2}CV^2$ $\frac{U_P}{P_S} = \frac{C_p}{C_S} = \frac{9}{3} = 3:1$	1½

24. (a) A uniform electric field E of 500 N/C is directed along +x axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. *2*

CBSE 23 C

24. (a) A uniform electric field E of 500 N/C is directed along +x axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. *2*

CBSE 23 C

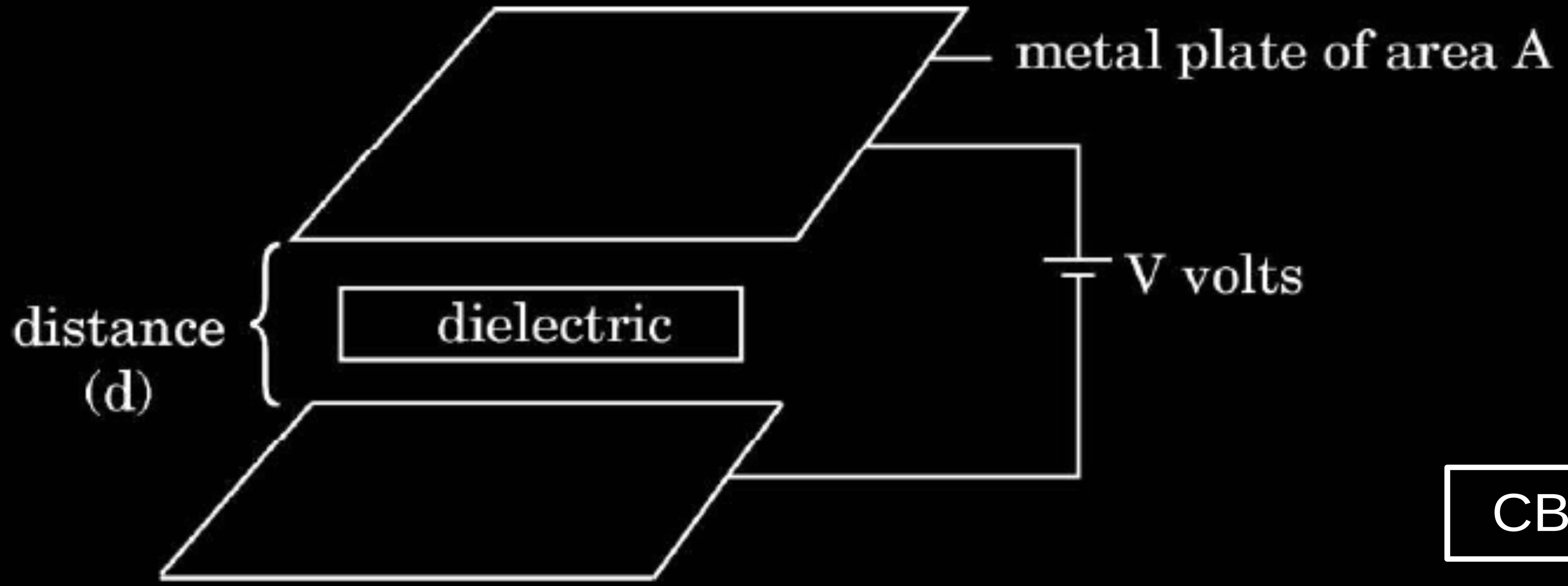
(a) (i) $V_{OA} = E\,(x_2-x_1)$	½
$V_{OA} = 500 \times 0 = 0$ volt	½
(ii) $V_{OB} = -E\,(x_2-x_1)$	½
$V_{OB} = -500 \times (4 \times 10^{-2})$	
$= -20$ V	½

(b) Three point charges $1\ \mu C$, $-1\ \mu C$ and $2\ \mu C$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. CBSE 23 C 2

(b) Three point charges 1 μC, –1 μC and 2 μC are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. CBSE 23 C 2

Initial electrostatic potential energy of the system $U_i = \frac{k}{r}[1\times(-1)+(-1)\times 2+(1)\times(2)]\times 10^{-12}$ $= \frac{9\times 10^9}{1}[-1-2+2]\times 10^{-12}$ $= -9\times 10^{-3}\,J$	1
Now $A_1B_1 = B_1C_1 = A_1C_1 = \frac{1}{2}$ m Final electrostatic potential energy of the system $U_f = \frac{-9\times 10^{-9}}{1/2} = -18\times 10^{-3}$ J	½
Amount of work done W= U_f - U_i W= $-18\times 10^{-3} + 9\times 10^{-3} = -9\times 10^{-3}\,J$	½

34.



CBSE 23 C

A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

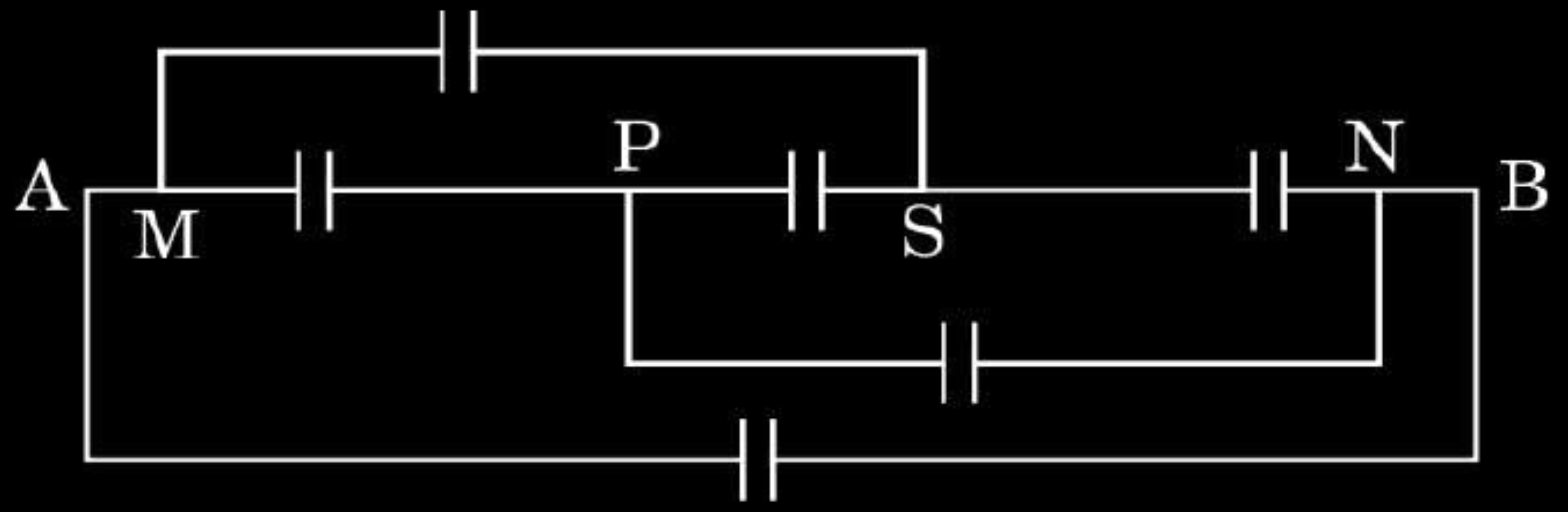
(a) The capacitance of a parallel plate capacitor increases from 10 μF to 80 μF on introducing a dielectric medium between the plates. Find the dielectric constant of the medium. *1*

(b) n capacitors, each of capacitance C, are connected in series. Find the equivalent capacitance of the combination. *1*

(c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer. *2*

OR

(c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C. *2*



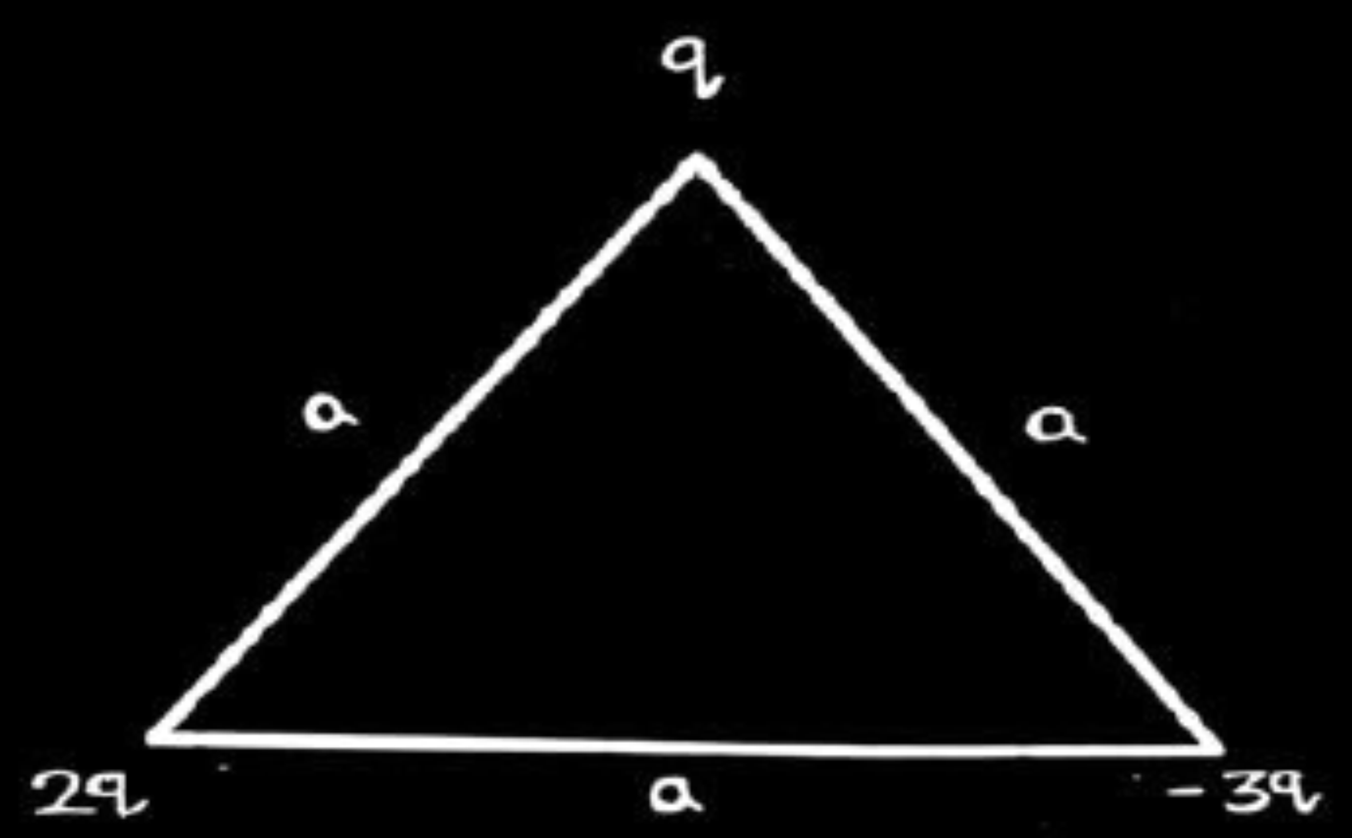
CBSE 23

(a) $K = \dfrac{C}{C_0}$	½
$K = \dfrac{80\mu F}{10\mu F} = 8$	½
(b) $\dfrac{1}{C_S} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots\ldots\ldots + \dfrac{1}{C_n}$	½
$\dfrac{1}{C_S} = \dfrac{n}{C}$	
$C_S = \dfrac{C}{n}$	½
(c) Charge is constant	
$Q_1 = Q_2$	
$C_2 = KC_1$	½
$C_1 V_1 = K\, C_1 V_2$	
$V_2 = \dfrac{V_1}{K}$ Potential diff decreases by a factor (1/K)	½

$U_2 = \dfrac{1}{2}\dfrac{Q^2}{C_2}$	½
$= \dfrac{1}{2}\dfrac{Q^2}{kC_1} = \dfrac{1}{k}\left(\dfrac{Q^2}{2C_2}\right)$	½
$U_2 = \dfrac{U_1}{K}$	½
Energy reduces by a factor of 1/K.	
OR	
For calculating effective capacitance = 2 C.	2

24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q and –3q placed at the vertices of an equilateral triangle of side a. CBSE 23 *2*

24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q and –3q placed at the vertices of an equilateral triangle of side a. CBSE 23 *2*

 **OR a similar diagram with different order of charges**	
$U = \frac{1}{4\pi \in_0} . \frac{q_1 q_2}{r}$	½
$U = \frac{1}{4\pi \in_0}\left[\frac{2q^2}{a} - \frac{6q^2}{a} - \frac{3q^2}{a}\right]$	1
$U = \frac{1}{4\pi \in_0}\frac{\left(-7q^2\right)}{a}$	½

(b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium. *2*

CBSE 23

(b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium. *2*

CBSE 23

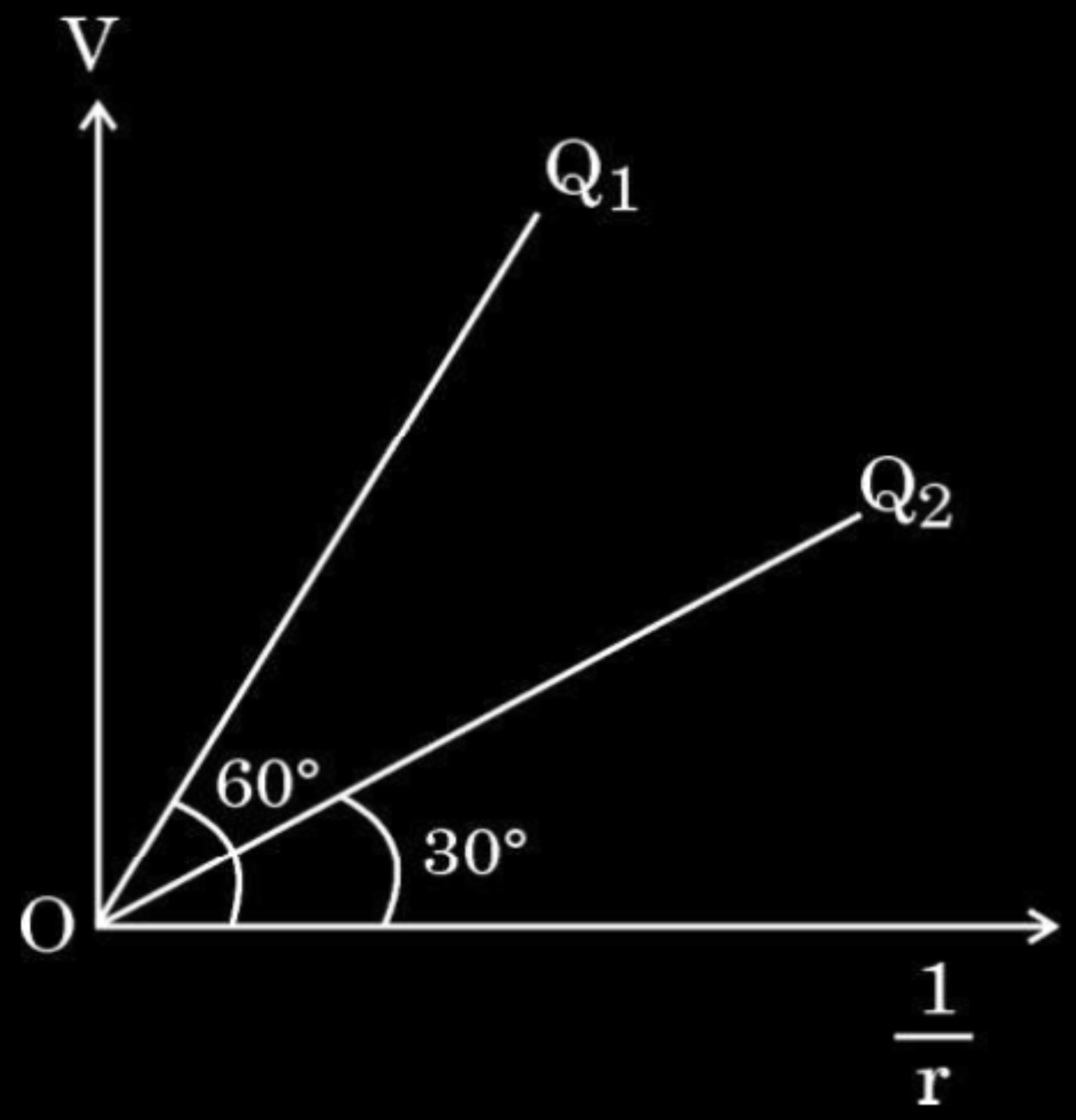
According to law of conservation of charge	
$q_i = q_f$	½
$q_1 + q_2 = q_1' + q_2' = Q$	
When two balls are connected with wire	
$V_1 = V_2$	½
$\frac{kq_1'}{r_1} = \frac{kq_2'}{r_2}$ or $\frac{q_1'}{r_1} = \frac{q_2'}{r_2}$	
$q_1' r_2 = q_2' r_1$	
$q_1' r_2 = (Q - q_1') r_1$	
$q_1' r_2 = Q r_1 - q_1' r_1$	
$q_1'(r_1 + r_2) = Q r_1$	
$q_1' = \frac{Q r_1}{r_1 + r_2} = \frac{(q_1 + q_2) r_1}{r_1 + r_2}$	½
$q_2' = Q - q_1'$	
$= Q - \frac{Q r_1}{r_1 + r_2}$	
$= \frac{Q r_2}{r_1 + r_2} = \frac{(q_1 + q_2) r_2}{r_1 + r_2}$	½

34. Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.

CBSE 23

Answer the following questions based on the above :

(a) Consider a uniformly charged thin conducting shell of radius R. Plot a graph showing the variation of $|\vec{E}|$ with distance r from the centre, for points $0 \leq r \leq 3R$. *1*

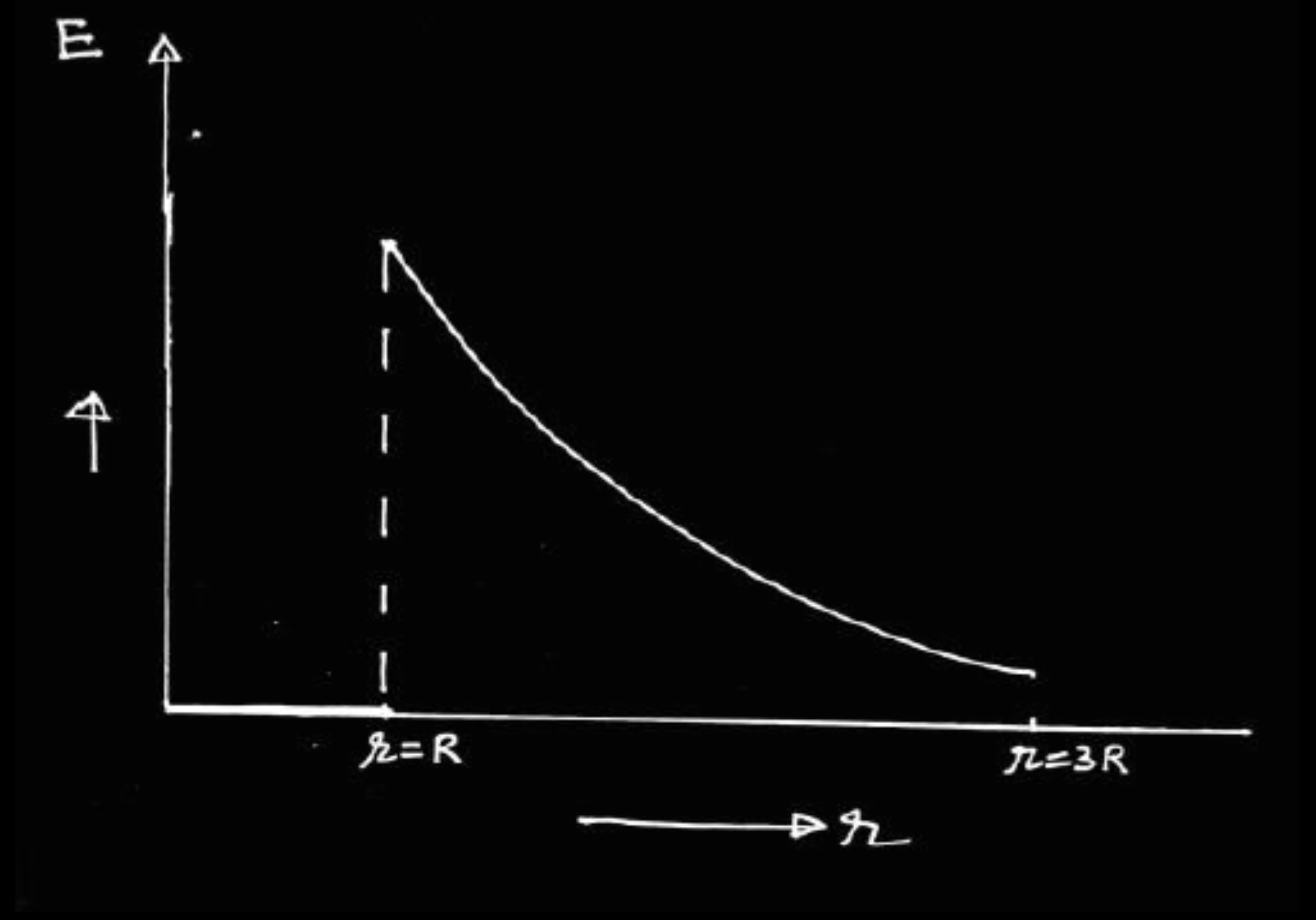


(b) The figure shows the variation of potential V with $\frac{1}{r}$ for two point charges Q_1 and Q_2, where V is the potential at a distance r due to a point charge. Find $\frac{Q_1}{Q_2}$. *1*

(c) An electric dipole of dipole moment of 6×10^{-7} C-m is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole. *2*

OR

(c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$ such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$. CBSE 23 *2*

(a) 	1
(b) $\because$ V= k Q/r	½
Slope of graph is proportional to Q	
$\dfrac{Q_1}{Q_2} = \dfrac{\tan 60^0}{\tan 30^0} = 3$	½
	½
(c) U= - p E cosθ	½
$\theta = 0^0$	
$U = -(6 \times 10^{-7}) \times (10^4)$	1
$U = -6 \times 10^{-3}$ J	
OR	
$\because$ Work done $W = -pE(\cos\theta_2 - \cos\theta_1)$	½
where $\theta_2 = 180^\circ$, $\theta_1 = 90^\circ$	½
$\Rightarrow W = -pE(\cos 180^\circ - \cos 90^\circ)$	½
$W = +pE$	½

(b) (i) Define electric potential at a point and write its SI unit.

(ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.

(iii) Two point charges + q and – q are located at points (3a, 0) and (0, 4a) respectively in x-y plane. A third charge Q is kept at the origin. Find the value of Q, in terms of q and a, so that the electrostatic potential energy of the system is zero. *5*

CBSE 23

(i) Electrical Potential – Electrostatic potential at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point. ½

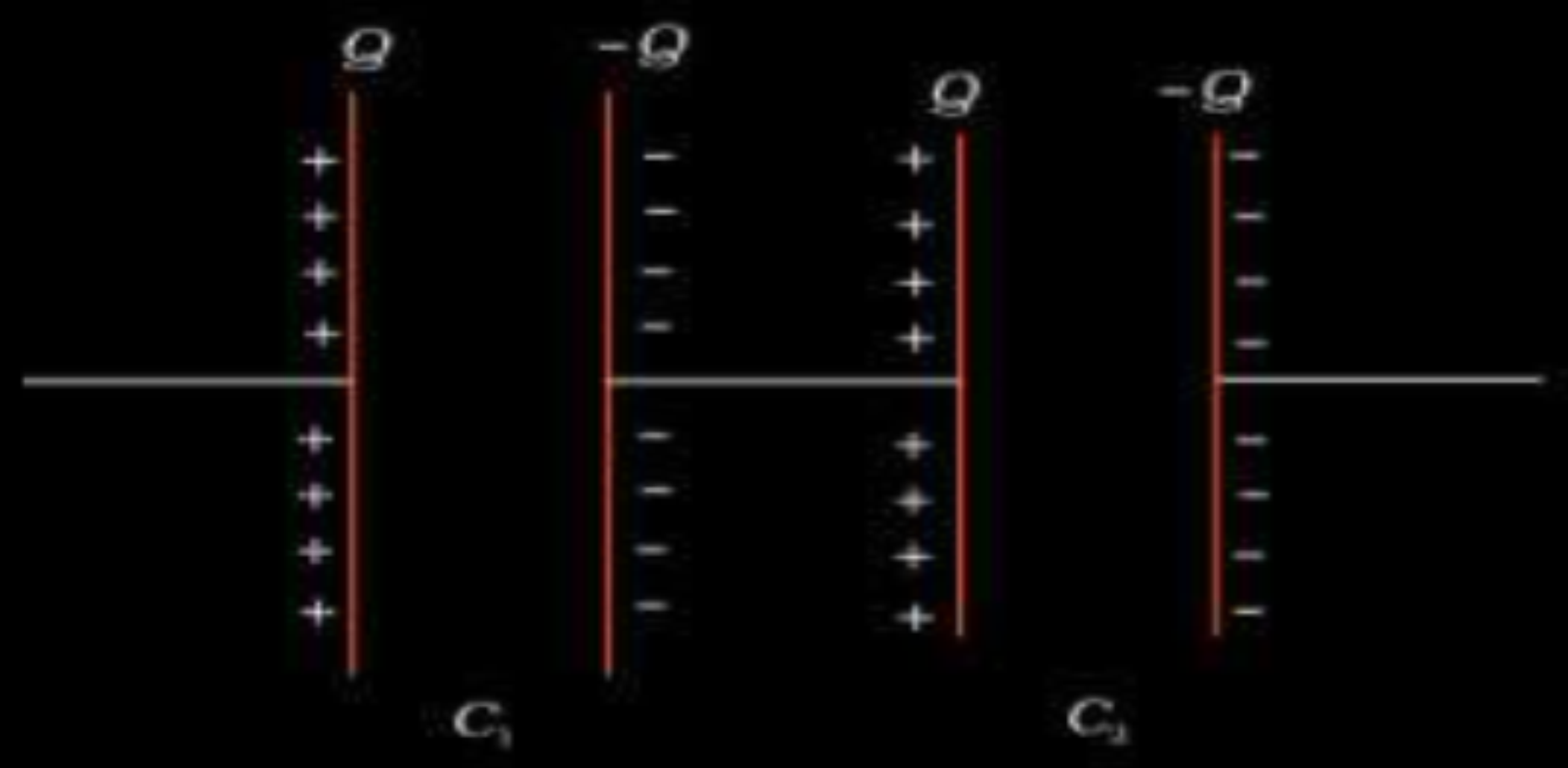
Alternatively:-

$V = \frac{Work\ Done}{q}$

$V = -\int \vec{E} \cdot \vec{dl}$

S.I. unit of electrostatic potential is volt. ½

Alternatively:-
S.I. unit is J/C.



½

$V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2}$ ½

$\frac{Q}{Ceq.} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$ ½

$\frac{1}{Ceq.} = \frac{1}{C_1} + \frac{1}{C_2}$ ½

(iii)

Potential energy of the system $= K\left[\frac{Q(-q)}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a}\right]$ ½

Potential energy of the system = 0 ½

$\Rightarrow K\left[\frac{-Qq}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a}\right] = 0$

$\Rightarrow \frac{-Q}{4} + \frac{Q}{3} - \frac{q}{5} = 0$

$\Rightarrow +\frac{Q}{12} - \frac{q}{5} = 0$ 1

$\Rightarrow Q = +\frac{12q}{5}$

34. A beam of electrons moving horizontally with a velocity of 3×10^7 m/s enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.

CBSE 23

Answer the following questions based on the above :

(a) How long does an electron take to strike the edge ? *1*

(b) What is the shape of the path followed by the electron and why ? *1*

(c) Find the potential difference applied. *2*

OR

(c) Find the magnitude and direction of the magnetic field which should be created in the space between the plates so that the electron beam goes straight undeviated. *2*

(a) Electron strikes the edge after travelling 3 cm horizontally (along x-axis). ½

$S_x = v_x \times t$

$3 \times 10^{-2} = (3 \times 10^7) \times t$ ½

$t = 10^{-9}$ s

(b) Shape of the path is parabola. ½

Reason: Force/acceleration is in a fixed direction perpendicular to the initial velocity. ½

(c) Along y-direction

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$$ ½

$a_y = -10^{16}$ m/s^2

Magnitude of acceleration $(a_y) = \frac{eE}{m} = \frac{e}{m}\left(\frac{V}{l}\right)$ ½

$$V = \frac{10^{16} \times 9.1 \times 10^{-31} \times 1 \times 10^{-2}}{1.6 \times 10^{-19}}$$ ½

½

$V = 568.75$ V

(c) $|qE| = |qvB|$; $B = \frac{E}{v} = (\frac{ma_y}{e})(\frac{1}{v})$ ½

Along y-direction

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$$ ½

$a_y = -10^{16}$ m/s^2

$$B = \left(\frac{9.1 \times 10^{-31} \times 10^{16}}{1.6 \times 10^{-19}}\right) \times \left(\frac{1}{3 \times 10^7}\right)$$ ½

$B = 1.9 \times 10^{-3}\, T$ ½

Direction of magnetic field will be out of the plane of the paper.

28. Three point charges Q_1 ($-15\ \mu C$), Q_2 ($10\ \mu C$) and Q_3 ($16\ \mu C$) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges. CBSE 23 *3*

28. Three point charges Q_1 (–15 μC), Q_2 (10 μC) and Q_3 (16 μC) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges. CBSE 23 *3*

$U_{Q1Q2} = \frac{1}{4\pi \in_o}\frac{Q_1Q_2}{r_{12}}$	½
$= \frac{9\times10^9\times(-15\times10^{-6})\times(10\times10^{-6})}{3\times10^{-2}}$	
$= -45$ J	½
$U_{Q2Q3} = \frac{1}{4\pi \in_o}\frac{Q_2Q_3}{r_{23}}$	
$= \frac{9\times10^9\times(10\times10^{-6})\times(16\times10^{-6})}{4\times10^{-2}}$	
$= 36$ J	½
$U_{Q1Q3} = \frac{1}{4\pi \in_o}\frac{Q_1Q_3}{r_{13}}$	
$= \frac{9\times10^9\times(-15\times10^{-6})\times(16\times10^{-6})}{5\times10^{-2}}$	
$= -43.2$ J	½
$U_{net} = U_{Q1Q2} + U_{Q2Q3} + U_{Q1Q3}$	½
$= -45+36-43.2$	
$= -52.2$ J	½
Alternatively	
$U = k(\frac{Q_1Q_2}{r_{12}} + \frac{Q_2Q_3}{r_{23}} + \frac{Q_1Q_3}{r_{13}})$	1
$= \frac{9\times10^9\times(-15\times10^{-6})\times(10\times10^{-6})}{3\times10^{-2}} + \frac{9\times10^9\times(10\times10^{-6})\times(16\times10^{-6})}{4\times10^{-2}} + \frac{9\times10^9\times(-15\times10^{-6})\times(16\times10^{-6})}{5\times10^{-2}}$	1
$=(-45 +36-43.2)$ J	
$= 52.2$ J	1

30. A 100 μF capacitor is charged by a 12 V battery.

(a) How much electrostatic energy is stored by the capacitor ?

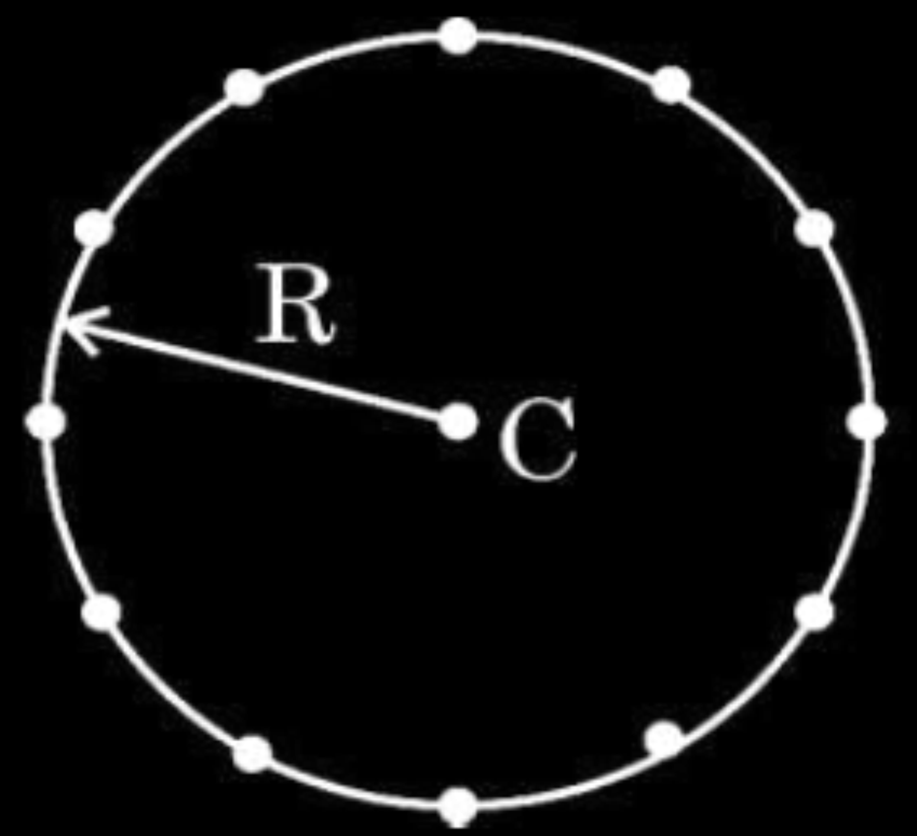
(b) The capacitor is disconnected from the battery and connected in parallel to another uncharged 100 μF capacitor. What is the electrostatic energy stored by the system ? CBSE 23 *3*

C= 100 $\mu F = 100\times10^{-6} F, V = 12V$	
a) U = $\frac{1}{2}CV^2$	½
$= \frac{1}{2}\times100\times10^{-6}\times(12)^2$	
$=\frac{1}{2}\times10^{-4}\times144$	½
$= 72\times10^{-4}$ J = 7.2 m J	
b) $C_{eq} = C_1 + C_2$	½
$= 200\ \mu F$	
$Q = CV$	
$= 100\times10^{-6}\times12$	½
$= 12\times10^{-4} C$	
$U = \frac{Q^2}{2C_{eq}} = \frac{(12\times10^{-4})^2}{2\times200\times10^{-6}}$	½
$= \frac{144}{4}\times10^{-4}$	
$= 36\times10^{-4}$J = 3.6 m J	½

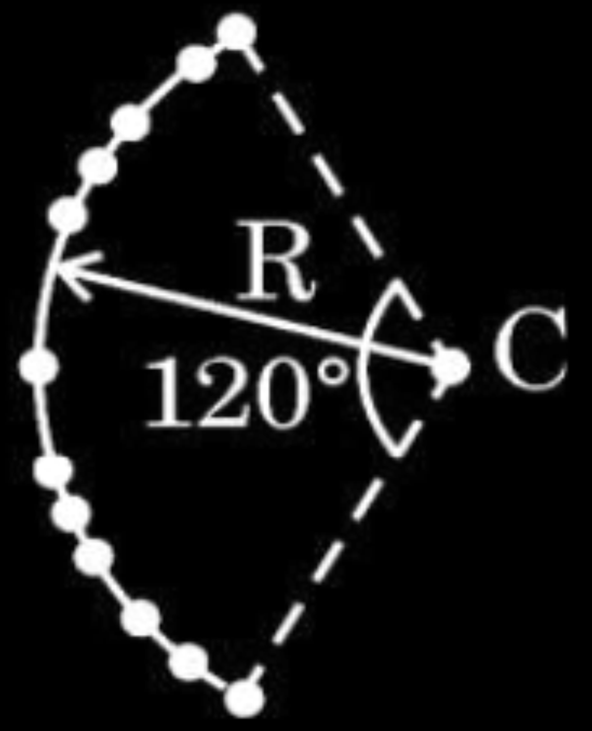
26. (a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i). Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle.

(b) If the charges are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii), find electric potential at the centre C. *3*

CBSE 23



(i)



(ii)

(a) Electric potential due to point charge	
$V = \frac{kq}{R}$	½
Value of each charge = - q , Total charge = -12q	
Total potential V = $\frac{k(-12q)}{R}$	
$V = \frac{-12kq}{R} = \frac{-12q}{4\pi \in_0 R}$	½
By symmetry the resultant of all electric field vectors becomes zero. So electric field is zero.	1
(b) Electric potential is a scalar quantity and does not depend on placement of charges	
Therefore $V = \frac{-12kq}{R} = \frac{-12q}{4\pi \in_0 R}$	1

18. **Assertion (A) :** Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R) : Electrostatic force is a conservative force. CBSE 23 **1**

18. **Assertion (A) :** Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R) : Electrostatic force is a conservative force. CBSE 23 **1**

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

28. (a) Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces. CBSE 23 **3**

28. (a) Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces. CBSE 23 **3**

When connected by a conducting wire both spheres will be at the same potential.	½
$\therefore \quad k\dfrac{q_1}{a} = k\dfrac{q_2}{b}$	½
$\therefore \dfrac{q_1}{q_2} = \dfrac{a}{b}$	½
$\dfrac{E_1}{E_2} = \dfrac{k\dfrac{q_1}{a^2}}{k\dfrac{q_2}{b^2}}$	1
$\dfrac{E_1}{E_2} = \dfrac{b}{a}$	½

(b) A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V. The battery is disconnected and an uncharged capacitor (B) of capacitance 2C is connected across A. Find the ratio of

(i) final charges on A and B.

(ii) total electrostatic energy stored in A and B finally and that stored in A initially. CBSE 23

(b) A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V. The battery is disconnected and an uncharged capacitor (B) of capacitance 2C is connected across A. Find the ratio of

(i) final charges on A and B.

(ii) total electrostatic energy stored in A and B finally and that stored in A initially.

CBSE 23

i)	Initially $Q = CV$	
	Finally $q_A = C_A V_1$ & $q_B = C_B V_1$	½
	$\dfrac{q_A}{q_B} = \dfrac{C_A}{C_B} = \dfrac{1}{2}$	½
ii)	$q_A + q_b = Q$	
	$\therefore \; q_A = \dfrac{Q}{3}$ & $q_B = \dfrac{2Q}{3}$	½
	$\dfrac{U_f}{U_i} = \dfrac{U_A + U_B}{U_{Ai}}$	
	$= \dfrac{\dfrac{q_A^2}{2C_A} + \dfrac{q_B^2}{2C_B}}{\dfrac{Q^2}{2C_A}}$	1
	$= \dfrac{1}{3}$	½

Alternatively , Common potential	
$V_1 = \dfrac{Q_1 + Q_2}{C_1 + C_2}$	
$= \dfrac{Q}{3C} = \dfrac{V}{3}$ $\quad \left[\because \dfrac{Q}{C} = V\right]$	1
$\dfrac{U_f}{U_i} = \dfrac{\frac{1}{2} C_{eq} V_1^2}{\frac{1}{2} C_A V^2}$	½
$= \dfrac{\frac{1}{2} 3C \times \left(\frac{V}{3}\right)^2}{\frac{1}{2} C V^2} = \dfrac{1}{3}$	½

(b) (i) Consider two identical point charges located at points (0, 0) and (a, 0).

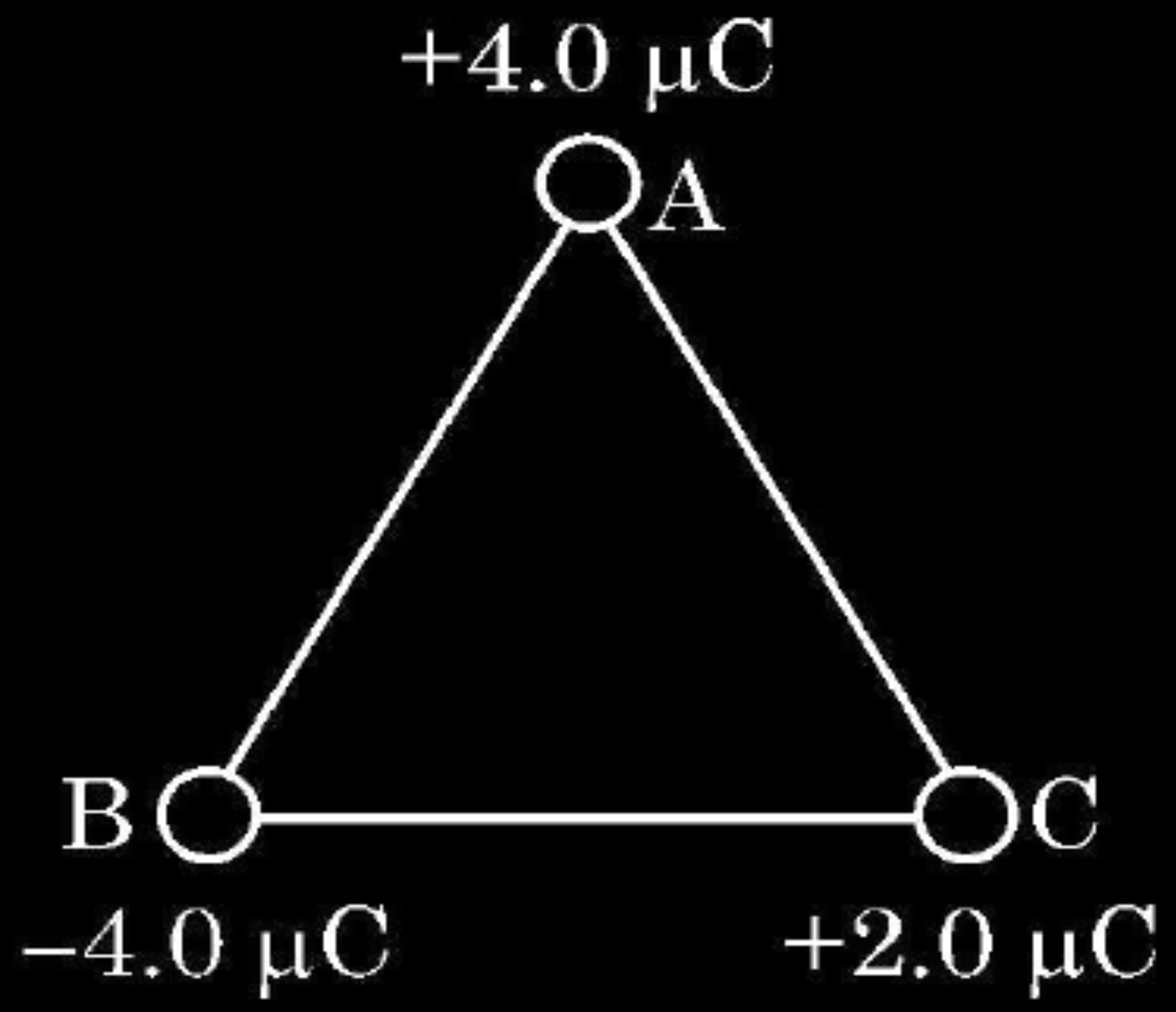
(1) Is there a point on the line joining them at which the electric field is zero ?

(2) Is there a point on the line joining them at which the electric potential is zero ?

Justify your answers for each case.

(ii) State the significance of negative value of electrostatic potential energy of a system of charges.

Three charges are placed at the corners of an equilateral triangle ABC of side 2.0 m as shown in figure. Calculate the electric potential energy of the system of three charges.



CBSE 23

(i) (1) Yes , electric field is zero at mid point.

Electric field being a vector quantity , its resultant is zero. ½

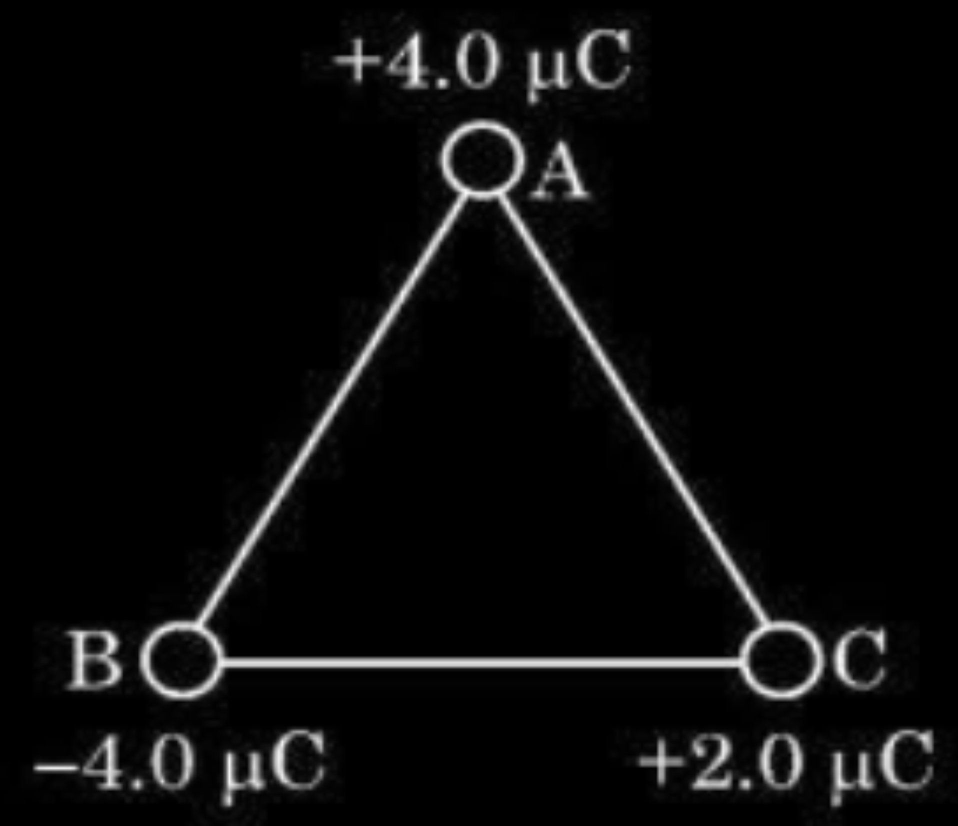
(2) No, potential cannot be zero on line joining the charges. ½

Electric potential being a scalar quantity, the net potential due to two identical charges cannot be zero. ½

(ii) Negative value of electrostatic potential energy of a system signifies that the system has attractive forces. ½

Alternatively 1

Give full credit , if a candidate writes the system is stable /bound.



$$U = \frac{1}{4\pi \in_0} \times \frac{q_1 q_2}{r}$$

½

$$U = \frac{1}{4\pi \in_0}\left[\frac{q_A q_B}{r} + \frac{q_B q_c}{r} + \frac{q_c q_A}{r}\right]$$

½

$$= \frac{9\times10^9}{2}[-16-8+8]\times10^{-12}$$

½

$$= -7.2\times10^{-2}\,J$$

½

2. A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to **1**

(a) $\frac{1}{x^2}$ (b) $\frac{1}{x^3}$

(c) $\frac{1}{x^4}$ (d) $\frac{1}{x^{1/2}}$

CBSE 23

2. A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to **1**

(a) $\frac{1}{x^2}$ (b) $\frac{1}{x^3}$

(c) $\frac{1}{x^4}$ (d) $\frac{1}{x^{1/2}}$

CBSE 23

(a) $\frac{1}{x^2}$

12. The capacitors, each of 4 μF are to be connected in such a way that the effective capacitance of the combination is 6 μF. This can be achieved by connecting **1**

(A) All three in parallel

(B) All three in series

(C) Two of them connected in series and the combination in parallel to the third.

(D) Two of them connected in parallel and the combination in series to the third.

CBSE 23

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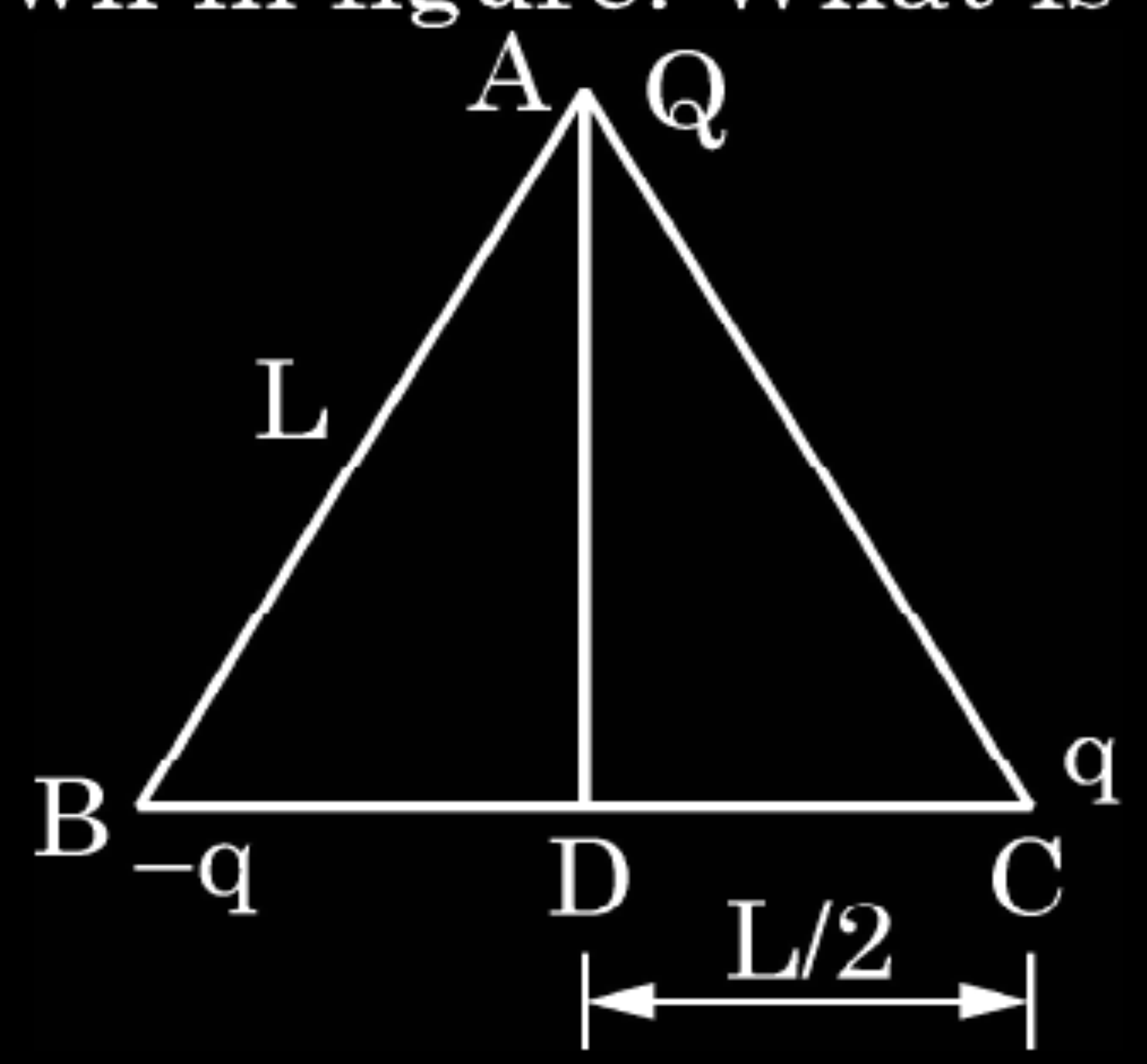
CBSE 23

(C) Two of them connected in series and the combination in parallel to the third

22. Three point charges Q, q and – q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is **2**

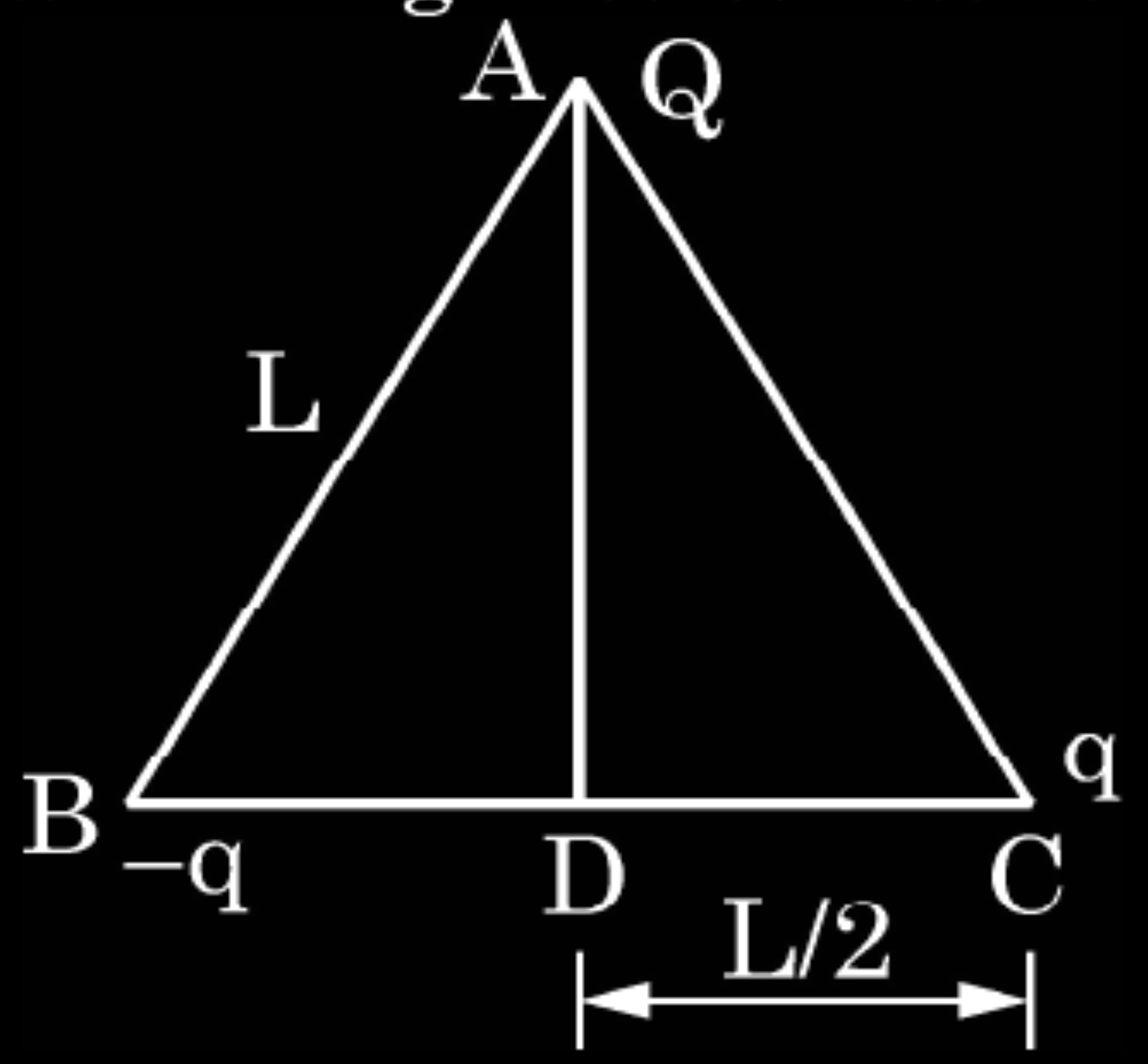


CBSE 23

(i) the electrostatic potential energy of the arrangement ? and

(ii) the potential at point D ?

22. Three point charges Q, q and – q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is **2**



CBSE 23

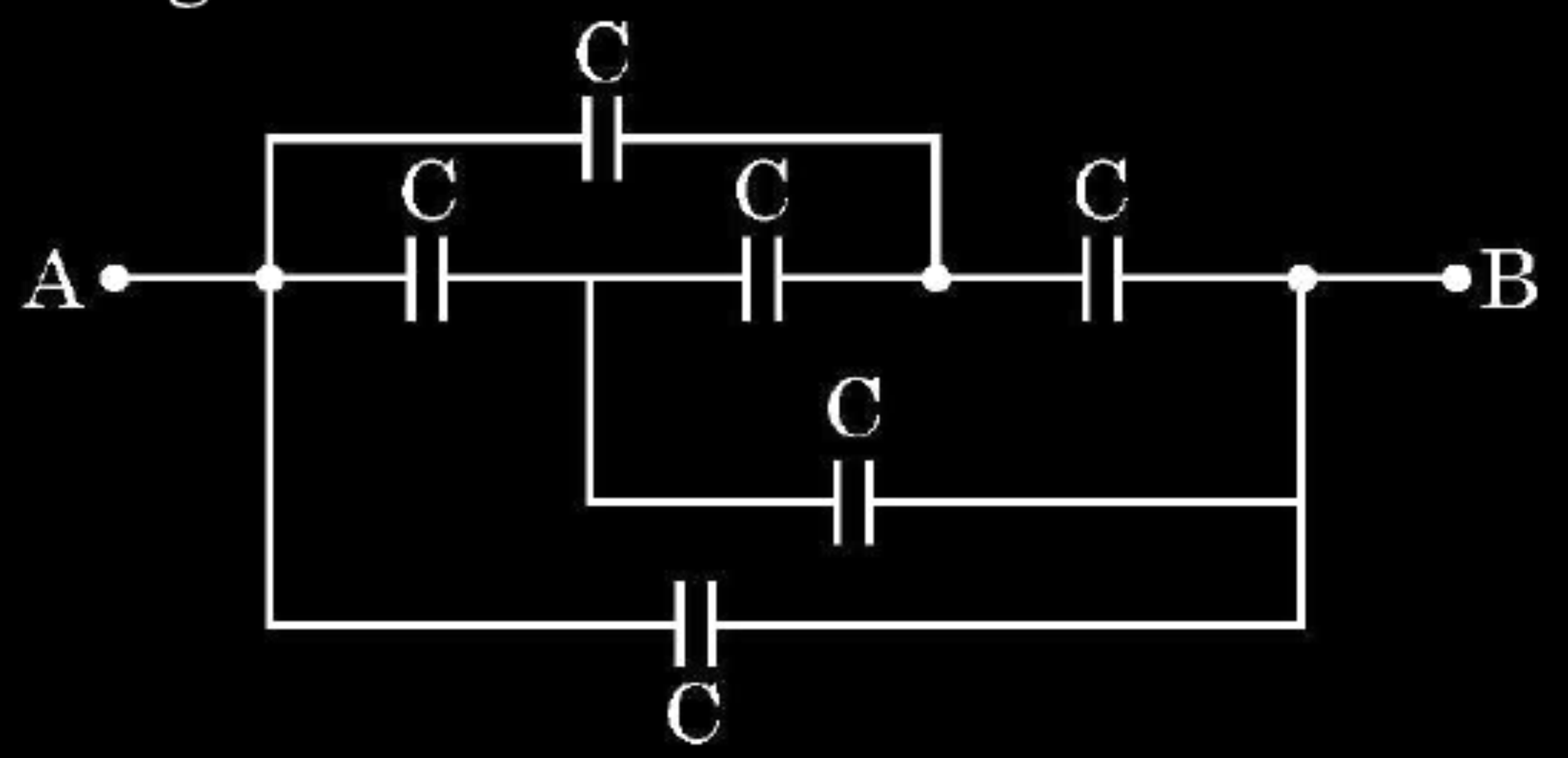
(i) the electrostatic potential energy of the arrangement ? and

(ii) the potential at point D ?

(i) $U = -\frac{kQq}{L} + \frac{kQq}{L} - \frac{kq^2}{L}$	½
$U = -\frac{kq^2}{L}$	½
(ii) $V = V_{DA} + V_{DB} + V_{DC}$	
$V = \frac{2KQ}{L\sqrt{3}} - \frac{2Kq}{L} + \frac{2Kq}{L}$	½
$V = \frac{2KQ}{L\sqrt{3}}$	½

35. A capacitor is a system of two conductors separated by an insulator. The two conductors have equal and opposite charges with a potential difference between them. The capacitance of a capacitor depends on the geometrical configuration (shape, size and separation) of the system and also on the nature of the insulator separating the two conductors. They are used to store charges. Like resistors, capacitors can be arranged in series or parallel or a combination of both to obtain desired value of capacitance. **4**

(i) Find the equivalent capacitance between points A and B in the given diagram.

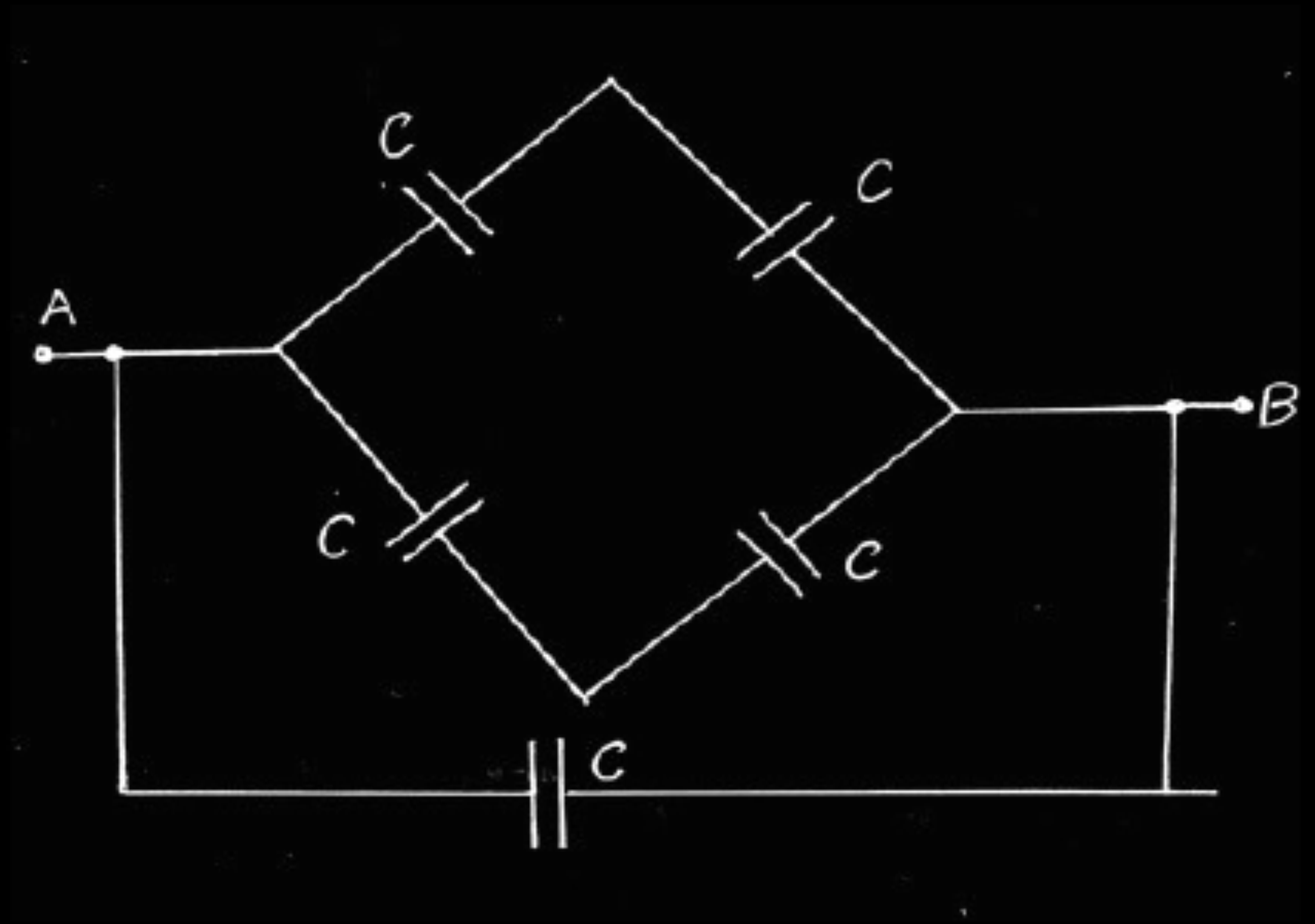


CBSE 23

(ii) A dielectric slab is inserted between the plates of a parallel plate capacitor. The electric field between the plates decreases. Explain.

(iii) A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.

(i)



½

$$C_{net} = C + C$$

$$= 2C$$ ½

(**Note:** Give full credit to a student if 2C written directly)

(ii) Within the dielectric slab, induced electric field due to polarization, decreases the electric field. 1

Alternatively: $\mathbf{E} = \mathbf{E_0} - \mathbf{E}_p$

Alternatively: $\mathbf{E} = \dfrac{\mathbf{E_0}}{K}$

(iii)

(a) $$V' = \frac{Q_{Total}}{C_{eqi}} = \frac{Q}{3C}$$ ½

$$V' = \frac{Q}{3C} = \frac{V}{3}$$ ½

(b)

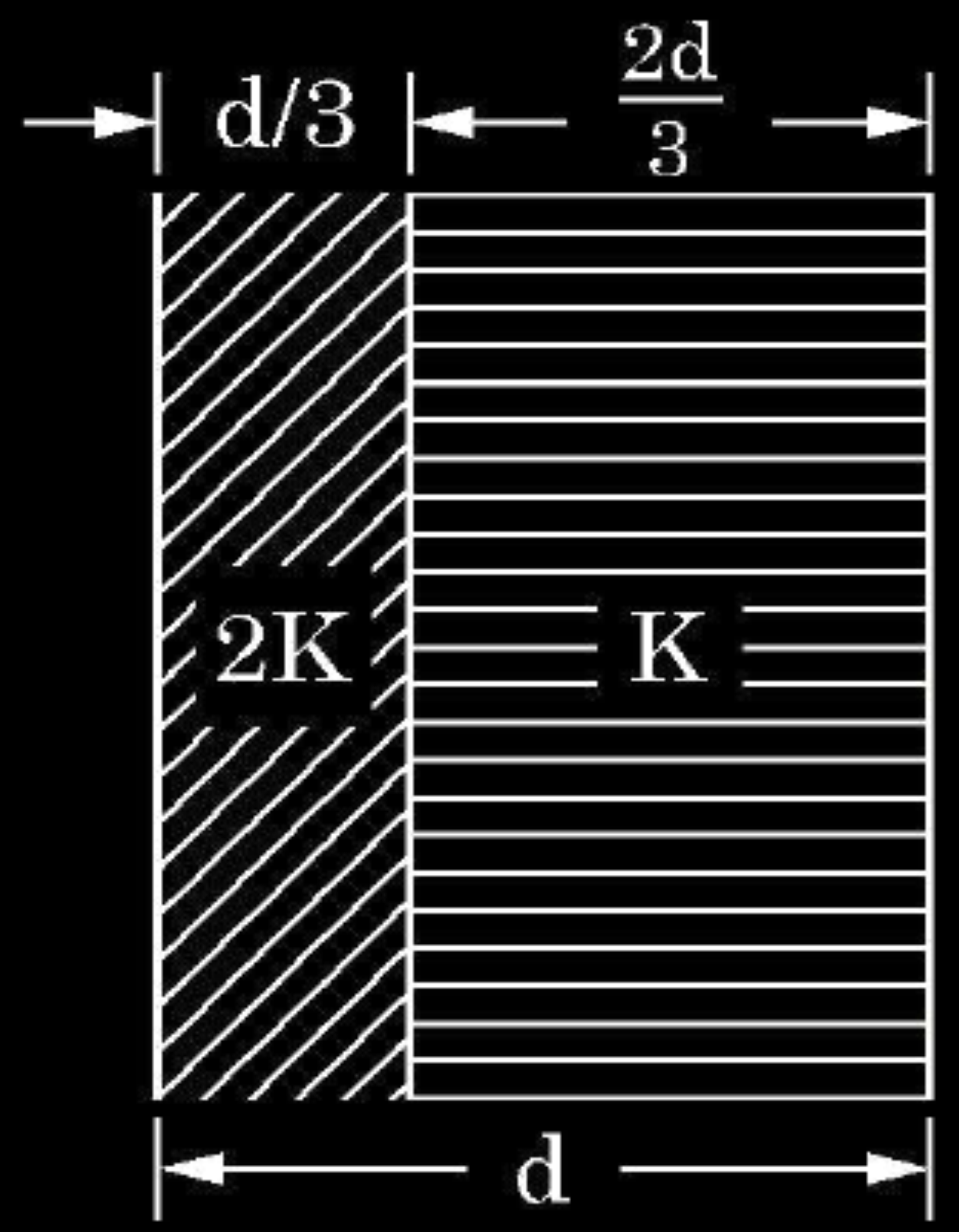
$$Q_A' = C \times \frac{V}{3} = \frac{Q}{3}$$ ½

$$Q_A = CV = Q$$

Charge lost by capacitor A is

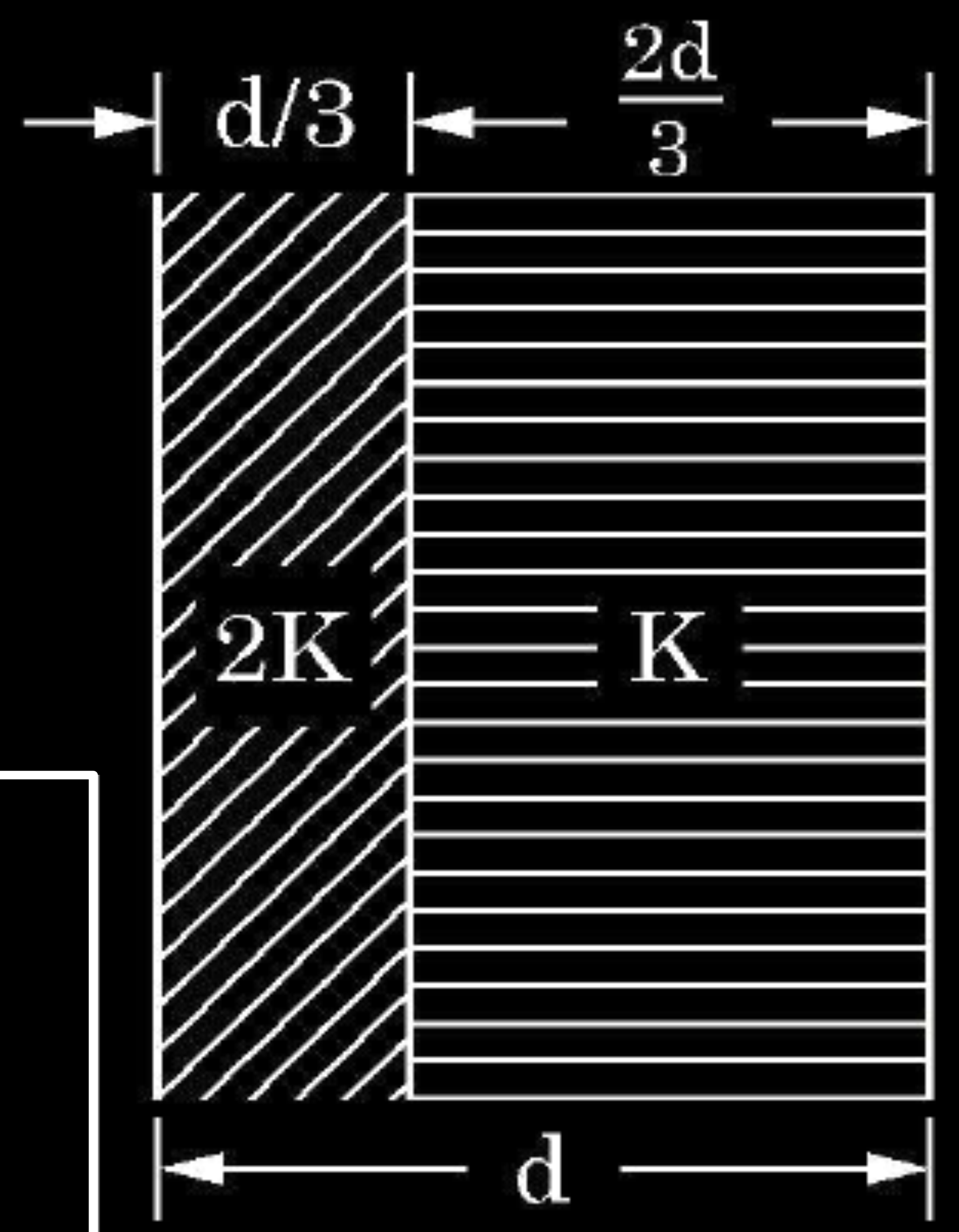
$$\Delta Q = Q - \frac{Q}{3} = \frac{2Q}{3}$$ ½

(iii) Two slabs of dielectric constants 2K and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.



CBSE 23

(iii) Two slabs of dielectric constants 2K and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.



CBSE 23

Capacitance of left portion, $C_1 = \dfrac{6K \in_0 A}{d}$	½
Capacitance of right portion, $C_2 = \dfrac{3K \in_0 A}{2d}$	½
As the capacitors are in series $\dfrac{1}{C_{eqi}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$	½
$\dfrac{1}{C_{eqi}} = \dfrac{d}{6K \in_O A} + \dfrac{2d}{3K \in_O A} = \dfrac{5d}{6KA \in_O}$ $C_{eqi} = \dfrac{6KA \in_O}{5d}$	½

13. A point charge is placed at the centre of a hollow conducting sphere of internal radius 'r' and outer radius '2r'. The ratio of the surface charge density of the inner surface to that of the outer surface will be ________. **1**

CBSE 20

13. A point charge is placed at the centre of a hollow conducting sphere of internal radius ‘r’ and outer radius ‘2r’. The ratio of the surface charge density of the inner surface to that of the outer surface will be __4:1__. **1**

CBSE 20

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor. CBSE 20 **2**

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor. CBSE 20 **2**

Let the charge on the capacitor plates at any instant, during charging process be q, amount of work done to supply further charge dq to the capacitor

$dW = Vdq$ ½

where V is the potential difference and equals to $\frac{q}{C}$

Total work done to charge *the capacitor upto* charge Q

$$W = \int_0^Q Vdq$$

$$= \int_0^Q \frac{q}{C}dq = \frac{Q^2}{2C}\ \left(\frac{1}{2}CV^2 = \frac{1}{2}QV\right)$$ ½

Since *Energy stored = work done*

$$\Rightarrow U = \frac{Q^2}{2C}\ \left(\frac{1}{2}CV^2 = \frac{1}{2}QV\right)$$ ½

Energy density: Electrical energy stored per unit volume is known as energy density. ½

Alternatively:

Energy density $= \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\frac{\sigma^2}{\varepsilon_0}$

33. (a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.

(b) If an external electric field (E) is applied on the system, write the expression for the total energy of this system. **3**

CBSE 20

33. (a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.

(b) If an external electric field (E) is applied on the system, write the expression for the total energy of this system. **3**

CBSE 20

(a) Work done in bringing the charge q_2, from infinity, to a point $= q_2 \times$ potential at the point due to charge q_1 $= q_2 \times \frac{1}{4\pi\varepsilon_0} \frac{q_1}{r_{12}}$	½
$\therefore$ potential energy of the system $= \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r_{12}}$	1
(b)Let the potentials, at two points, due to an external electric field (E) be V_1 and V_2 respectively. Now the total energy of the system is: $\left[q_1 V_1 + q_2 V_2 + \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r_{12}}\right]$	1½

37. (a) Write two important characteristics of equipotential surfaces.

(b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r(in m) as

$$E = 10\,r + 5$$

CBSE 20

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

37. (a) Write two important characteristics of equipotential surfaces.

(b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r(in m) as

$$E = 10\,r + 5$$

CBSE 20

By how much does the electric potential increase in moving from point at r = 1 m to a point at r = 10 m.

(a) For equipotential surfaces (i) Potential has the same value at all points on the surface. (ii) Electric field is normal to the equipotential surface at all points (iii) Work done in moving any charge between any two points on the equipotential surface is zero (any two)	1+1

(b) $E = 10r+5$ $dV = -E\cdot dr$ $\int dV = -\int_1^{10} \vec{E}\, d\vec{r}$	½
$= -\int_1^{10} (10r+5)dr$ $V = -\left[\int_1^{10} 10r\,dr + \int_1^{10} 5dr\right]$	½
$V = 10\left[\frac{r^2}{2}\right]_1^{10} + 5(r)_1^{10}$ $V = -5[100-1]+5[10-1]$ $V = -5\times 99 + 5\times 9 = -540V$	½

12. A proton released from rest in an electric field, will start moving towards a region of _______ potential in the field. **1**

CBSE 20

12. A proton released from rest in an electric field, will start moving towards a region of _______ potential in the field. **1**

CBSE 20

Decreasing/Lower

16. The work done in moving a charge particle between two points in an uniform electric field, does not depend on the path followed by the particle. Why ? 1

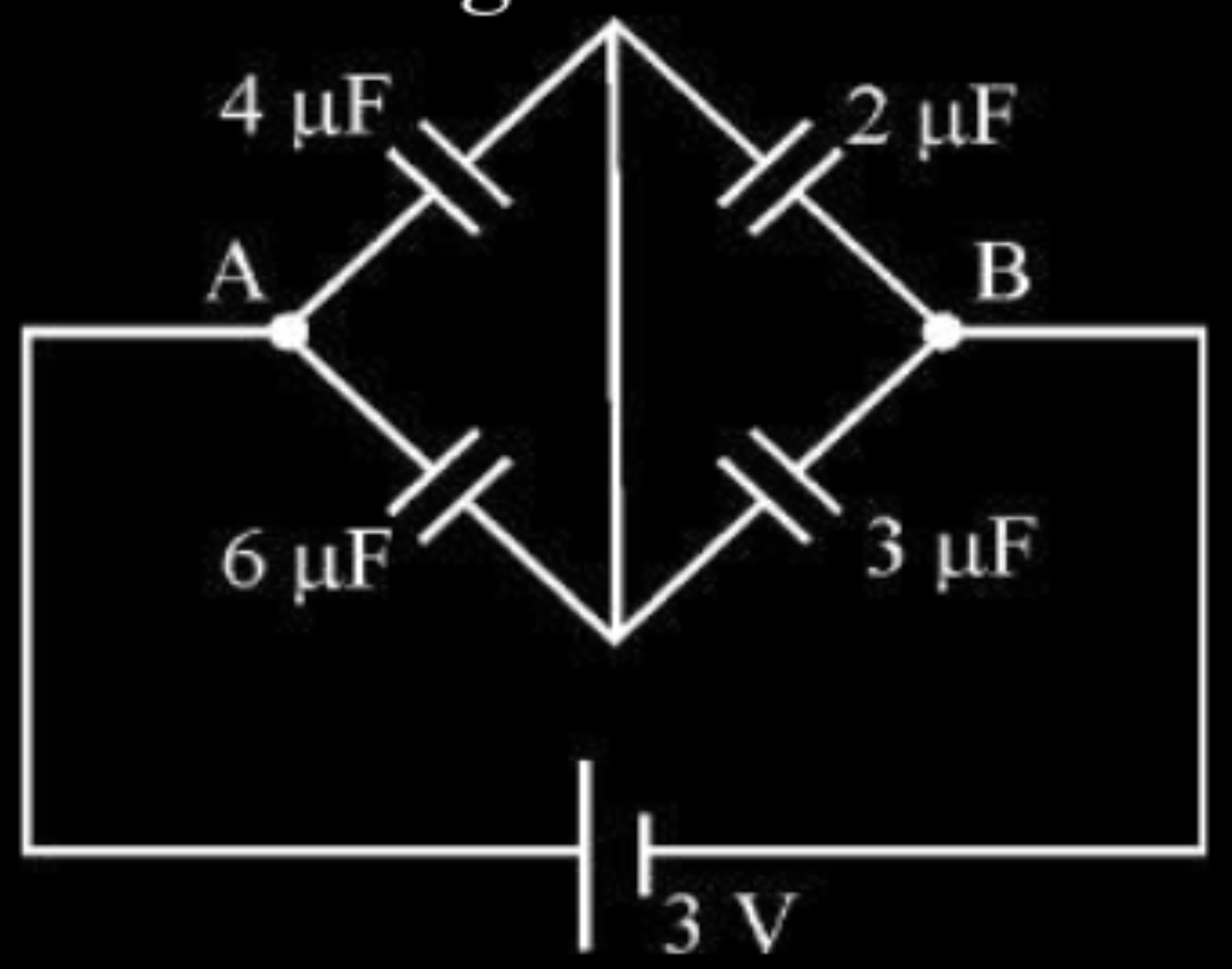
CBSE 20

16. The work done in moving a charge particle between two points in an uniform electric field, does not depend on the path followed by the particle. Why ? 1

CBSE 20

Because the electrostatic force is conservative in nature **Alternatively:-** Electric field is conservative in nature / work done by or against the electric field does not depend upon the path followed.	1

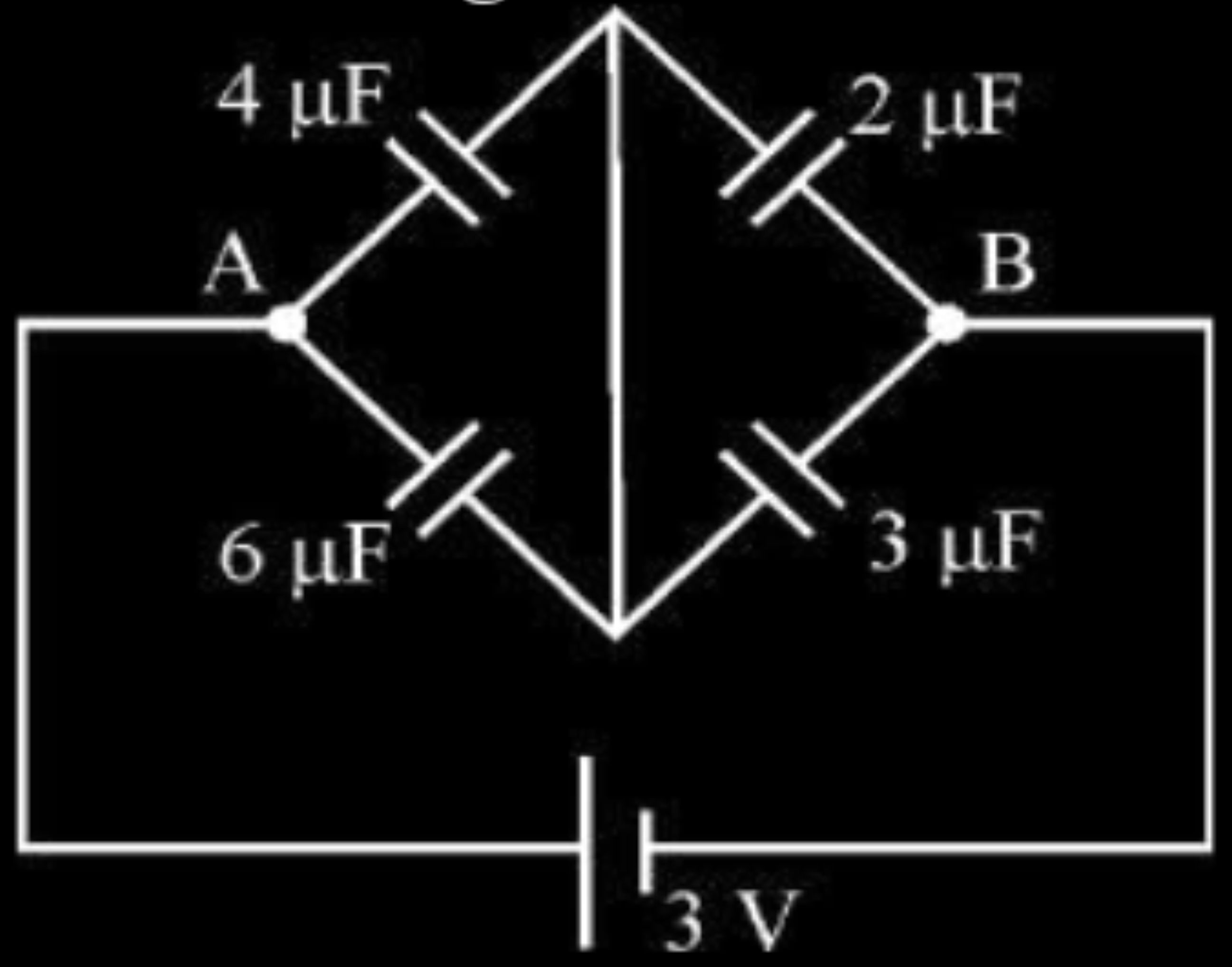
21. Find the total charge stored in the network of capacitors connected between A and B as shown in figure :



CBSE 20

2

21. Find the total charge stored in the network of capacitors connected between A and B as shown in figure :



CBSE 20

2

(i)	Net Capacitance =>	$\frac{1}{C_{net}} = \frac{1}{10} + \frac{1}{5}$ $C_{net} = \frac{10}{3}\,\mu F$	½ ½
(ii)	$q = C_{net}\,V = \frac{10}{3} \times 3 = 10\,\mu C$		½+ ½

1. If a positive charge is displaced against the electric field in which it was situated, then *1*

(A) work will be done by the electric field on the charge.

(B) the intensity of the electric field decreases.

(C) energy of the system will decrease.

(D) energy will be provided by external source displacing the charge.

CBSE 20

1. If a positive charge is displaced against the electric field in which it was situated, then *1*

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CBSE 20

(D)

energy will be provided by external source displacing the charge.

3. Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be *1*

CBSE 20

(A) $\dfrac{C_1}{C_2}$

(B) $\sqrt{\dfrac{C_1}{C_2}}$

(C) $\sqrt{\dfrac{C_2}{C_1}}$

(D) $\dfrac{C_2}{C_1}$

3. Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be *1*

CBSE 20

(A) $\frac{C_1}{C_2}$

(B) $\sqrt{\frac{C_1}{C_2}}$

(C) $\sqrt{\frac{C_2}{C_1}}$

(D) $\frac{C_2}{C_1}$

(A)

$\frac{C_1}{C_2}$

(a) Derive an expression for the energy stored in a parallel plate capacitor of capacitance C when charged up to voltage V. How is this energy stored in the capacitor ?

(b) A capacitor of capacitance 1 μF is charged by connecting a battery of negligible internal resistance and emf 10 V across it. Calculate the amount of charge supplied by the battery in charging the capacitor fully. CBSE 20 5

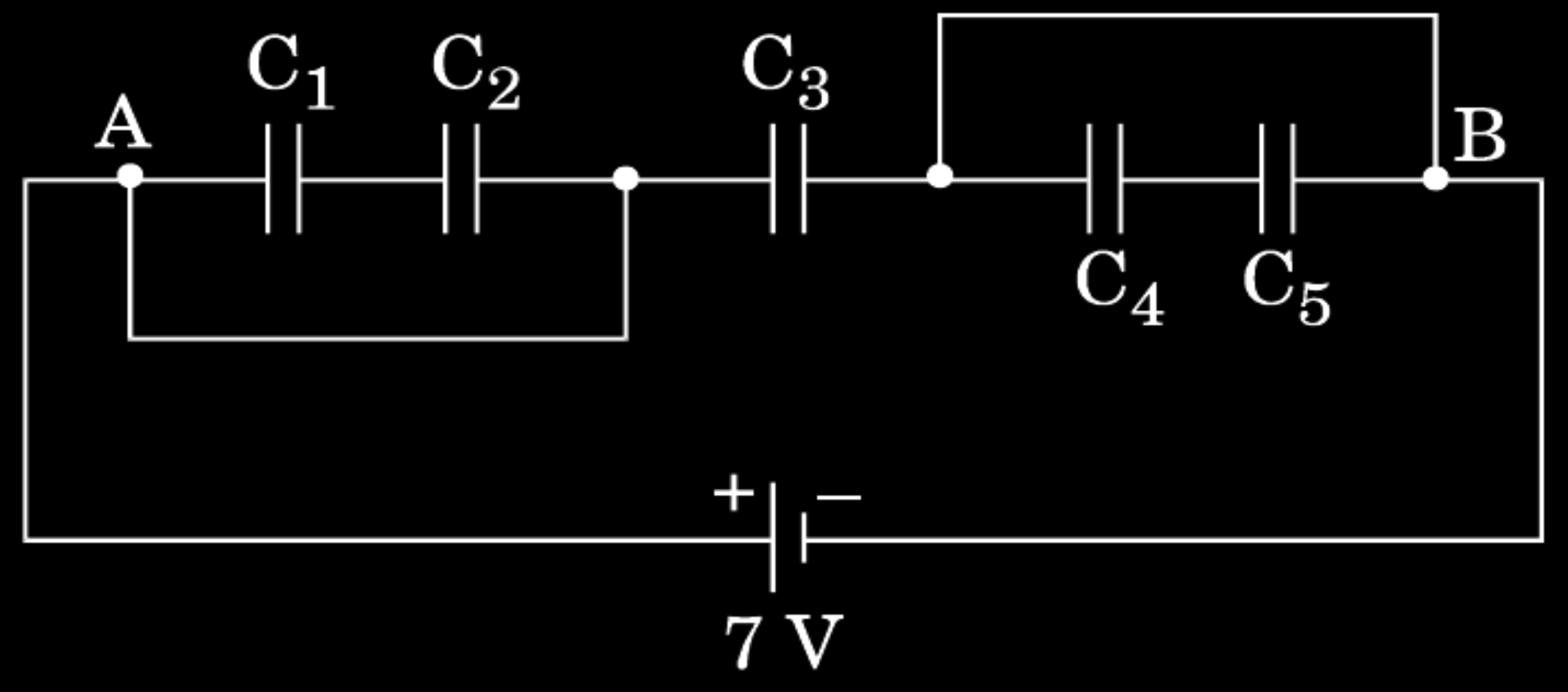
a) Work done in adding a charge dq $= dW$	
$= Vdq$	½
$= \frac{q}{c} dq$	½
∴Total Amount of work(W)in charging a capacitor	
$W = \int dW = \frac{1}{C} \int_0^Q q dq$	½
$W = \frac{Q^2}{2C}$	½
$= \frac{(CV)^2}{2C} = \frac{1}{2} CV^2$	½
The electrostatic Energy/ potential energy is stored in the electric field between the plates.	½
b) $C = 1\mu F = 1 \times 10^{-6} F;$ $V = 10\ volt$	
$Q = CV$	1
$= 1 \times 10^{-6} \times 10$	½
$= 10^{-5} coulomb$	½

In the figure given below, find the

(a) equivalent capacitance of the network between points A and B.
Given : $C_1 = C_5 = 8\ \mu F$, $C_2 = C_3 = C_4 = 4\ \mu F$.

(b) maximum charge supplied by the battery, and

(c) total energy stored in the network. *3*



CBSE 20

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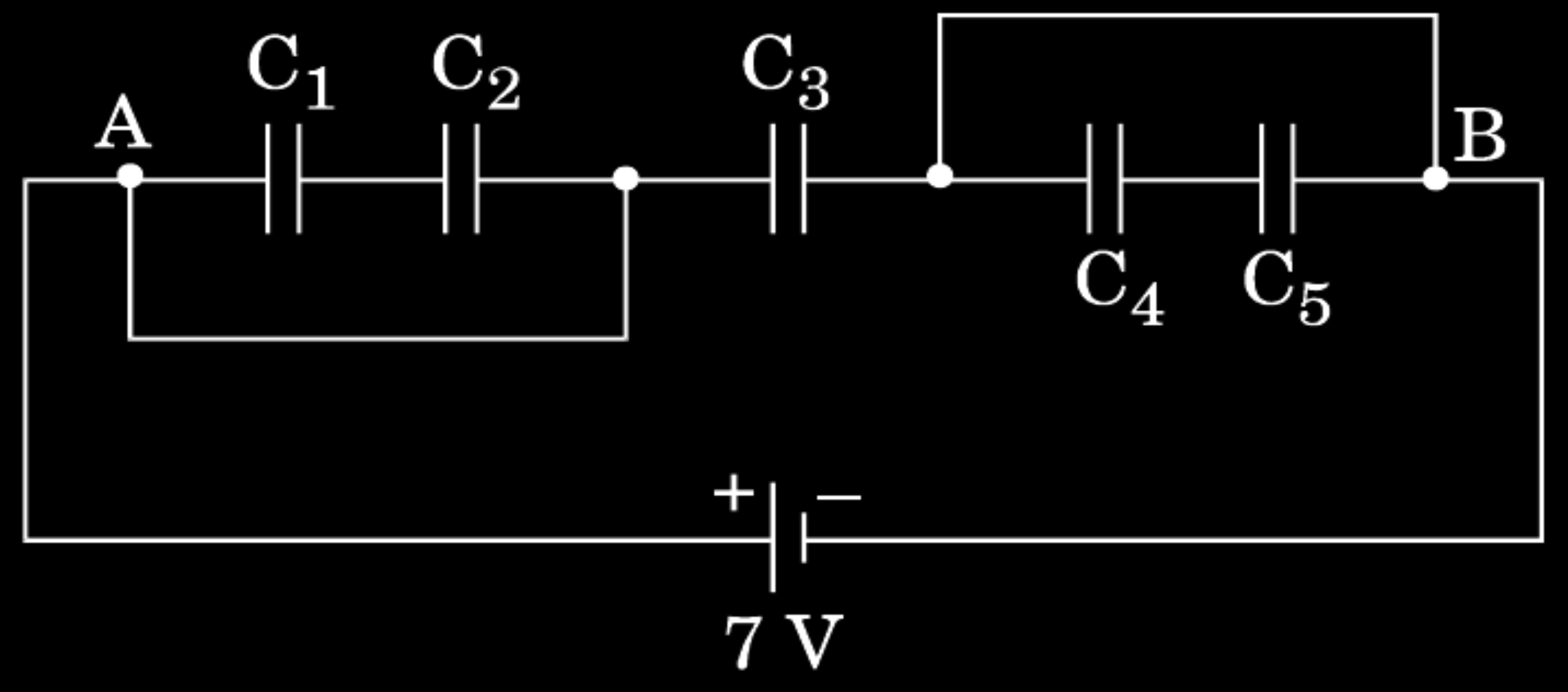


CBSE 20

(a) $C = C_4 = 4\mu F$ (as C_1, C_2, C_4, C_5 are short circuited)

(b) $Q = CV = 4 \times 7\ \mu C$

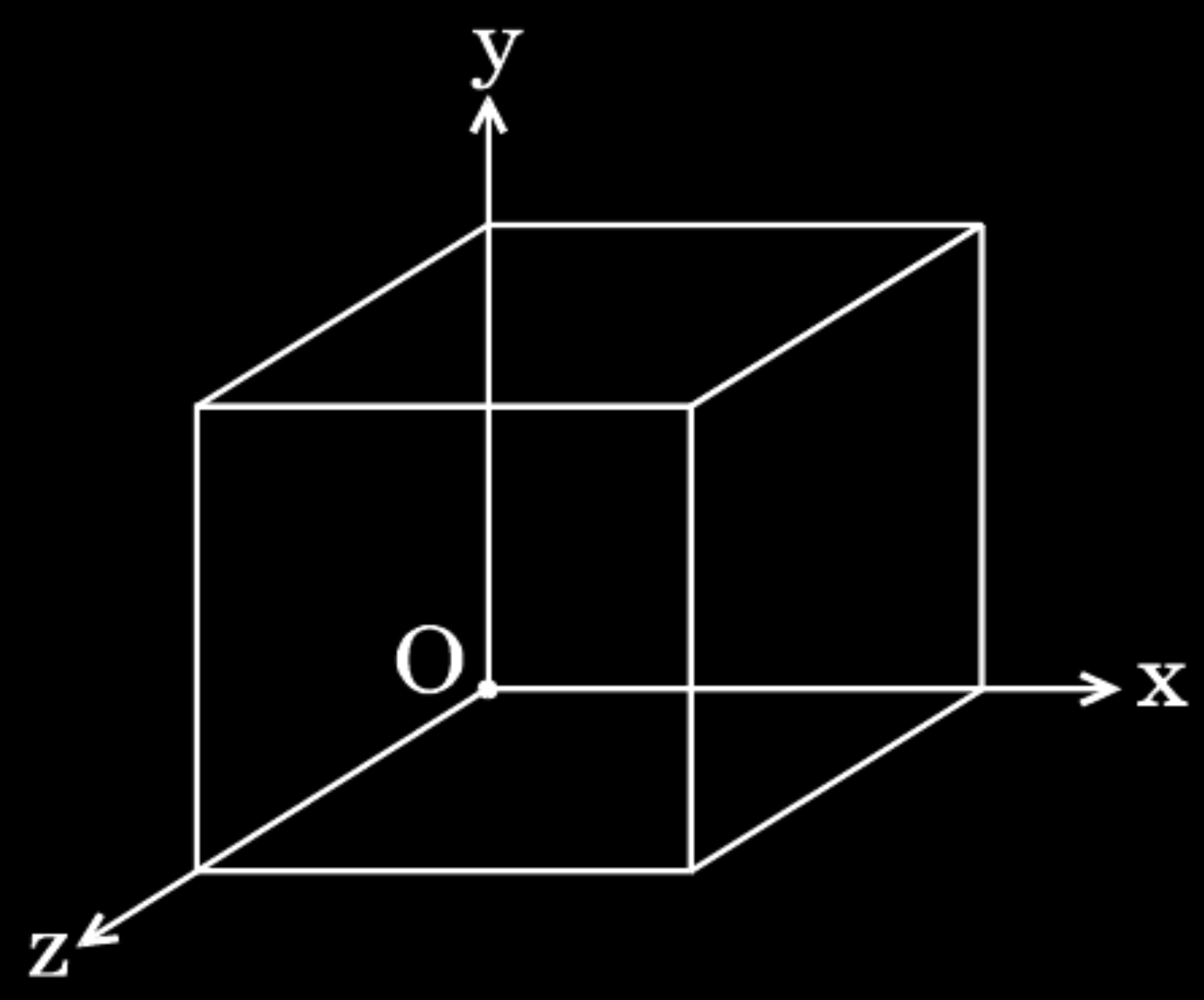
$= 28\ \mu C$

(c) $\mathcal{U} = \frac{1}{2} CV^2$

$$= \frac{1}{2} \times 4 \times 10^{-6} \times 7 \times 7 = 98 \times 10^{-6} J$$

(a) Two point charges q_1 and q_2 are kept r distance apart in a uniform external electric field $\vec{E}$. Find the amount of work done in assembling this system of charges.

(b) A cube of side 20 cm is kept in a region as shown in the figure. An electric field $\vec{E}$ exists in the region such that the potential at a point is given by $V = 10x + 5$, where V is in volt and x is in m.



CBSE 20

Find the

(i) electric field $\vec{E}$, and

(ii) total electric flux through the cube. 5

The work done in bringing charge q_1 from infinity to r_1 is $$W_1 = q_1 V_1$$	½
The work done in bringing charge q_2 from infinity to r_2 is $$W_2 = q_2 V_2$$	½
Work done in moving q_2 against the field due to q_1 $$W_3 = \frac{1}{4\pi\epsilon_o}\frac{q_1 q_2}{r}$$	½
Hence total work done is $W = W_1 + W_2 + W_3$ $$W = q_1 V_1 + q_2 V_2 + \frac{1}{4\pi\epsilon_o}\frac{q_1 q_2}{r}$$ [V_1 and V_2 are potentials at the two points in the electric field]	½
(b) (i) $$E = \frac{-dV}{dx} = -\frac{d}{dx}(10x+5)$$	1
$$\therefore \vec{E} = -10\hat{\imath}\ N/C$$	½
(ii) Electric flux through the cube, ϕ =sum of electric flux through 6 faces Electric flux through faces perpendicular Y and Z axis = 0 $$\because E \text{ is along x axis}$$	½
Electric flux through faces perpendicular to x axis $$= \phi_1 + \phi_2$$	½
$$= 10 \times (0.2)^2 - 10 \times (0.2)^2$$ $$=0$$	½

2. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is *1*

(A) qLE

(B) zero

(C) – qLE

(D) – 2 qEL

CBSE 20

2. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is *1*

(A) qLE

(B) zero

(C) – qLE

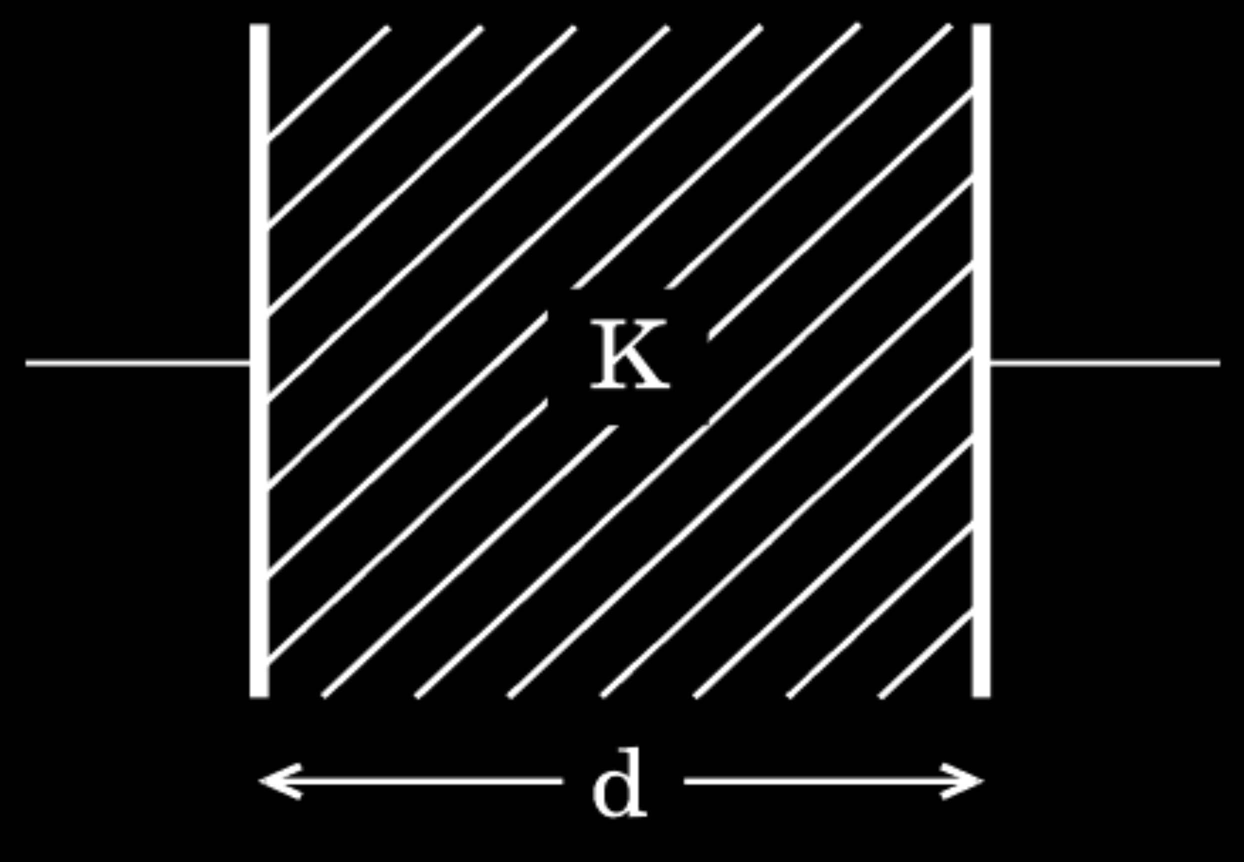
(D) – 2 qEL

CBSE 20

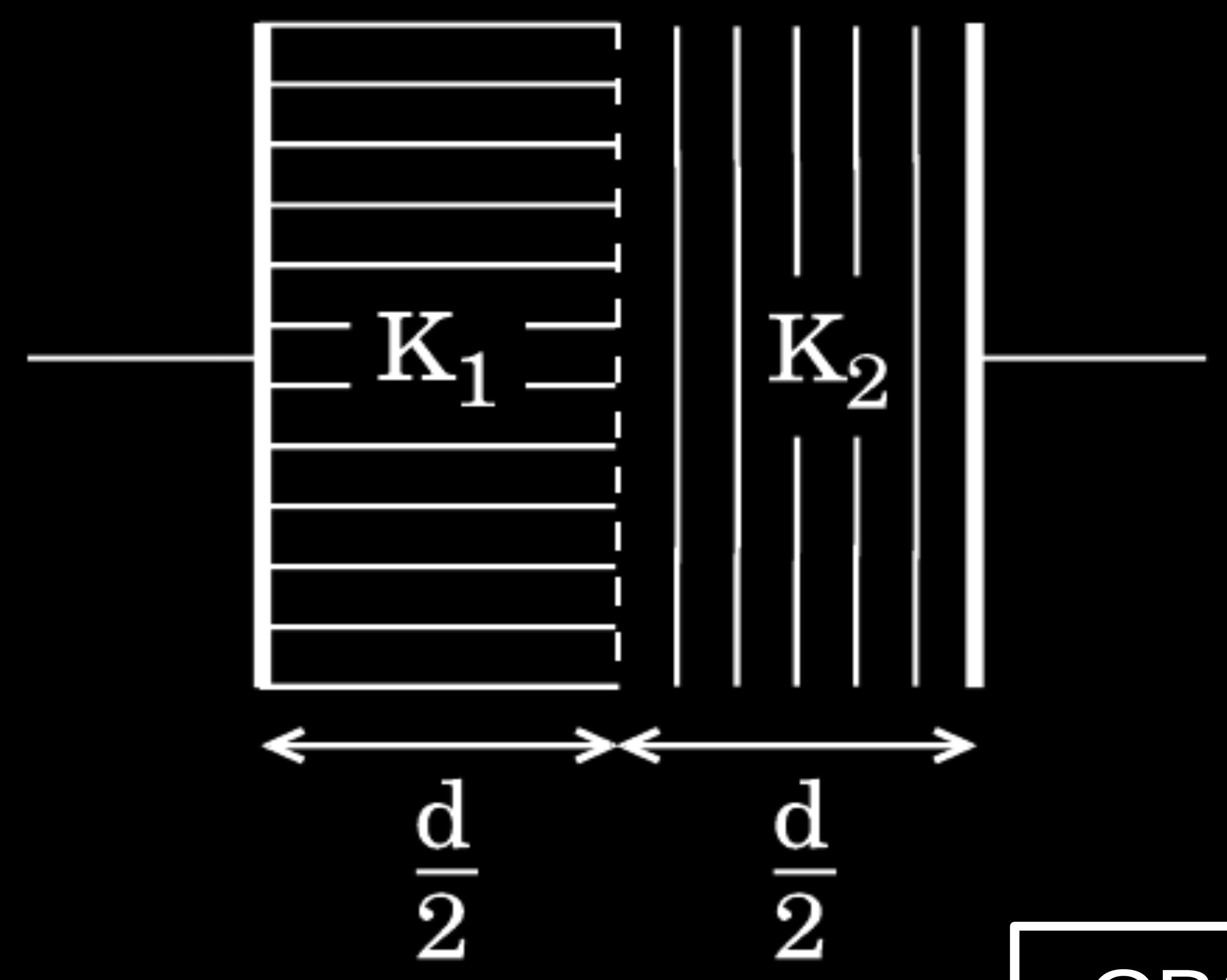
(C)

$-qLE$

22. The space between the plates of a parallel plate capacitor is completely filled in two ways. In the first case, it is filled with a slab of dielectric constant K. In the second case, it is filled with two slabs of equal thickness and dielectric constants K_1 and K_2 respectively as shown in the figure. The capacitance of the capacitor is same in the two cases. Obtain the relationship between K, K_1 and K_2. *2*



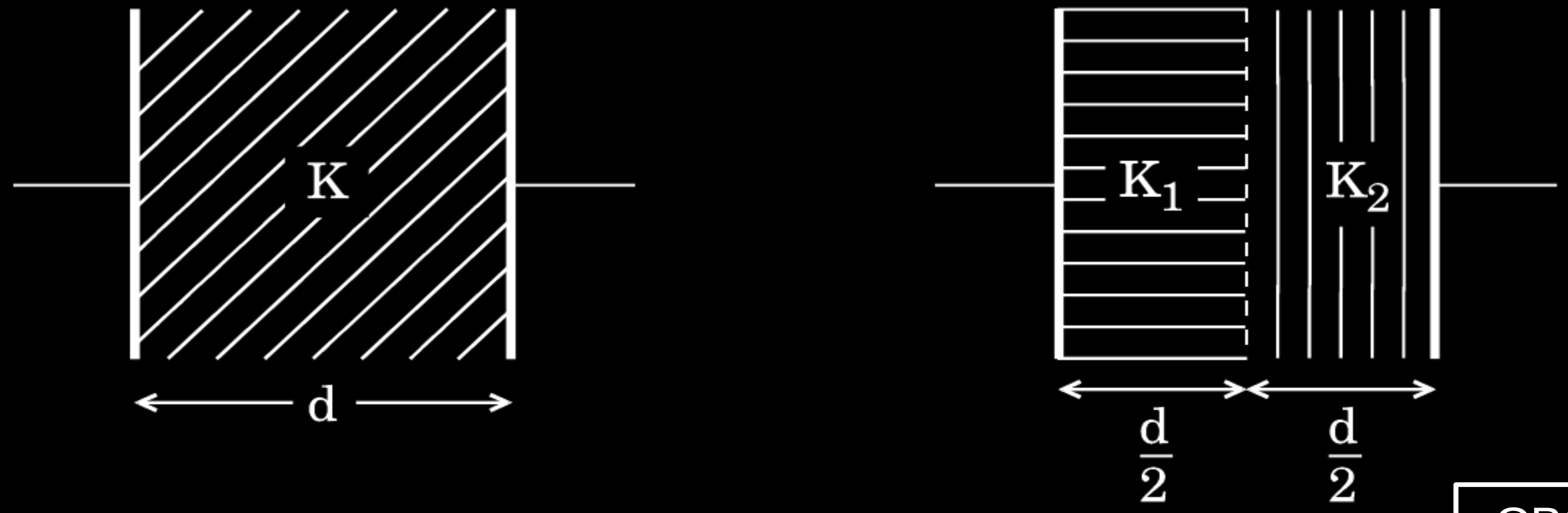
(Case 1)



(Case 2)

CBSE 20

22. The space between the plates of a parallel plate capacitor is completely filled in two ways. In the first case, it is filled with a slab of dielectric constant K. In the second case, it is filled with two slabs of equal thickness and dielectric constants K_1 and K_2 respectively as shown in the figure. The capacitance of the capacitor is same in the two cases. Obtain the relationship between K, K_1 and K_2. *2*



(Case 1) (Case 2)

CBSE 20

$C_1 = \dfrac{k\epsilon_0 A}{d}$	½
Capacitor are connected in series $C_2 = \dfrac{C'C''}{C' + C''} = \left(\dfrac{2K_1K_2}{K_1 + K_2}\right)\dfrac{\epsilon_0\ A}{d}$	1
$C_1 = C_2$	
$K = \dfrac{2K_1K_2}{K_1 + K_2}$	½

(a) Find the expression for the potential energy of a system of two point charges q_1 and q_2 located at $\vec{r}_1$ and $\vec{r}_2$, respectively in an external electric field $\vec{E}$.

(b) Draw equipotential surfaces due to an isolated point charge $(-q)$ and depict the electric field lines.

(c) Three point charges $+1\ \mu C$, $-1\ \mu C$ and $+2\ \mu C$ are initially infinite distance apart. Calculate the work done in assembling these charges at the vertices of an equilateral triangle of side 10 cm. *5*

CBSE 20

(a) Work done in bringing q from infinity against the field $E= q_1 V\|\vec{r_1}\|$	1
Work done on q_2 against the field $E= q_2 V\|\vec{r_2}\|$ Work done on q_2 against the field due to q_1 $= \dfrac{q_1 q_2}{4\pi\epsilon_0 (r_{12})}$	1
Potential energy of the system= Total work done in assembling the system	½
$= q_1 V(\vec{r_1}) + q_2 V(\vec{r_2}) + \dfrac{q_1 q_2}{4\pi\epsilon_0 r_{12}}$	½

b)

c) Work done= charge in potential energy	1
$= \dfrac{kq_1q_2}{r_{12}} + \dfrac{kq_1q_3}{r_{13}} + \dfrac{kq_2q_{31}}{r_{23}}$	½
$= \dfrac{9\times10^{9}\times10^{-12}}{0.1}[1\ \times\ -1+\ -1\times2+1\ \times2]$ $= 9\times10^{-2}[-1-2+2]$ $= -9\times10^{-2}J$	½

Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor ?

CBSE 19

Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor ? CBSE 19

$E = 0$ inside the conductor & has no tangential component on the surface. $\therefore$No work is done in moving charge inside or on the surface of the conductor. &Potential is constant. [Even if a student writes “because $E = 0$ inside the conductor” - award full marks Or No work is done in moving a charge inside the conductor - award full marks.]	½ ½

Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination. CBSE 19

Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination.

CBSE 19

$C_s = 6\ pF$	½
$E_s = \frac{1}{2} C_s V^2 = \frac{1}{2} \times 6 \times 10^{-12} \times 50 \times 50 = 7500 \times 10^{-12}$ J 2	
$= 7.5 \times 10^{-9}$ J	½
$C_p = 24\ pF$	½
$E_p = \frac{1}{2} C_p V^2 = \frac{1}{2} \times 24 \times 50 \times 50 \times 10^{-12}$	
$= 3 \times 10^{-8}$ J	½

A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R ($R >> r$), such that their surface charge densities are equal. Derive the expression for the potential at the common centre.

CBSE 19

A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R (R >> r), such that their surface charge densities are equal. Derive the expression for the potential at the common centre.

CBSE 10

$$Q = q_1 + q_2$$

$$= 4\pi\sigma\left(r^2 + R^2\right)$$

Potential at common centre

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{r} + \frac{q_2}{R}\right]$$ ½

$$= \frac{1}{4\pi\varepsilon_0} \times \left[\frac{4\pi r^2\sigma}{r} + \frac{4\pi R^2\sigma}{R}\right]$$ ½

$$= \frac{(r + R)\sigma}{\varepsilon_0}$$ ½

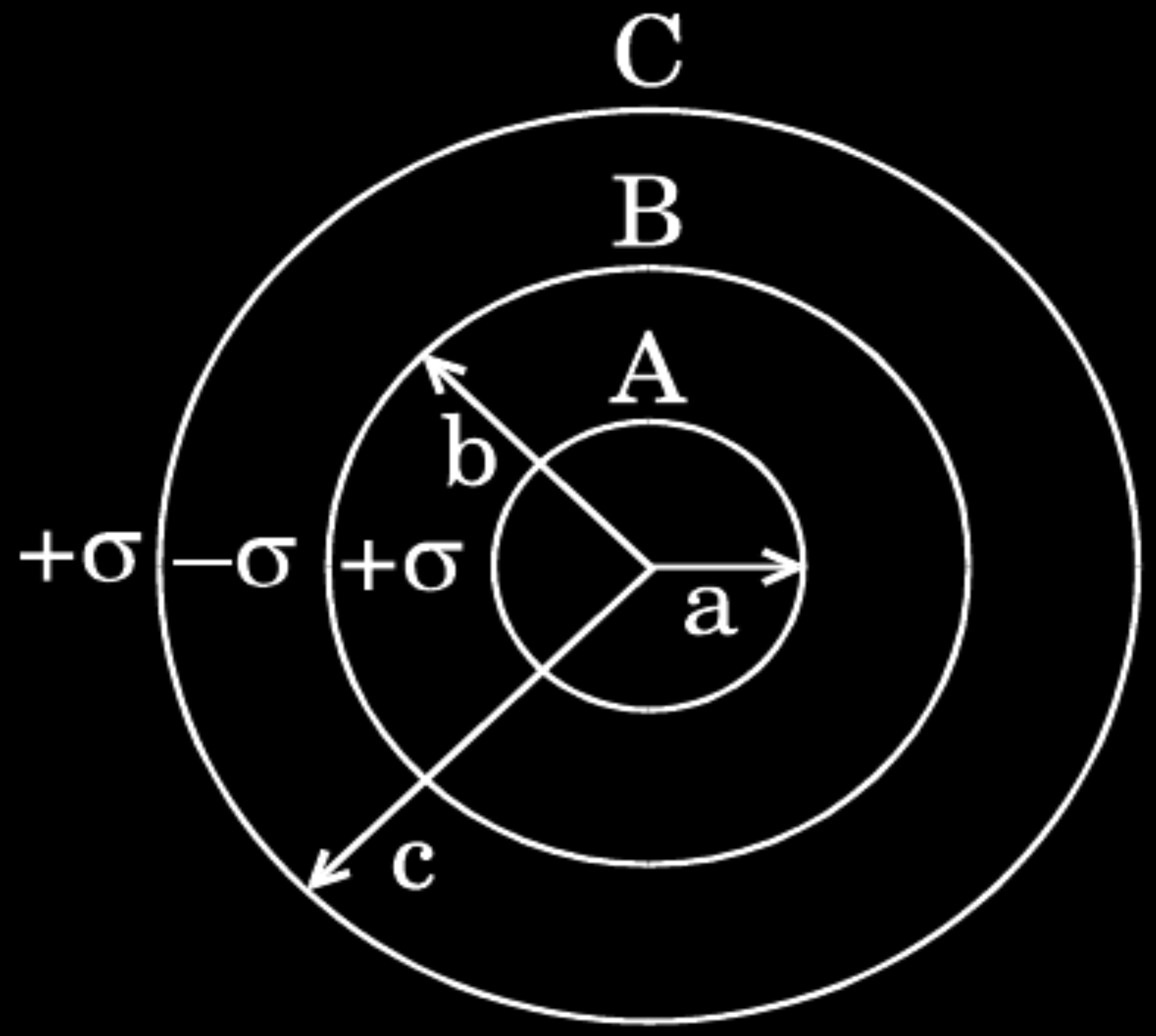
$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{Q(r + R)}{r^2 + R^2}\right]$$ ½

Three concentric metallic shells A, B and C of radii a, b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively as shown.

(a) Obtain the expressions for the potential of three shells A, B and C.

(b) If shells A and C are at the same potential, obtain the relation between a, b and c.

CBSE 19



$$V_A = \frac{kQ_A}{a} + \frac{kQ_B}{b} + \frac{kQ_C}{c}$$

$$V_A = \frac{1}{4\pi\varepsilon_0}\left[\frac{4\pi a^2\sigma}{a} - \frac{4\pi b^2\sigma}{b} + \frac{4\pi c^2\sigma}{c}\right] \qquad ½$$

$$= \frac{\sigma}{\varepsilon_0}[a - b + c] \qquad ½$$

$$V_B = \frac{kQ_A}{b} + \frac{kQ_B}{b} + \frac{kQ_C}{c}$$

$$V_B = \frac{1}{4\pi\varepsilon_0}\left[\frac{4\pi a^2\sigma}{b} - \frac{4\pi b^2\sigma}{b} + \frac{4\pi c^2\sigma}{c}\right] \qquad ½$$

½

$$V_B = \frac{\sigma}{\varepsilon_0}\left[\frac{a^2 - b^2}{b} + c\right]$$

$$V_C = \frac{kQ_A}{c} + \frac{kQ_B}{c} + \frac{kQ_C}{c}$$

$$V_C = \frac{1}{4\pi\varepsilon_0}\left[\frac{4\pi a^2\sigma}{c} - \frac{4\pi b^2\sigma}{c} + \frac{4\pi c^2\sigma}{c}\right] \qquad ½$$

$$= \frac{\sigma}{\varepsilon_0}\left[\frac{a^2 - b^2 + c^2}{c}\right]$$

$$V_A = V_C$$

$$a - b + c = \frac{a^2 - b^2}{c} + c \qquad ½$$

$$c = a + b$$

A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell ?

CBSE 19

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CBSE 19

Surface charge density on inner surface $= -\dfrac{Q}{4\pi R_1^2}$	½
Surface charge density on Outer surface $= +\dfrac{Q}{4\pi R_2^2}$	½

(a) When a parallel plate capacitor is connected across a dc battery, explain briefly how the capacitor gets charged.

(b) A parallel plate capacitor of capacitance 'C' is charged to 'V' volt by a battery. After some time the battery is disconnected and the distance between the plates is doubled. Now a slab of dielectric constant $1 < k < 2$ is introduced to fill the space between the plates. How will the following be affected ?

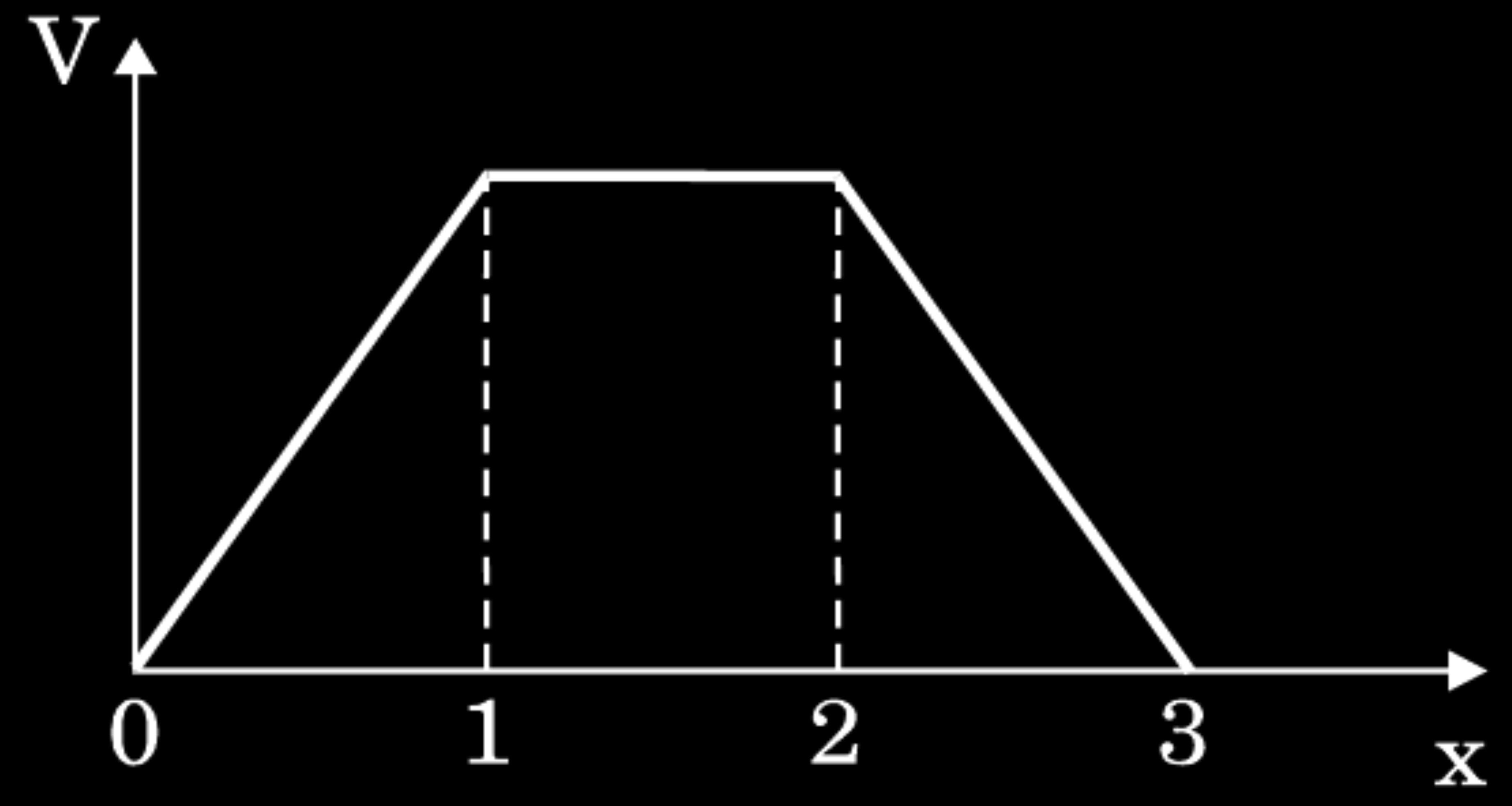
(i) The electric field between the plates of the capacitor.

(ii) The energy stored in the capacitor.

CBSE 19

Justify your answer in each case.

(c) The electric potential as a function of distance 'x' is shown in the figure. Draw a graph of the electric field E as a function of x.



a) Charging of capacitor with dc battery whenever parallel plate capacitor is connected with dc source, plates start acquiring charge in accordance with the terminals of the battery till potential difference across the plate becomes equal to terminal potential of dc battery. 1

Note : Any other relevant explanation may also be accepted.

b) i) The electric field between the plates of parallel plate capacitor

$$E_0 = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$ ½

If dielectric is inserted

$$E' = \frac{Q}{\varepsilon_0 A.K} = \frac{\varepsilon_0}{K}$$ ½

So, the electric field intensity decreases to 1/K times. ½

ii) Since Energy stores in the capacitor

$$U = \frac{Q^2}{2C} = \frac{Q^2 d}{2\varepsilon_0 A} \quad \text{...} (1)$$ ½

Similarly

$$U' = \frac{Q^1}{2C'} = \frac{Q^2 d_1}{2K\varepsilon_0 A}$$

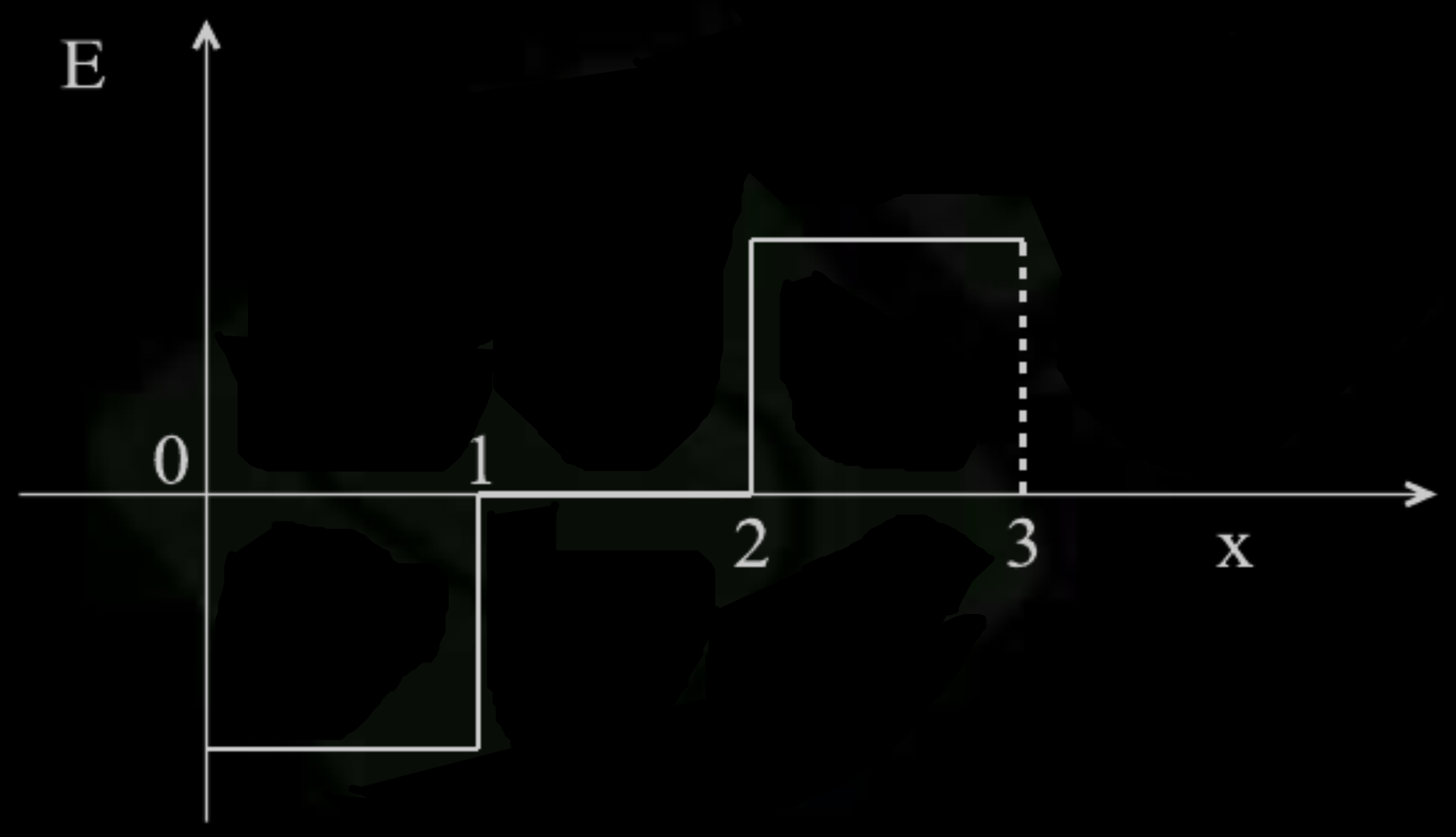
$$= \frac{2}{K}\left(\frac{Q^2 d_1}{2\varepsilon_0 A}\right)$$

$$= \frac{2U}{K}$$ ½

i<k<2

Therefore energy stored between the plates increases ½

iii)

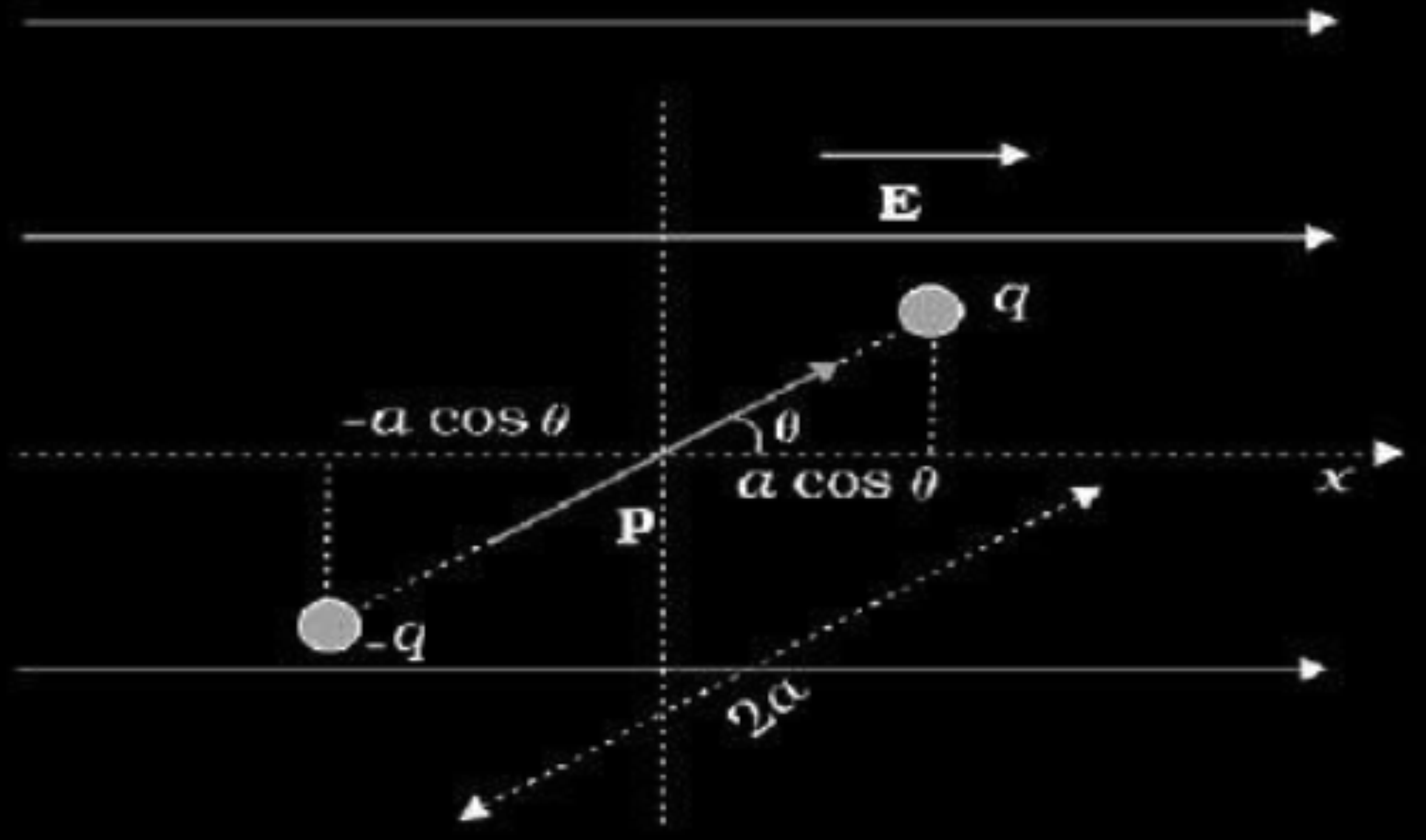


(a) Derive an expression for the potential energy of an electric dipole in a uniform electric field. Explain conditions for stable and unstable equilibrium.

(b) Is the electrostatic potential necessarily zero at a point where the electric field is zero ? Give an example to support your answer.

CBSE 19

a) Since torque acting on dipole



$$\vec{\tau} = \vec{p} \times \vec{E}$$ ½

$$\vec{\tau} = pE\sin\theta.\hat{n}$$ ½

work done $d\omega = \tau.d\theta$

$= pE\sin\theta d\theta$

$$w = \int_{\theta_1}^{\theta_2} dw \quad pE\int_{\theta_1}^{\theta_2} \sin\theta\, d\theta$$ ½

$$w = pE[-\cos\theta]_{\theta_1}^{\theta_2}$$

$$= pE[\cos\theta_1 - \cos\theta_2]$$

if $\theta_1 = 0, \theta_2 = \theta$

$$w = pE(1-\cos\theta)$$ ½

Conditions-

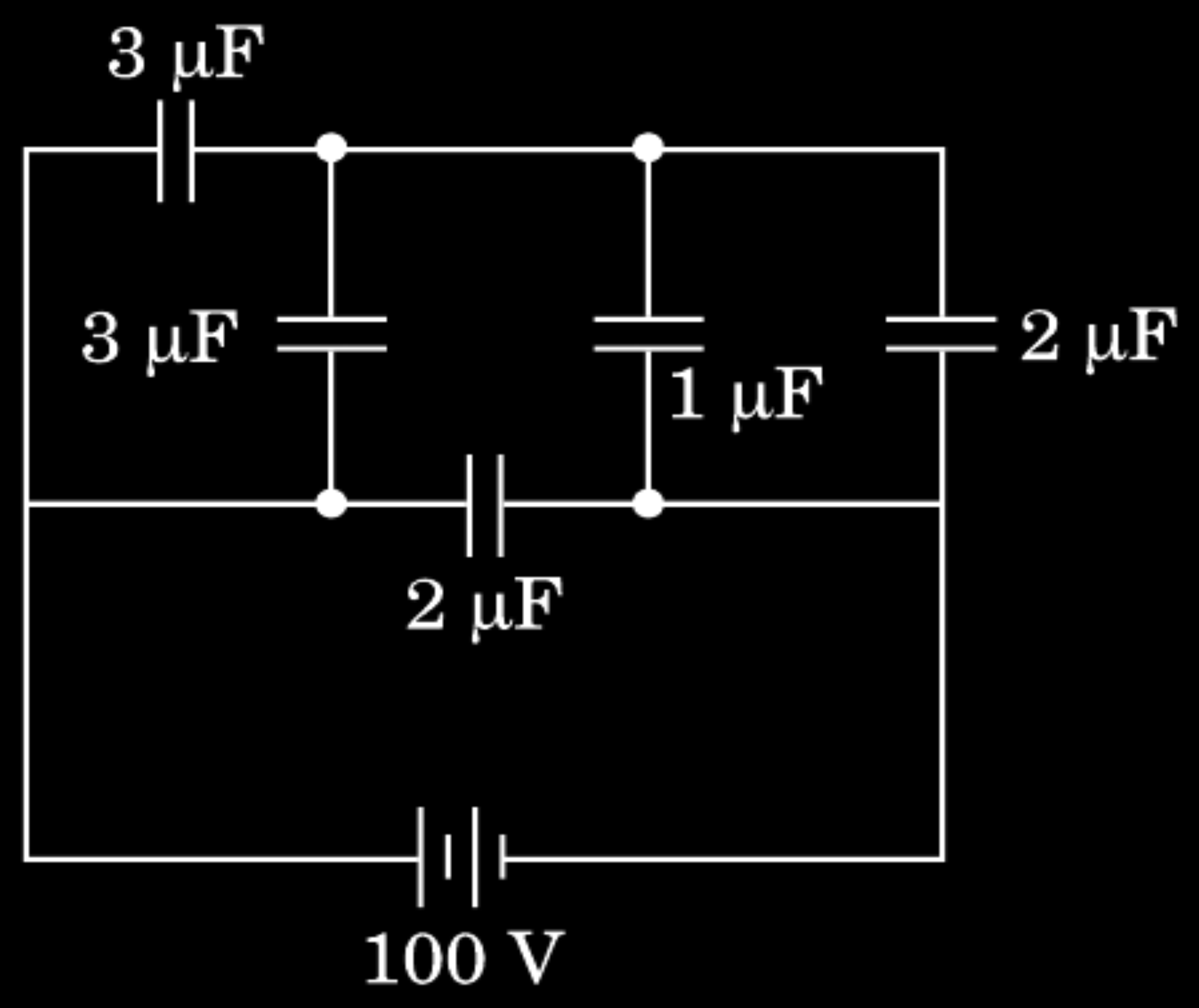
For stable equilibrium - When electric dipole is parallel to electric field. 1

For unstable equilibrium - Anti Parallel to electric field. 1

b) No. ½

Inside equipotential surface ½

The figure shows a network of five capacitors connected to a 100 V supply. Calculate the total energy stored in the network.



CBSE 19

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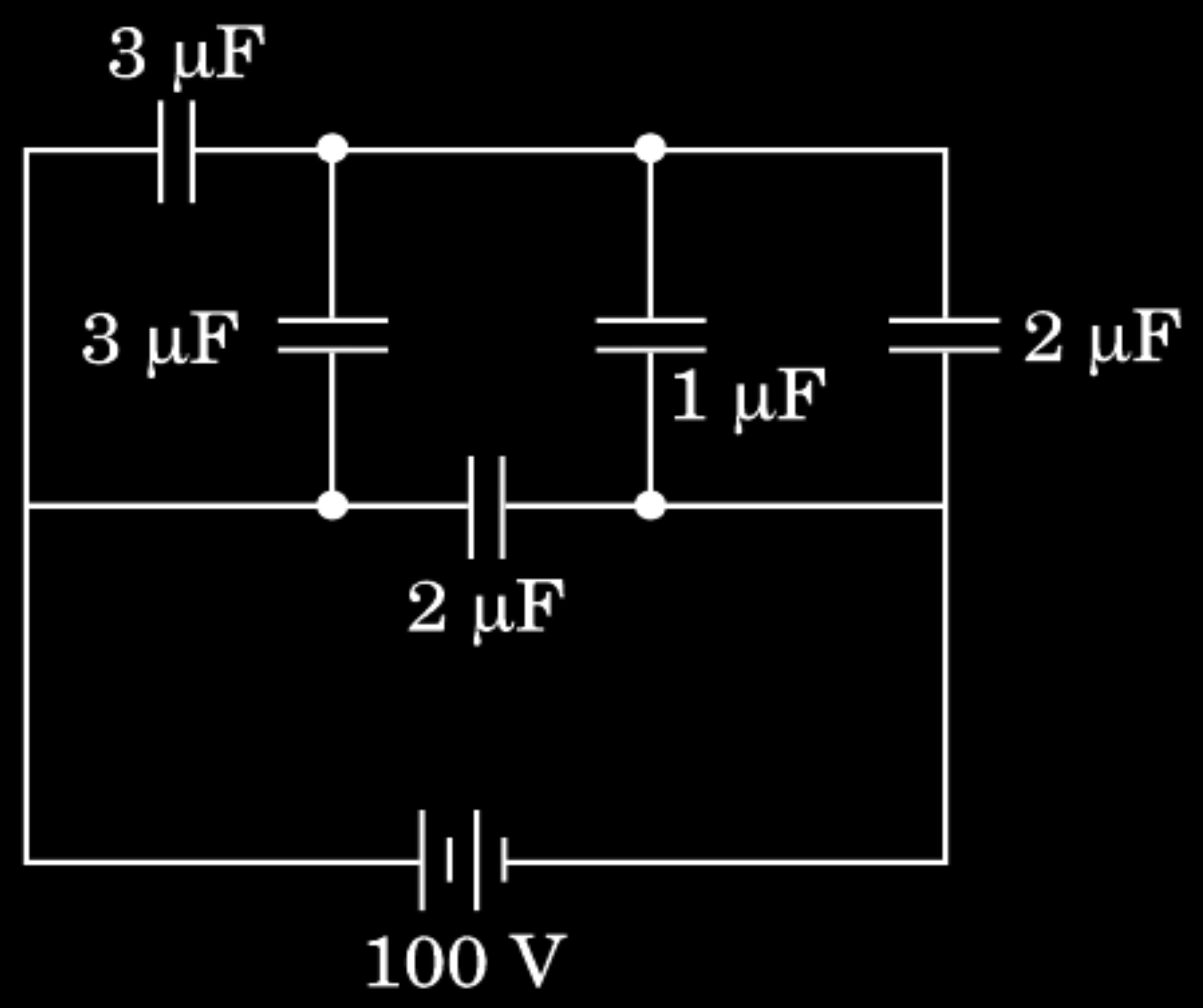

CBSE 19

i)	Equivalent capacitance of two 3μF capacitor in parallel $C_1 = 3+3 = 6\mu F$	
	Similarly equivalent capacitance of 1μF and 2μF in parallel	
	$C_2 = 1 + 2 = 3\mu F$	½
	Equivalent capacitance of	
	C_1 and C_2 in series	
	$C_{12} = \frac{6\times 3}{6+3} = 2\mu F$	
	Net capacitance	
	$C = 2 + 2 = 4\mu F$	½
ii)	Energy stored $E = \frac{1}{2} CV^2$	½
	$= \frac{1}{2} \times 4 \times 10^{-6} \times (100)^2$	
	$= 0.02$ J	½

[Note: Award the 1 mark for correct calculation in part (ii) if the value of C_{eq} obtained in part (i) is correct]

(a) Explain briefly, using a proper diagram, the difference in behaviour of a conductor and a dielectric in the presence of external electric field. CBSE 19

(b) Define the term polarization of a dielectric and write the expression for a linear isotropic dielectric in terms of electric field.

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CBSE 19

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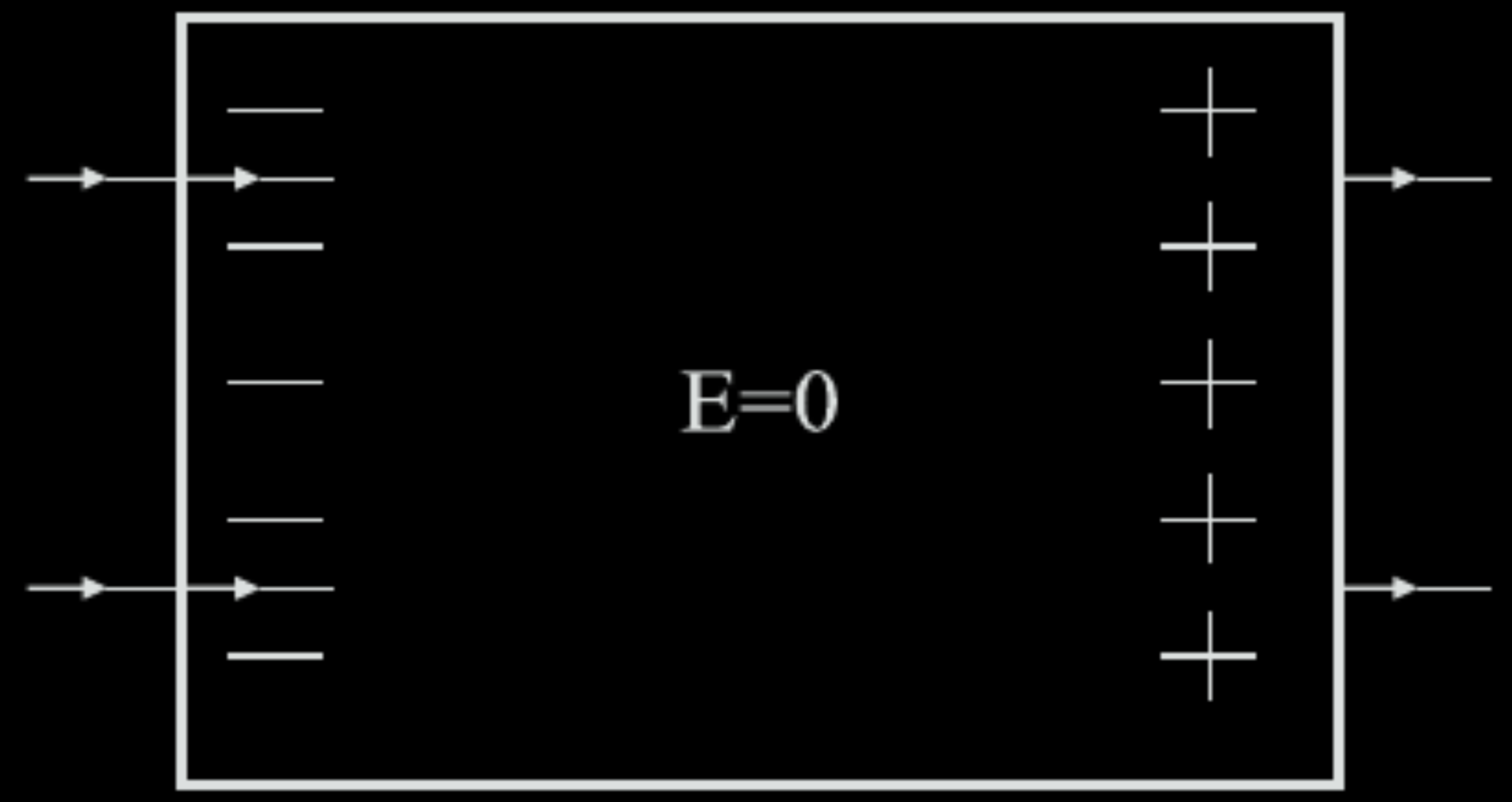
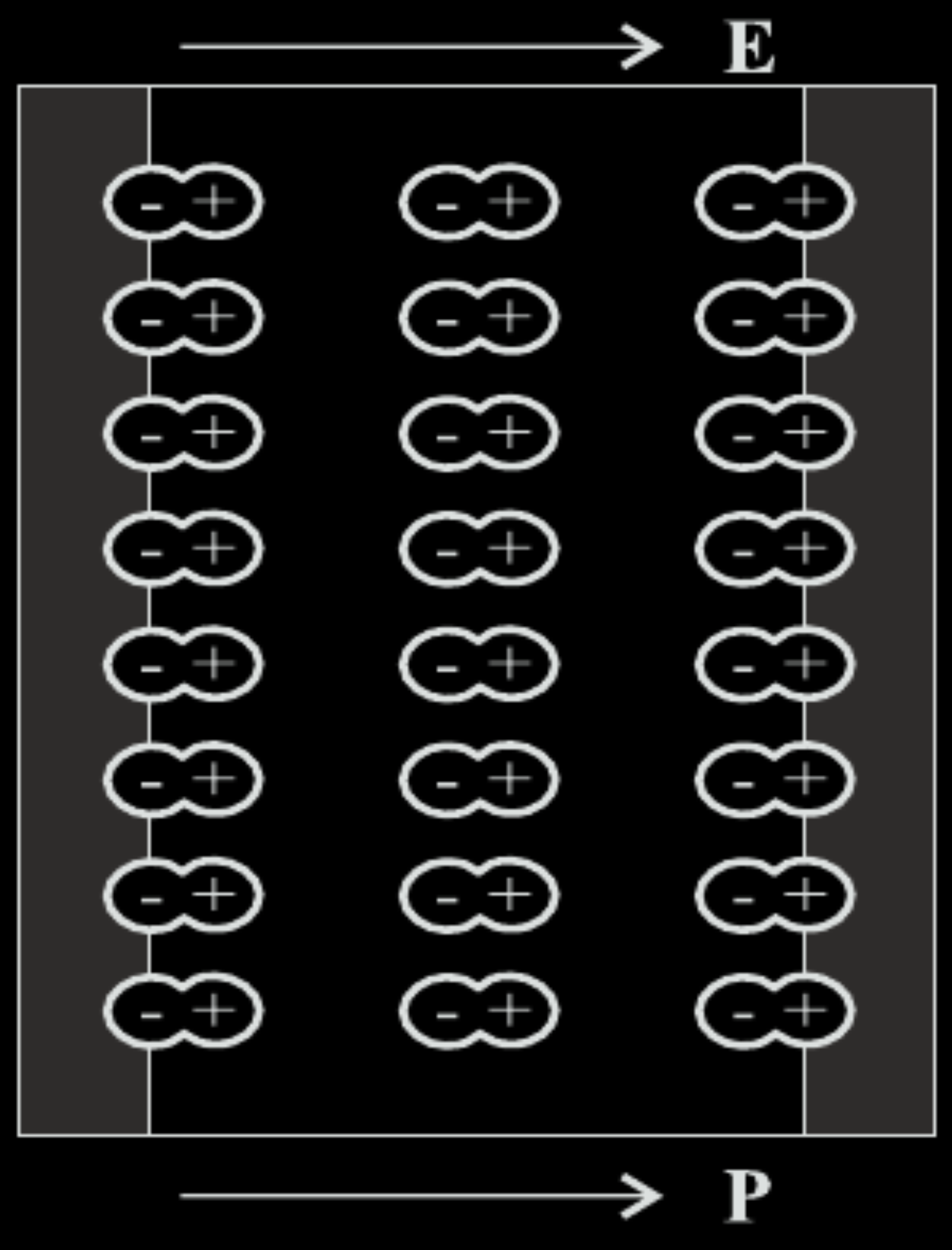


½



For conductor

Due to induction the free electrons collect on the left face of slab creating equal positive charge on the right face. Internal electric field is equal and opposite to external field; hence net electric field (inside the conductor) is zero. ½

½

For dielectiric

Due to alignment of atomic dipoles (permanent or induced) along $\vec{E}$, the net electric field within the dielectric decreases. ½

ii) The net dipole moment developed per unit volume in the presence of external electric field is called polarization vector $\vec{P}$. ½

Expression ∴ ½

$$\vec{P} = X_e \vec{E}$$

Draw equipotential surfaces for an electric dipole.

CBSE 19

Draw equipotential surfaces for an electric dipole.

CBSE 19

(Even if a student mentions or draws equatorial plane, award 1 mark.)

1

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.

(a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.

(b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not ? Explain. CBSE 19

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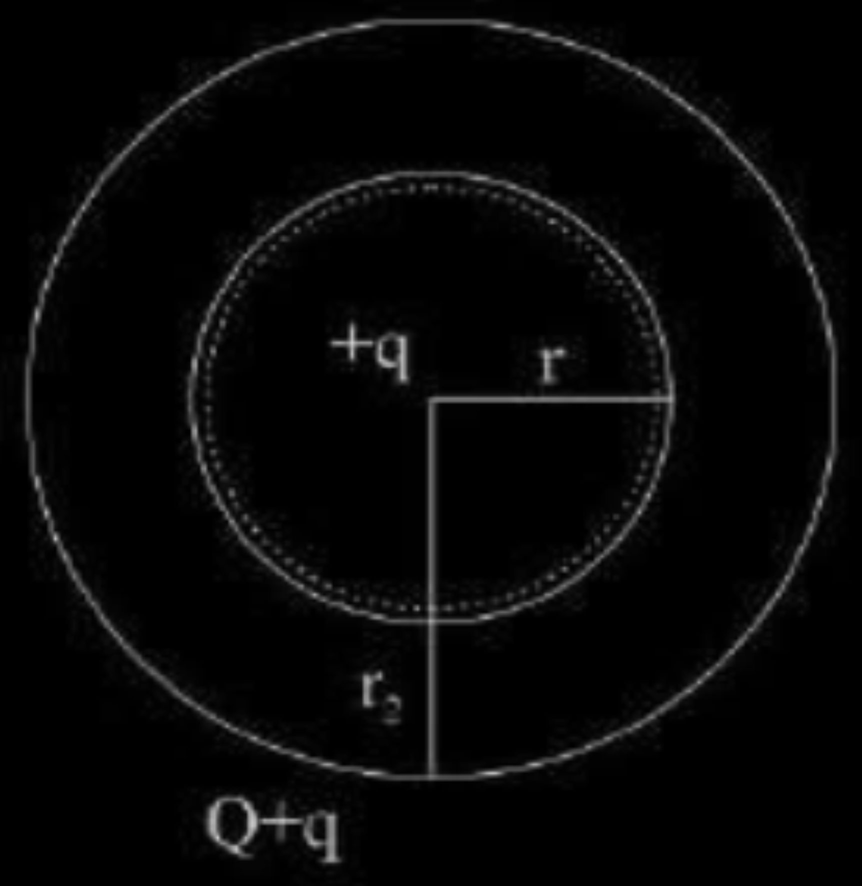
(a) Diagram



½

The surface charge density on inner surface of the shell is $\sigma_1 = -\dfrac{q}{4\pi r_1^2}$ 1

The surface charge density on outer shell is $\sigma_2 = \dfrac{Q+q}{4\pi r_2^2}$ ½

(b) Consider a Gaussian surface inside the shell, net flux is zero since $q_{net} = 0$. According to Gauss's law it is independent of shape and size of shell. 1

A 200 μF parallel plate capacitor having plate separation of 5 mm is charged by a 100 V dc source. It remains connected to the source. Using an insulated handle, the distance between the plates is doubled and a dielectric slab of thickness 5 mm and dielectric constant 10 is introduced between the plates. Explain with reason, how the (i) capacitance, (ii) electric field between the plates, (iii) energy density of the capacitor will change ?

CBSE 19

Dielectric slab of thickness 5mm is equivalent to an air capacitor of thickness $=\frac{5}{10}mm$.	½
Effective separation between the plates with air in between is $=(5+0.50)mm=5.5mm$	
(i) Effective new capacitance $=200\mu F\times\frac{5mm}{5.5mm}=\frac{2000}{11}\mu F$ $\approx 182\mu F$	1
(ii) Effective new electric field $=\frac{100\,V}{5.5\times10^{-3}m}=\frac{20000}{11}$ $\approx 18182\,V/m$	½
(iii) $\frac{New\ energy\ stored}{Original\ energy\ stored}=\frac{\frac{1}{2}C'V^2}{\frac{1}{2}CV^2}=\frac{C'}{C}=\frac{10}{11}$	1
New Energy density will be $\left(\frac{10}{11}\right)^2$ of the original energy density $=\frac{100}{121}$ of the original energy density.	
Note: If the student writes $C=\frac{A\varepsilon_0}{d}$ $C_m=\frac{KA\varepsilon_0}{d}$ $E'=\frac{V}{d}$ $U=\frac{1}{2}\varepsilon_0E^2$ Award full marks	

(a) Describe briefly the process of transferring the charge between the two plates of a parallel plate capacitor when connected to a battery. Derive an expression for the energy stored in a capacitor.

(b) A parallel plate capacitor is charged by a battery to a potential difference V. It is disconnected from battery and then connected to another uncharged capacitor of the same capacitance. Calculate the ratio of the energy stored in the combination to the initial energy on the single capacitor.

CBSE 19

The electrons are transferred to the positive terminal of the battery from the metallic plate connected to the positive terminal, leaving behind positive charge on it. Similarly, the electrons move on to the second plate from negative terminal, hence it gets negatively charged. Process continuous till the potential difference between two plates equals the potential of the battery. ½

[Note: award this $\frac{1}{2}$ mark, If the student writes, there will be no transfer of charge between the plates]

Let ‘dw’ be the work done by the battery in increasing the charge on the capacitor from q to (q+ dq). ½

$$dW = V\ dq$$

Where V = $\frac{q}{c}$ ½

$\therefore$ dW = $\frac{q}{c}dq$ ½

Total work done in changing up the capacitor

$$W = \int dw = \int_0^Q \frac{q}{c}dq$$ ½

$\therefore$ W= $\frac{Q^2}{2C}$ ½

Hence energy stored = W= $\frac{Q^2}{2C}$ (=$\frac{1}{2}CV^2$ = $\frac{1}{2}$ QV)

(b) Charge stored on the capacitor q=CV

When it is connected to the uncharged capacitor of same capacitance, sharing of charge takes place between the two capacitor till the potential of both the capacitor becomes $\frac{V}{2}$ ½

½

Energy stored on the combination $(u_2) = \frac{1}{2} C \left(\frac{V}{2}\right)^2 + \frac{1}{2} C \left(\frac{V}{2}\right)^2 = \frac{CV^2}{4}$

Energy stored on single capacitor before connecting ½

$U_1 = \frac{1}{2} CV^2$

Ratio of energy stored in the combination to that in the single capacitor

½

$$\frac{U_2}{U_1} = \frac{CV^2/4}{CV^2/2} = 1:2$$

An electron is accelerated through a potential difference V. Write the expression for its final speed, if it was initially at rest. CBSE 18

Two point charges q and –q are located at points (0, 0, –a) and (0, 0, a) respectively.

(a) Find the electrostatic potential at (0, 0, z) and (x, y, 0)

(b) How much work is done in moving a small test charge from the point (5, 0, 0) to (–7, 0, 0) along the x-axis ?

(c) How would your answer change if the path of the test charge between the same points is not along the x-axis but along any other random path ?

(d) If the above point charges are now placed in the same positions in a uniform external electric field $\vec{E}$, what would be the potential energy of the charge system in its orientation of unstable equilibrium ?

Justify your answer in each case.

CBSE 18

A capacitor of capacitance C_1 is charged to a potential V_1 while another capacitor of capacitance C_2 is charged to a potential difference V_2. The capacitors are now disconnected from their respective charging batteries and connected in parallel to each other.

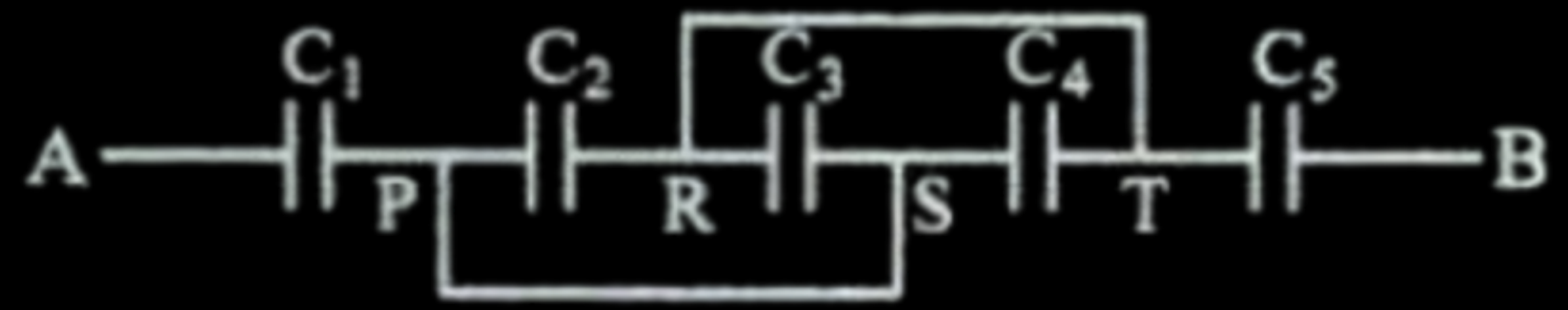
(a) Find the total energy stored in the two capacitors before they are connected.

(b) Find the total energy stored in the parallel combination of the two capacitors.

(c) Explain the reason for the difference of energy in parallel combination in comparison to the total energy before they are connected.

CBSE 18

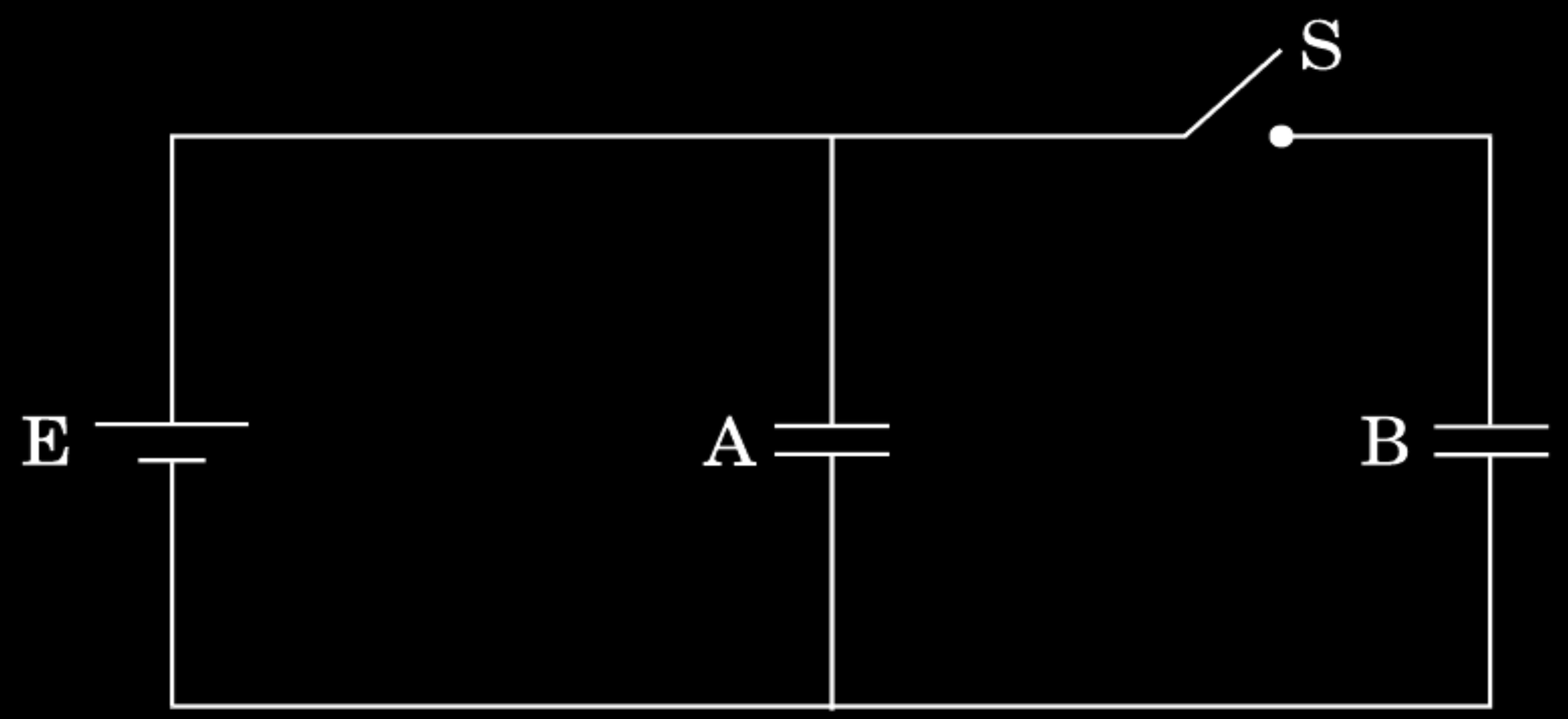
(i) Find equivalent capacitance between A and B in the combination given below. Each capacitor is of 2 μF capacitance.



(ii) If a dc source of 7 V is connected across AB, how much charge is drawn from the source and what is the energy stored in the network ?

CBSE 17

Two identical parallel plate capacitors A and B are connected to a battery of V volts with the switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K. Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric.



CBSE 17

Energy stored = $\frac{1}{2}\ CV^2\ (=\frac{1}{2}\frac{Q^2}{C})$	½
Net capacitance with switch S closed = $C + C = 2C$	½
$\therefore$ Energy stored = $\dfrac{1}{2} \times 2C \times V^2 = CV^2$	½
After the switch S is opened, capacitance of each capacitor= KC	
$\therefore$ Energy stored in capacitor A = $\dfrac{1}{2}KCV^2$	
For capacitor B,	
Energy stored = $\frac{1}{2}\frac{Q^2}{KC} = \frac{1}{2}\frac{C^2V^2}{KC} = \frac{1}{2}\frac{CV^2}{K}$	½
$\therefore$ Total Energy stored = $\dfrac{1}{2}KCV^2 + \dfrac{1}{2}\dfrac{CV^2}{K} = \dfrac{1}{2}CV^2\left(K + \dfrac{1}{K}\right)$	
$= \dfrac{1}{2}CV^2\left(\dfrac{K^2+1}{K}\right)$	½
$\therefore$ Required ratio = $\dfrac{2CV^2.K}{CV^2(K^2+1)} = \dfrac{2K}{(K^2+1)}$	½

A point charge +Q is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ positive, negative or zero ?

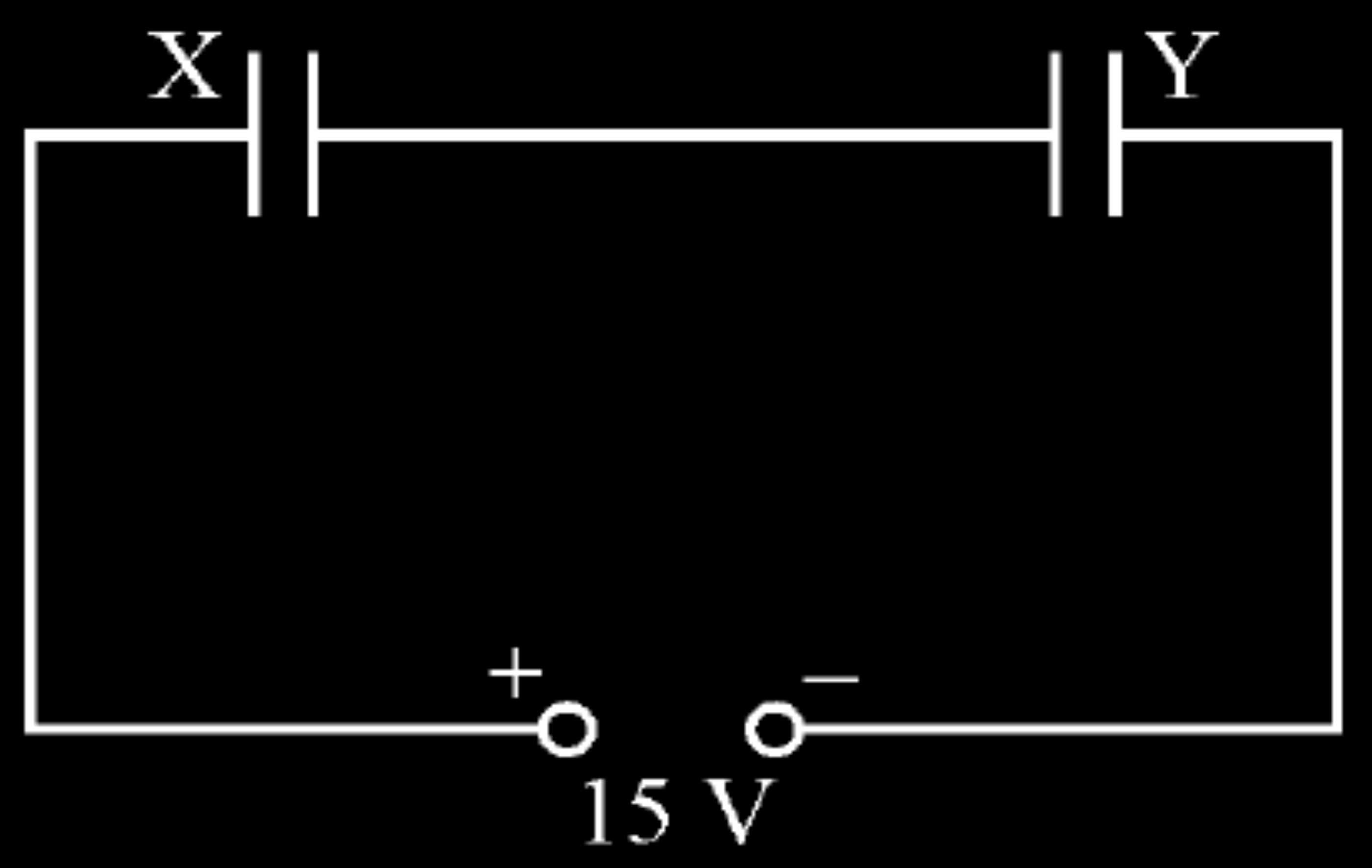
CBSE 16

A point charge +Q is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ positive, negative or zero ?

CBSE 16

Positive

Two parallel plate capacitors X and Y have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of $\varepsilon_r = 4$.



(i) Calculate capacitance of each capacitor if equivalent capacitance of the combination is 4 μF.

(ii) Calculate the potential difference between the plates of X and Y.

(iii) Estimate the ratio of electrostatic energy stored in X and Y.

CBSE 16

i)	Let $C_X = C$	
	$C_Y = 4C$ (as it has a dielectric medium of $\varepsilon_r = 4$	
	For series combination of two capacitors	
	$\frac{1}{C} = \frac{1}{C_X} + \frac{1}{C_Y}$	
	$\Rightarrow \frac{1}{4\mu F} = \frac{1}{C} + \frac{1}{4C}$	
	$\frac{1}{4\mu F} = \frac{5}{4C}$	
	$\Rightarrow$ C= $5\mu F$	
	Hence $C_X = 5\mu F$	½
	$C_Y = 20\mu F$	½

ii)	Total charge $Q = CV$	
	$= 4\mu F \times 15\ V = 60\mu C$	
	$V_X = \frac{Q}{C_X} = \frac{60\ \mu C}{5\mu F} = 12\ V$	½
	$V_Y = \frac{Q}{C_Y} = \frac{60\ \mu C}{20\mu F} = 3\ V$	½
iii)	$\frac{E_x}{E_y} = \frac{\frac{Q^2}{2C_X}}{\frac{Q^2}{2C_Y}} = \frac{C_Y}{C_X} = \frac{20}{5} = 4 : 1$	1
	(Also accept any other correct alternative method)	

What is electrostatic shielding ? How is this property used in actual practice ? Is the potential in the cavity of a charged conductor zero ?

CBSE 16

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CBSE 16

The field inside a conductor is zero.	½
Sensitive instruments are shielded from outside electrical influences by enclosing them in a hollow conductor .	1
(any other relevant answer.)	
Potential inside the cavity is not zero/ potential is constant.	½

(a) A parallel plate capacitor (C_1) having charge Q is connected, to an identical uncharged capacitor C_2 in series. What would be the charge accummulated on the capacitor C_2 ?

(b) Three identical capacitors each of capacitance $3\mu F$ are connected, in tern, in series and in parallel combination to the common source of V volt. Find out the ratio of the energies stored in two configurations. CBSE 16

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a) Zero	½
b) We have $C_{series} = \frac{3\mu F}{3} = 1\ \mu F$	½
Also, $C_{parallel} = (3+3+3) = 9\mu F$	½
Energy stored = $\frac{1}{2}CV^2$	
∴Energy in series combination = $\frac{1}{2} 1 \times 10^{-6} \times V^2$	½
Energy in parallel combination = $\frac{1}{2} 9 \times 10^{-6} \times V^2$	½
∴Ratio = 1:9	½

What is the amount of work done in moving a point charge Q around a circular arc of radius ‘r’ at the centre of which another point charge ‘q’ is located ?

CBSE 16

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CBSE 16

Zero / No work done / None

(a) Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field ?

(b) A capacitor of capacitance C is charged fully by connecting it to a battery of emf E. It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change ?

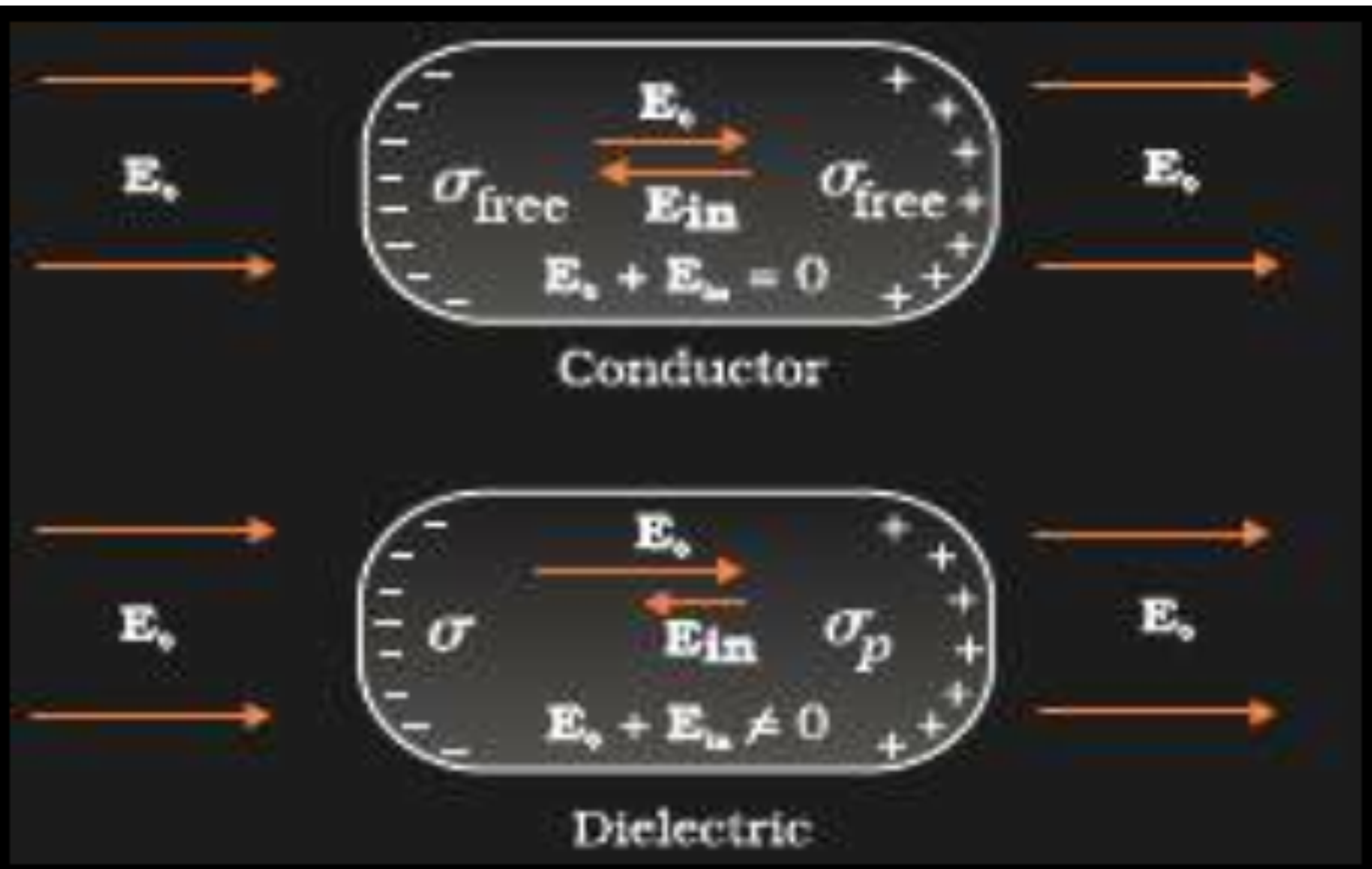
(i) charge stored by the capacitor.

(ii) field strength between the plates.

(iii) energy stored by the capacitor.

CBSE 16

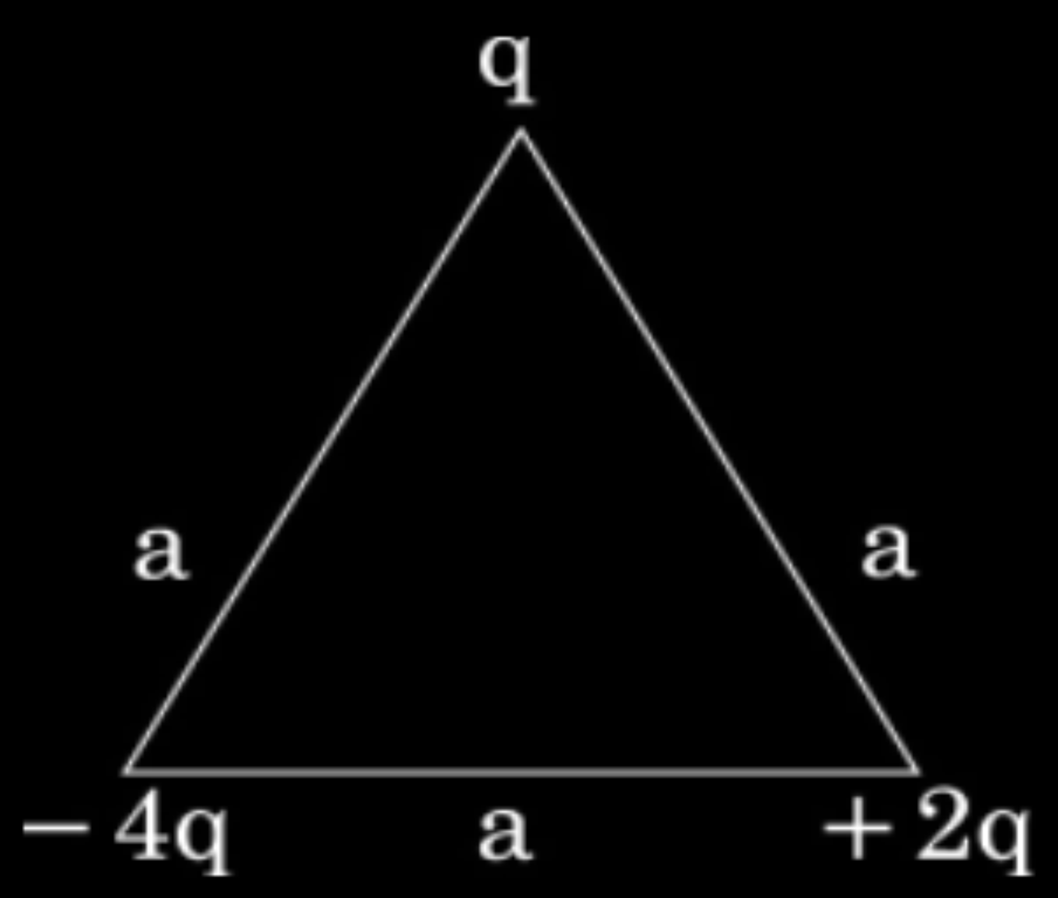
Justify your answer in each case.

a)  No electric field inside a conductor . (Give full credit to diagram. Give ½ mark if explanation only is given without diagram)	½ + ½
Induced electric field ,due to polarisation of dielectric, is in opposite direction to the applied field. $$E_{net} = E_0 - E_\rho$$	1
(b) (i) Charge remains same, as after disconnecting capacitor no transfer of charge take place.	½ + ½
(ii) Electric field, $E = \frac{\sigma}{\epsilon_0} = \frac{q}{\epsilon_0 A}$ remain same, as there is no change in charge.	½ + ½
(iii)Energy stored $= \frac{q^2}{2C} = \frac{q^2}{2\left(\frac{\epsilon_0 A}{d}\right)} = \frac{q^2 d}{2\epsilon_0 A}$	½
a. Energy will be doubled as separation between the plates(d) is doubled.	½

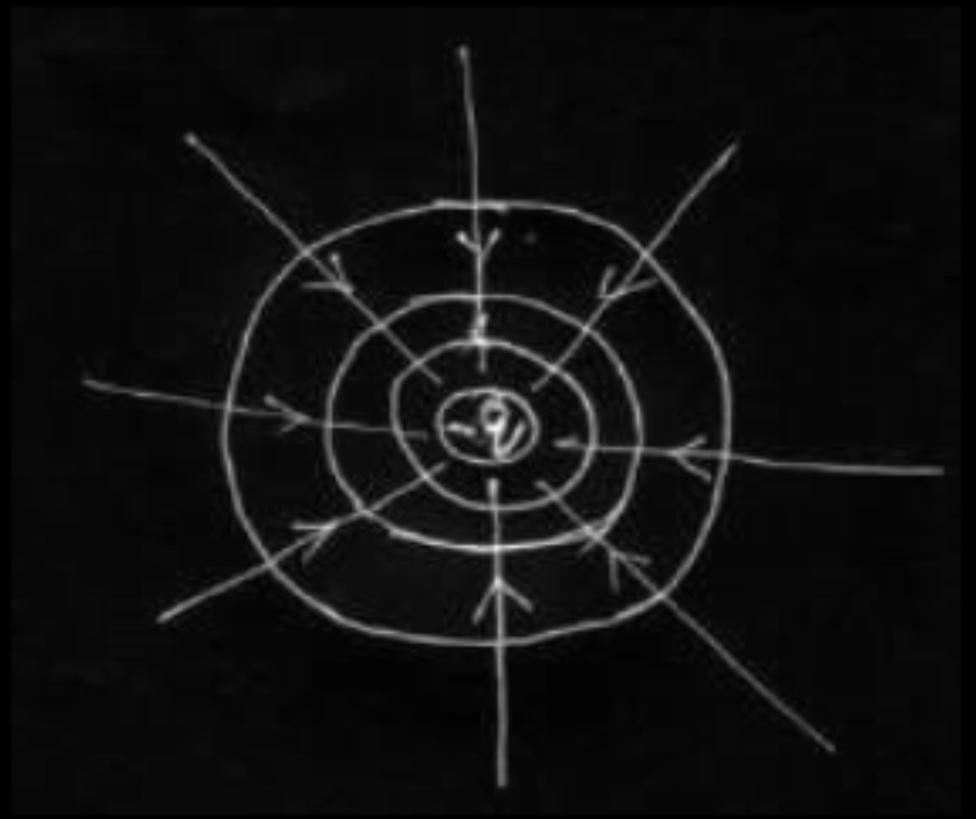
(a) Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at that point.

Draw a sketch of equipotential surfaces due to a single charge $(-q)$, depicting the electric field lines due to the charge.

(b) Obtain an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side 'a' as shown below.



CBSE 16

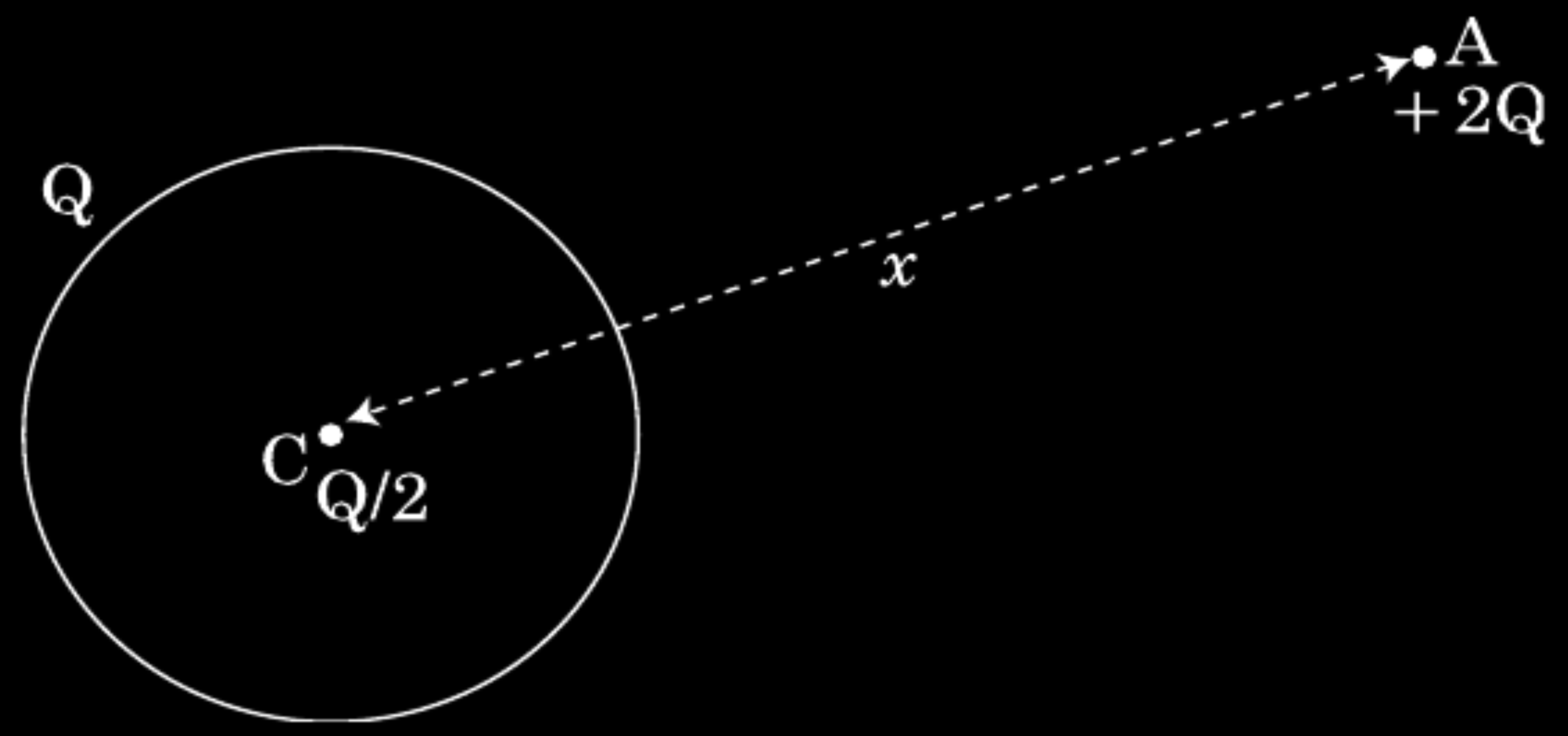
(a) If the field is not normal to an equipotential surface, it would have a non zero component along the surface. This would imply that work would have to be done to move a charge on the surface which is contradictory to the definition of equipotential surface.	1 ½
(Alternatively,	
Work done to move a charge dq, on a surface, can be expressed as	
$dW = dq(\vec{E}.\overrightarrow{dr})$	½
But $dW=0$ on an equipotential surface	½
$\therefore \vec{E} \perp \overrightarrow{dr}$)	½
Equipotential surfaces for a charge –q 	½ +½
(b) Work done to dissociate the system = -Potential energy of the system	½
$= \frac{-1}{4\pi\epsilon_0}\left[\frac{(-4q)(q)}{a} + \frac{(2q)(q)}{a} + \frac{(-4q)(2q)}{a}\right]$	1
$= -\frac{1}{4\pi\epsilon_0 a}[-4q^2 + 2q^2 - 8q^2]$	½
$= +\left[\frac{10q^2}{4\pi\epsilon_0 a}\right]$	½

A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge Q/2 is placed at the centre C and another charge +2Q is placed outside the shell at A at a distance x from the centre as shown in the figure.

(i) Find the electric flux through the shell.

(ii) State the law used.

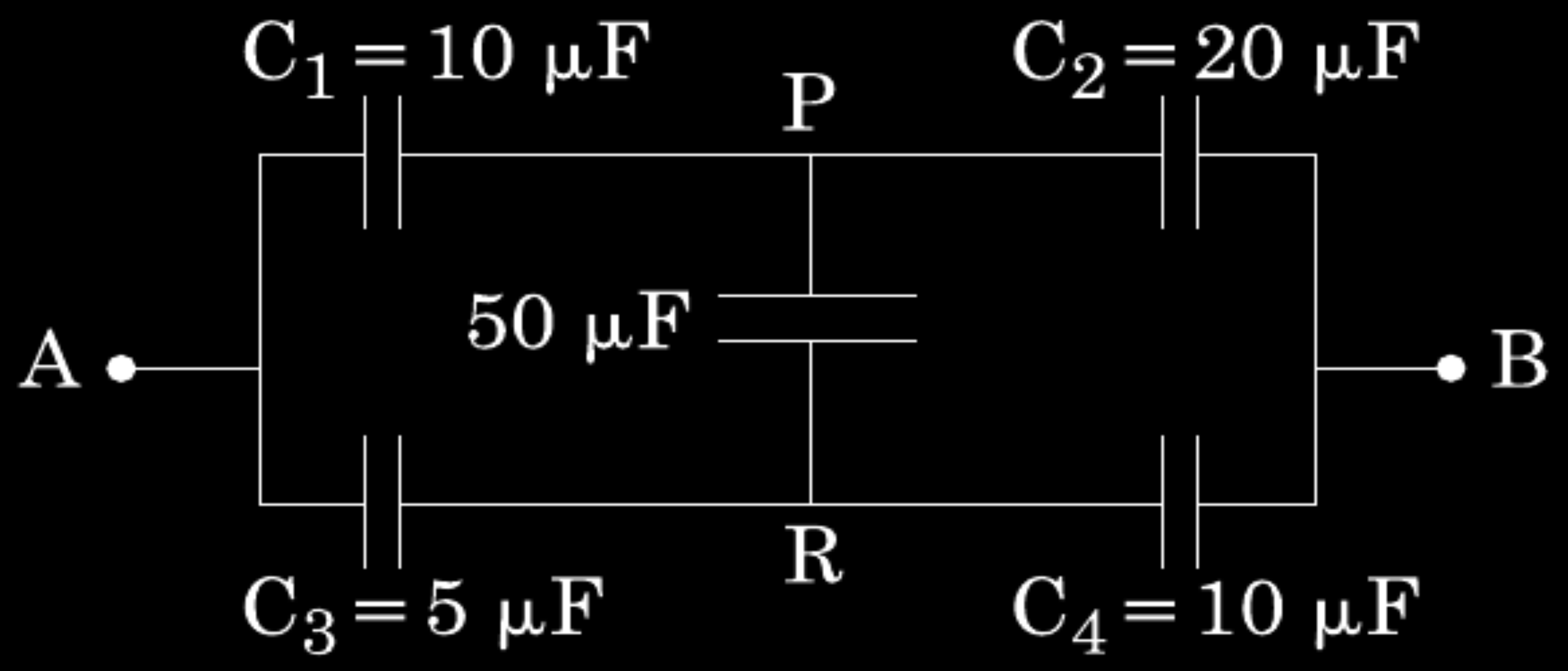
(iii) Find the force on the charges at the centre C of the shell and at the point A.



CBSE 16

(i) Electric flux through a Gaussian surface, $\varphi = \frac{total\ \ enclosed\ \ charge}{\epsilon_0}$	½
Net charge enclosed inside the shell q=0 ∴ Electric flux through the shell $\frac{q}{\epsilon_o}$=0 Award ½ mark even when the student writes - Electric flux through the shell is zero as electric field inside the shell is zero.	½
(ii) Gauss Law- Electric flux through a Gaussian surface is $^1/_{\epsilon_0}$ times the net charge enclosed with in it. Alternatively, $\oint \vec{E}.\overrightarrow{dS} = \frac{q}{E_0}$	1
(iii) Force on the charge at the centre i.e. Charget $^Q/_2 = 0$	½
$F_A = \frac{1}{4\pi E_0} \frac{2Q \times \left(Q + {^Q/_2}\right)}{x^2}$ $= \frac{1}{4\pi E_0} \frac{3Q^2}{x^2}$	½

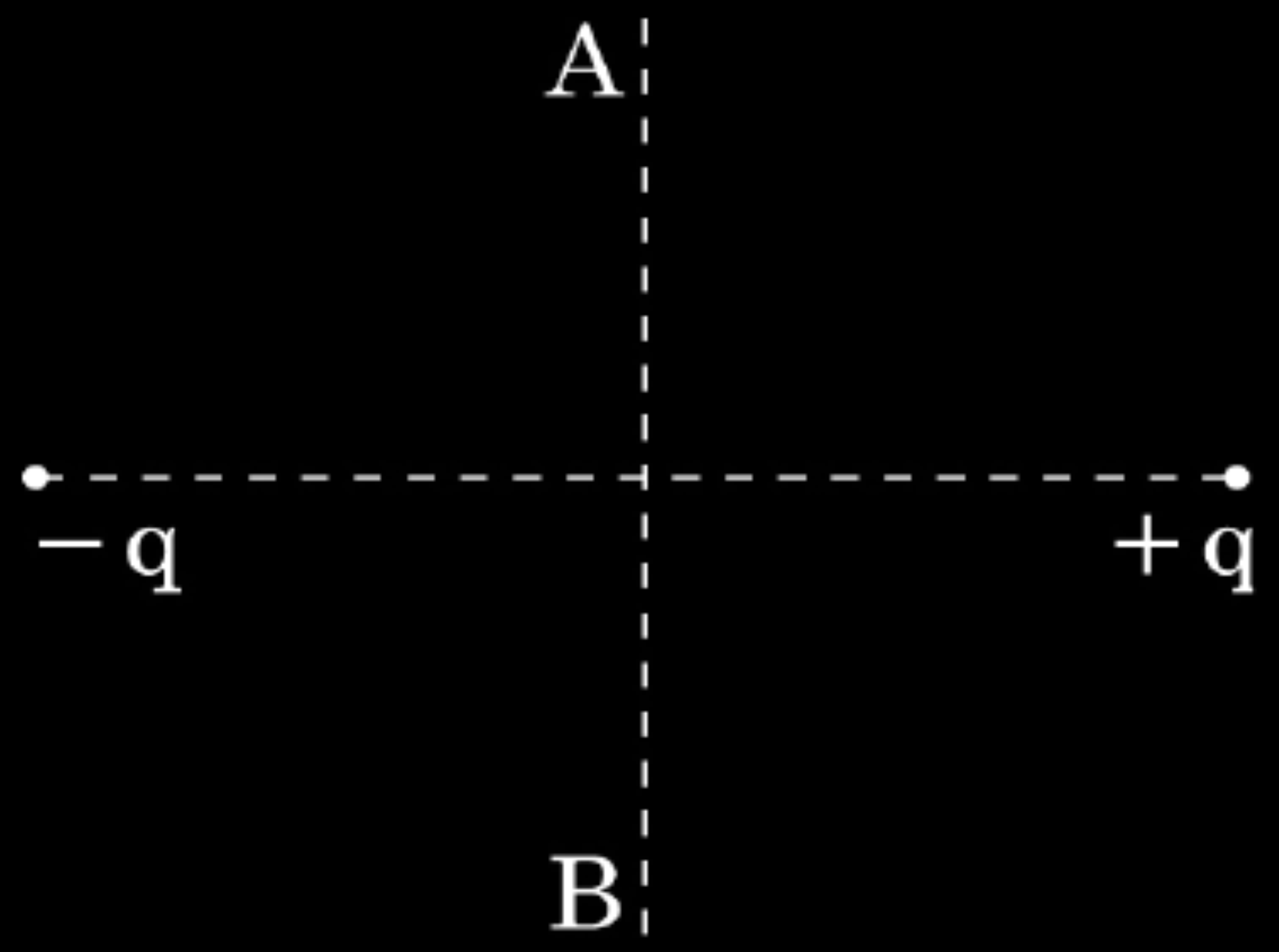
Calculate the equivalent capacitance between points A and B in the circuit below. If a battery of 10 V is connected across A and B, calculate the charge drawn from the battery by the circuit.



CBSE 16

$\because \frac{C_1}{C_2} = \frac{C_3}{C_4}$	½
This is the condition of balance so there will be no current across PR (50 μF capacitor)	
Now C_1 and C_2 are in series	
$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{10 \times 20}{10+20} = \frac{200}{30} = \frac{20}{3}$ μF	½
$\because$ C_3 and C_4 are in series	
$C_{34} = \frac{C_3 C_4}{C_3 + C_4} = \frac{5 \times 10}{5+10} = \frac{50}{15} = \frac{10}{3}$ μF	½
Equivalent capacitance between A and B is	
$C_{AB} = C_{12} + C_{34} = \frac{20}{3} + \frac{10}{3} = 10$ μF	½
Charge drawn from battery (q) = CV	½
= 10 x 10 μC	
= 100 μC or 10^{-4}C	½

A charge ‘q’ is moved from a point A above a dipole of dipole moment ‘p’ to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



CBSE 16

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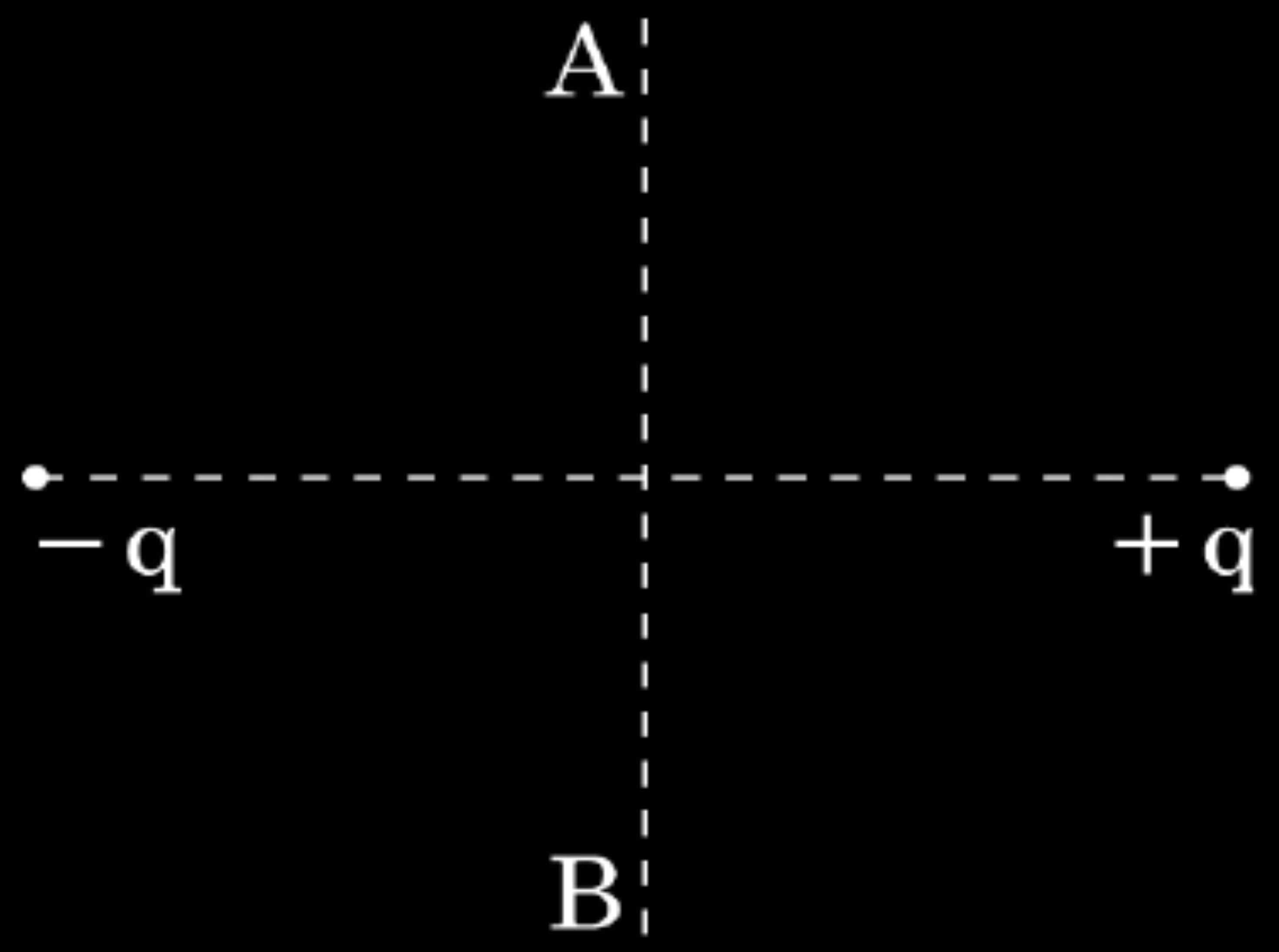

CBSE 16

No work is done /

$W = qV_{AB} = q \times 0 = 0$

CBSE 16

Define an equipotential surface. Draw equipotential surfaces :

(i) in the case of a single point charge and

(ii) in a constant electric field in Z-direction.

Why the equipotential surfaces about a single charge are not equidistant ?

(iii) Can electric field exist tangential to an equipotential surface ? Give reason.

CBSE 16

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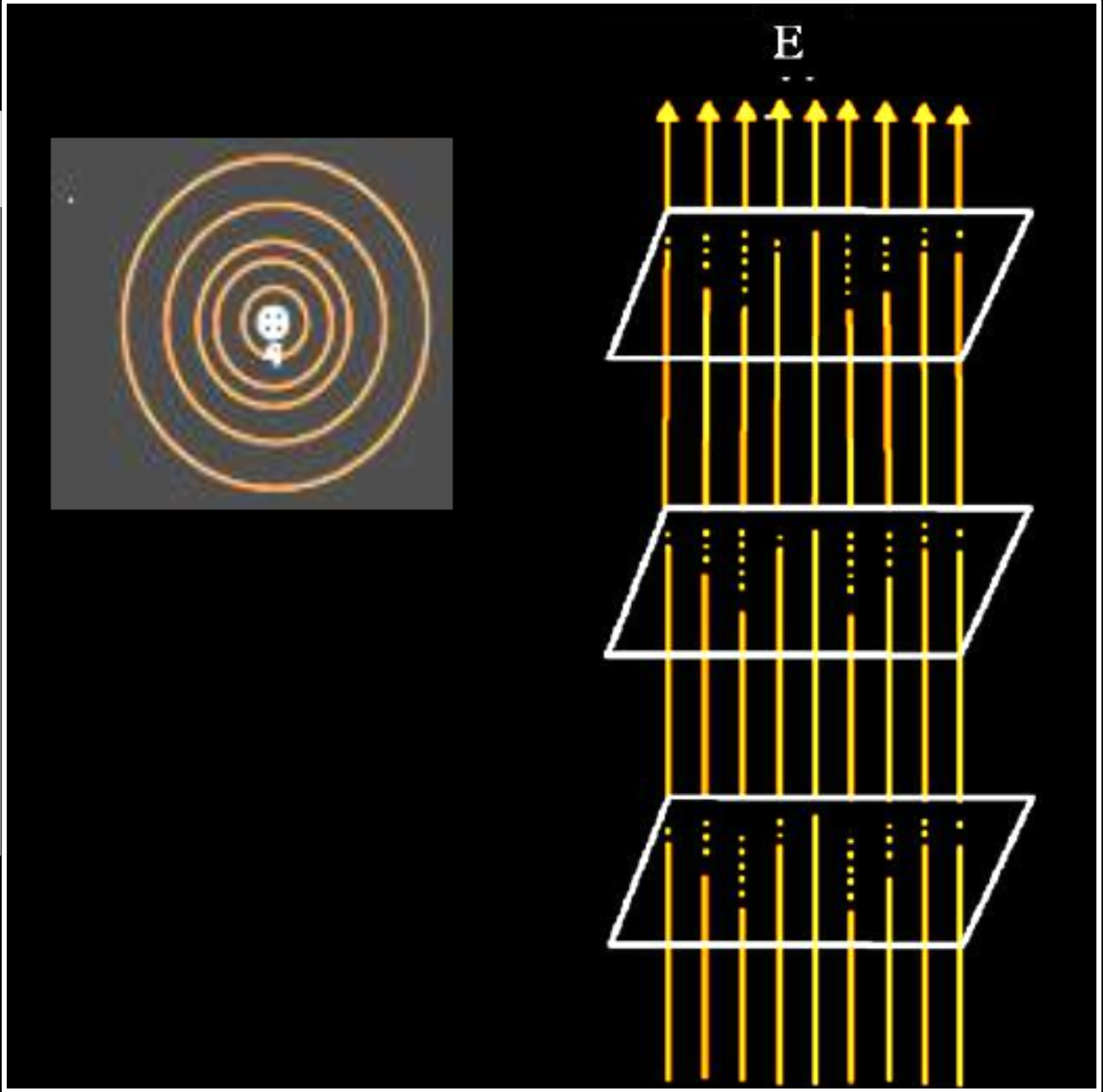
Surface with a constant value of potential at all points on the surface.	½
ii. 	
$V \propto \frac{1}{r}$	½
iii.No	½
If the field lines are tangential, work will be done in moving a charge on the surface which goes against the definition of equipotential surface.	½



(ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1 : 2 so that the energy stored in the two cases becomes the same. CBSE 16

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CBSE 16

$$U_s = \frac{1}{2} C_s V_s^2$$ ½

$$U_p = \frac{1}{2} C_p V_p^2 U$$ ½

$$\Rightarrow \frac{V_{series}}{V_{parallel}} = \sqrt{\frac{C_{equivalent\ parallel}}{C_{equivalent\ series}}}$$ ½

$$= \sqrt{\frac{\frac{C_1+C_2}{C_1C_2}}{C_1+C_2}}$$ ½

$$= \frac{C_1+C_2}{\sqrt{C_1C_2}} = \frac{3}{\sqrt{2}}$$

(i) If two similar large plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$ are separated by a distance d in air, find the expressions for

(a) field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.

(b) the potential difference between the plates.

(c) the capacitance of the capacitor so formed.

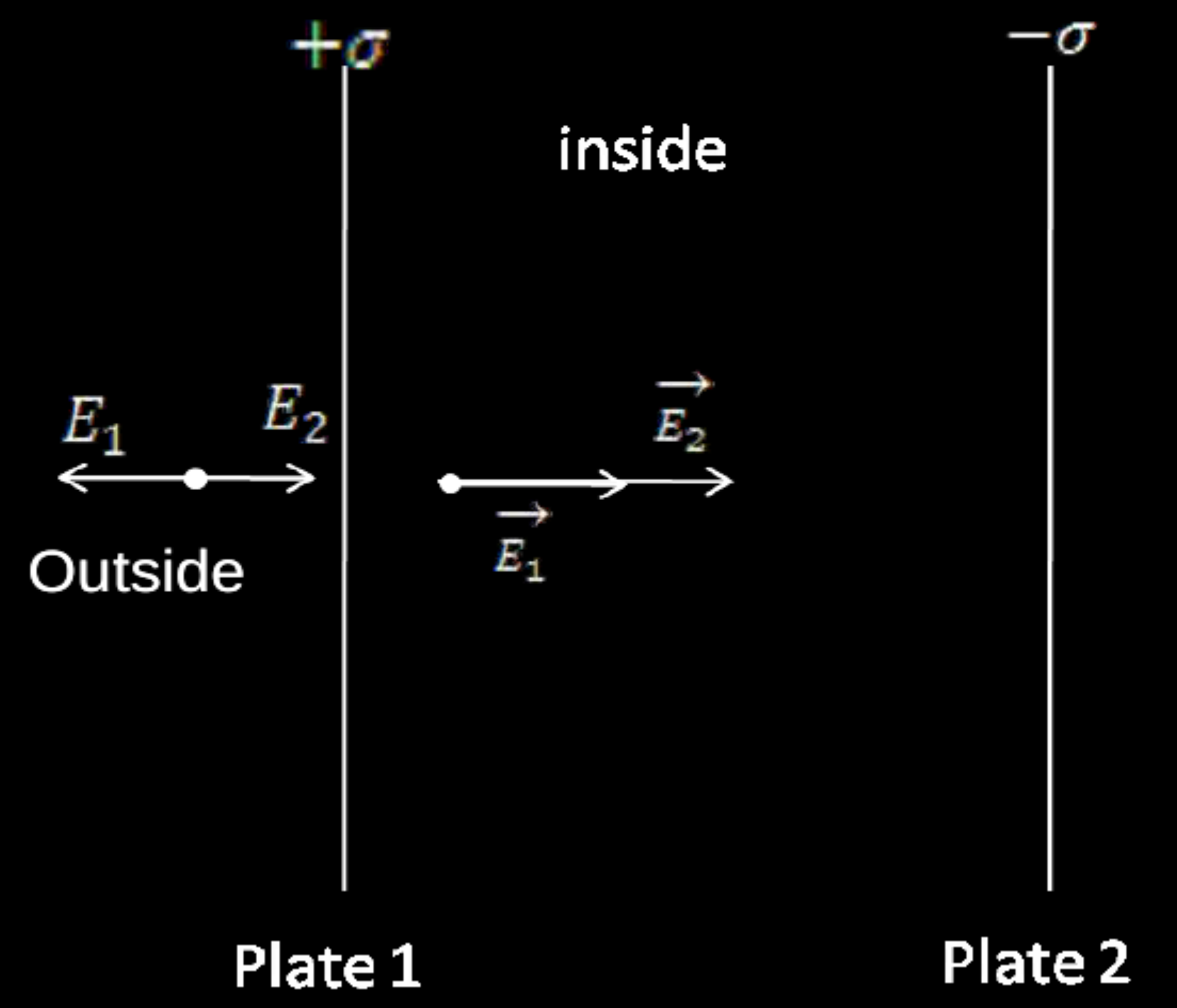
(ii) Two metallic spheres of radii R and 2R are charged so that both of these have same surface charge density σ. If they are connected to each other with a conducting wire, in which direction will the charge flow and why ?

CBSE 16

Inside

$\vec{E} = \vec{E_1} + \vec{E_2}$

$= \frac{\sigma+\sigma}{2E_0} = \frac{\sigma}{E_0}$ ½ + ½



Outside

$\vec{E} = \vec{E_2} - \vec{E_1}$

$= \frac{\sigma-\sigma}{2\epsilon_0} = 0$ ½

b. Potential difference between plates

$$V = Ed = \frac{1}{\epsilon_o}\frac{Qd}{A}$$ ½ + ½

c. Capacitance

½ + ½

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

ii. As potential on and inside a charged sphere is given

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r} = \frac{1}{4\pi\epsilon_0}\cdot\frac{4\pi r^2\sigma}{r}$$ ½

$\therefore, V \propto r$

½

Hence, the bigger sphere will be at higher potential, so charge will flow from bigger sphere to smaller sphere.